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Life Cycle Inventories of Detergents

Subproject “Detergents”

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Preface to Version v2.0

This report corresponds to the original report (Zah & Hischier (2004)) that has been updated with the changes reported in Frischknecht et al. (2006). The data for the crude palm oil resp. palm kernel oil production have been deleted from this report – herefore new data have been established within the framework of the biofuels project (Jungbluth et al. (2007)). In addition, errors in the datasets of the two “Fluorescent Whitening Agents” have been changed.

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Summary

This report of ecoinvent (report no. 12) contains the information concerning the most important ingredients contained in detergents.

Detergents and their ingredients have been the subject of public environmental interest since almost ten years. Several studies over these years have recorded ecological data on the manufacture of many of the chemicals contained in detergents. These studies were carried out at various institutes using differing basic data, leading to very non-uniform LCIs. A first harmonization in the field of detergents has been undertaken by EMPA at the end of the 90s (Dall'Acqua et al. (1999)), in the framework of a study by the Federal Environmental Agency (UBA, Berlin) and the Öko-Institut e.V. Freiburg (Germany). With the present project – ecoinvent – one other step into the direction of more harmonized data is done. This time, not only data from one sector (e.g. detergents industry) but from a multitude of different sectors are updated and harmonized, following the same rules (described in Frischknecht et al. (2007b)).

Detergents are mixed products that contain different types of substances. The individual components have quite distinct features within the washing process, although their effects sometimes influence each other synergistically. These substances may be divided into larger groups: tensides, builders, bleaches and auxiliary agents. The main part of this report describes the sources and assumptions used for establishing the datasets of the various types of detergent ingredients that are integrated into the database ecoinvent; in detail, these are:

- **Tensides:** LAS, FAS, AE, soap, esterquat and their respective precursors
- **Builders:** sodium silicates, sodium tripolyphosphates, polycarboxylates, zeolithes
- **Bleaches:** sodium perborates, sodium percarbonate
- **Auxiliary agents:** CMC, fluorescent whitening agents (FWA)

Furthermore, a specific dataset for steam used in the production process of these chemicals is reported.

All these datasets have been established in accordance with the quality criterias that have been fixed by ecoinvent. They are valuable for the situation in the year 2000 and they represent European conditions. One part of the datasets – although established according to the rules mentioned – is not shown on the level of a unit process

here in this report due to confidentiality reasons. The data behind these datasets are based on personal communication provided by the manufacturers in the framework of the mentioned former EMPA study about detergents. Nevertheless, all these datasets are established in a similar way like the other datasets (i.e. they use the same datasets for the various auxiliaries, fuels etc.) and therefore can be compared with those other data without any restrictions.

Within the update from ecoinvent data v1.01 to ecoinvent data v1.1, no substantial changes to the detergents data have been made. The cumulative datasets of the examined FWA have just been adapted according to the changes in the nomenclature of the metal resources.

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1 Introduction

Detergents and their ingredients have been the subject of public environmental interest since almost ten years, as can be seen e.g. in Giger et al. (1994). Individual chemicals have attracted various levels of attention because of their high level of environmental pollution. This has led to the industry making efforts towards ecological optimisation of detergents. Several research projects have been undertaken with this in mind, and have produced reliable basic information.

Several studies over the last ten years have recorded ecological data on the manufacture of many of the chemicals contained in detergents. These studies were carried out at various institutes using differing basic data. EMPA itself has also performed some Life Cycle Assessments, on the mandate of the SWI (Association of Swiss Soap and Detergent Manufacturers) and the relevant European manufacturers' associations (sector groups of CEFIC). Over the same period, the methodology of life cycle assessment has also developed further and is now an ISO Standard (ISO 14040 Series).

This heterogeneity necessarily leads to very non-uniform LCIs. However, this does not mean we should give preference to one or another methodology; within the context of the goals set at the time, each of these studies fulfil its purpose. The situation becomes problematic when results from the various individual studies are combined or compared. For example, if the complete life cycle of a detergent application is calculated on the basis of one particular equation, the end result must be questioned critically. The result is much more credible if all data are available in a harmonised form.

A first harmonization in the field of detergents has been undertaken by EMPA at the end of the 90s (Dall'Acqua et al. (1999)). The objective of that study, commissioned by the Federal Environmental Agency (UBA, Berlin) and the Öko-Institut e.V. Freiburg (Germany), was actually to prepare LCIs of chemicals contained in detergents in a harmonised and updated form. In addition to a new determination of process data in LCIs already published, basic data was replaced or supplemented by the appropriate modules from BUWAL 250 (Habersatter et al. (1998)) – e.g. data about energy production and consumption, transport data, and data on the manufacture of basic materials. Included in this first step of more harmonized data was also an adaption of parameter lists from the various sources used. For the first time, data were supplemented by a qualitative description of the data, which uses the format suggested in SPOLD 1997. All these details enhanced the comparability of individual LCIs.

With the present project – ecoinvent – one step further into the direction of more harmonized data is done. This time, not only data from one sector (e.g. detergents industry) but from a multitude of different sectors are updated and harmonized, following the same rules (described in Frischknecht et al. (2007b)). The influence of all these rules on the detergents data, collected by EMPA some years ago, is described in this report here. After a general introduction into the subject of detergents (chapter 2), the undertaken harmonization steps from the former EMPA publications to the ecoinvent project are described in chapter 3. The resulting datasets – sorted according to the different groups described in the following chapter – are then described in chapters 4 up to 7. The last data chapter (chapter 8) contains data for the production of steam – an often used energy carrier for the production of the various chemicals described in the preceding chapters.

2 Detergents

This chapter gives a brief overview of the application of detergent ingredients and of detergent use in the household and in commercial laundries. It is taken from the former EMPA report about detergents (Dall'Acqua et al. (1999)).

2.1 The Washing Process

Laundering and cleaning in aqueous wash liquors is a complex process in which numerous physical and chemical influences interact. A detailed description of the processes can be found in the literature e.g. in Oude (1992) or in Häussinger et al. (2000).

By “washing” we understand the removal of deposits which are difficult to dissolve in water and the sequestration of water-soluble soils. The washing result is influenced by factors such as textile characteristics, type of dirt, composition of the water, washing technique (mechanics, duration, temperature) and the composition of the detergents. These mutually influential factors can usually be altered only to a limited extent. The composition of the detergent is thus of major importance.

Water serves as a solvent for the detergent and for soluble salts in the dirt, and also as a transport medium for the dispersed or colloidal soil components. The composition of the water, in particular its natural salt content, can act negatively on the washing effect and the washing machine. The hardness of water caused by calcium and magnesium compounds can produce precipitates in the form of carbonate, or through a reaction with washing powder ingredients which may appear as irritating deposits. The sequestering agent and ion exchanger contained in detergents should bind these alkalis or heavy metal ions.

From the point of view of washing, the most important types of soil can be divided into groups. We can distinguish water-soluble substances (salts, sugar, urea, sweat), pigments (metals, carbonates, silicates, humus, soot), fats, proteins, carbohydrates and bleachable dyes. The removal of the dirt from the surface can be coupled to a transformation of the substance (i.e. a chemical reaction) or can take place without modification. Examples of the first case are redox reactions using whitening agents, where oxidisable substances (e.g. natural dyes) are broken down, and the breakdown of protein soiling by enzymes. In very many cases, however, the soil consists of substances that are not altered chemically and are therefore removed from the fibres by purely physical procedures. This is also reflected in the components of the detergents, the water-soluble sequestering agent and the water-insoluble ion exchanger.

The ability to remove dirt also depends on the type of textile substrate. Textile fibres that have a high content of calcium on the surface (e.g. cotton) differ substantially from synthetic textile fibres with low calcium content. The wetting ability and ability to wash out are also influenced by the level of hydrophobicity and hydrophilicity.

After being dissolved the dirt must be stabilised in the wash liquor and its redeposition prevented. This characteristic is described as the wash liquor's soil antiredeposition capability. The most effective principle is dispersion.

Detergents are mixed products that contain different types of substance. These may be divided into: tensides, builders, bleaches and auxiliary agents. The individual components have quite distinct features within the washing process, although their effects sometimes influence each other synergistically. This means that only the sum of all effects can ensure successful washing under different conditions.

2.2 Tensides

Tensides form the most important group of all detergent ingredients; they are contained in all types of detergents. They are surface-active substances (i.e. surfactants) and have a polar (=water-soluble, fat-insoluble) and a nonpolar (=water-insoluble, fat-soluble) part. They can therefore arrange themselves on the interface between polar and nonpolar substances and achieve thorough penetration of both phases. The structure of the water-insoluble part has a significant influence on the characteristics of the tensides. While tensides with relatively unbranched alkyl residues usually demonstrate good washing but less extensive wetting characteristics, comparably strongly branched tensides show good wetting but poorer washing effects.

Depending on the charge of the part of the molecule carrying the alkyl chain, we distinguish anion-active tensides (anionic tensides), nonionic tensides, cationic tensides and amphoteric tensides. Anionic tensides predominate in terms of quantity. Nonionic tensides have also achieved considerable significance. Cationic tensides are used almost exclusively in post-laundry treatment substances because of their incompatibility with anionic tensides. Amphoteric tensides have been of minor significance up to now.

Soap is one of the anionic tensides. It is not as significant in today's detergents as it was before synthetic tensides could be manufactured on a large scale. The disadvantage of sensitivity to water hardness is a further factor leading to reduced significance of soap in detergents. Today, soaps serve in detergents primarily as foam regulators. Linear alkylbenzene sulphonate (LAS) has largely replaced soap as an active component because of its economy and its advantageous properties. This is true not just for its washing but also its foaming properties. The foaming behaviour of a tenside is of great importance for its application. Although LAS is very foam-intensive, its behaviour can be kept under control through the addition of suitable foam regulators (e.g. soap). Because of its good solubility, LAS is also useful in the formulation of liquid products. Fatty alcohol sulphates (FAS) and sulphonates also possess a very favourable washing ability and are thus increasingly used today, in universal as well as specialised detergents.

A significant advantage of nonionic tensides over anionic tensides is that the hydrophobic and hydrophilic parts of the molecule can be better tailored to one another. Optimal properties of adsorption and washing effect, with the same hydrophobic residue, may be produced using an appropriate ethoxylation level (often shortened to EO level). The washing effect reaches a peak at a certain EO level and then decreases. The proportion of nonionic tensides in the whole tenside production has been increasing over the last few years. These are primarily fatty alcohol ethoxylates (AE). The increased use is largely due to the favourable washing properties, especially for synthetics and at low temperatures.

Cationic tensides show antagonistic behaviour to anionic tensides in terms of their ability to attach to solid bodies. Equimolar mixtures of anionic and cationic tensides are almost unadsorbed onto surfaces and are also not washing-effective. However, mixtures of nonionic tensides and cationic tensides are largely compatible. Such mixtures are partly used in special detergents with a scrooping effect (softeners). The tenside class of distearyl-dimethyl-ammonium chlorides (esterquats) are particularly used as fabric softeners.

2.3 Builders

Builders in detergents have central significance in the washing process. The function of these substances is primarily to bind the calcium and magnesium ions which stem partly from the water and partly from dirt or textiles; and to support the effect of the tenside. Builder substances include alkalis (such as sodium carbonate and sodium silicates), sequestering agents (sodium tripolyphosphate, EDTA, NTA), ion exchangers (zeolite A, SKS-6) and co-builders (Polycarboxylate).

As washing alkalis raise the pH value, dirt and textile fibres become more strongly negatively charged and thus their mutual electrostatic repulsion is increased. Apart from this, the hardness components of

the water are removed. The builders currently in use no longer remove the hardness formers from the water, but eliminate them through complex formation.

As a rule, temperature and the concentration of the sequestering agent are decisive factors in the elimination of high-value metal ions. For most sequestering agents, the binding ability decreases as the temperature increases. An extensive ability to bind earth alkali ions is important because of their high concentrations in water, compared to other high-value ions. Heavy metal ions, which have a very negative effect on the washing process even in the smallest quantities, must also be removed. Selectively active sequestering agents in small concentrations are generally used for this. In addition to the elimination of disturbing cations and the existence of a good washing effect, the dispersal of dirt to prevent greying are of the greatest significance for the washing process. Because of their adsorption by charged soil pigments, sequestering agents often act as a means of dispersal for these pigments.

Like the low molecular weight sequestering agents, the removal of disturbing polyvalent metal ions can also be achieved by ion exchangers. The insoluble zeolite A, a sodium aluminium silicate, has proved particularly useful for the washing process and is also of economic interest. Ion exchange is dependent on the size of the ions and their level of hydration. Besides Ca and Mg ions, Pb, Cu, Ag, Cd, Zn and Hg ions are also exchanged. In addition, it is advantageous to use combinations with water-soluble sequestering agents. For example, the builder system sodium tripolyphosphate / zeolite A guarantees excellent washing results, and largely prevents incrustations due to the precipitation of calcium and magnesium phosphate.

Co-builders (e.g. watersoluble polymeres like polycarboxylate), besides their function as sequestrating agents, show also a pronounced threshold effect. This means they have an adsorption effect on amorphous and crystalline particles and dirt also in low concentration. These polymeres therefore prevent greying of the laundry.

2.4 Bleaches

In chemical bleaching, stains that have an affinity to fibre and cannot be washed out are decoloured oxidatively through degradation of the chromophore systems. The bleaching effect achieved is determined by the type of whitening agent, its concentration and the time it spends in the washing process, the washing temperature and the stains which are to be bleached. Bleachable stains are usually of vegetable origin.

Two processes of oxidative whitening have been successful: peroxide and hypochlorite bleaching. In peroxide bleaching, which is most common in Europe, perhydroxyl anions arise as an active transition substance in the alkaline medium of hydrogen peroxide. Hydrogen peroxide is delivered by inorganic peroxides and peroxyhydrates, the most important representative of which is sodium perborate. This hydrolyses in water with the formation of hydrogen peroxide, but on the other hand in detergents it has excellent stability. Sodium percarbonate (actually peroxyhydrates) must be additionally stabilised in order to be stored.

To achieve a good whitening effect with sodium perborate at temperatures below 60 °C, bleach activators (TAED) are often used. These are chemicals which, with the whitening agent, form per-acids. At low temperatures these have a greater ability to oxidise than hydrogen peroxide.

Traces of copper, manganese and iron ions may enter the wash liquor through the dirt on the laundry, and also through tap water. These heavy metals can cause the breakdown of the bleaching agent, and thus chemical damage to the fibres in the washing process. Adding bleach stabilisers (magnesium silicate) stops most of the catalytic breakdown of perborate. A further way to eliminate heavy metal traces is the addition of selectively active sequestering agents of the EDTA type or organic sequestering agents.

2.5 Auxiliary Agents

Tensides, builders and whitening agents are the main components of modern detergents in terms of quantity, while the additives described below are used only in small quantities. However, they should not be overlooked in today's detergent formulations.

Enzymes:

Protein stains such as those from milk, cocoa, blood, egg yolk and grass are often as tough to remove from laundry as bleachable stains. Such impurities can usually be removed without difficulty in the washing process by protein-splitting enzymes. The effect of the enzymes rests on the enzymatic hydrolysis of peptide bonds. There are also fat-splitting enzymes.

Soil antiredeposition agents (Greying inhibitors):

Greying inhibitors act by attaching themselves to dirt particles and thus preventing soil and fibres re-attaching to one another. Classic greying inhibitors are carboxymethyl celluloses (CMC). These substances however work only on cellulose fibres such as cotton. Certain tensides and special nonionic polymers are particularly suitable for synthetic fibres. Modern detergents usually contain mixtures of anionic and nonionic polymers.

Foam regulators:

Increased foam formation in drum washing machines leads to overfoaming of the wash liquor, often with substantial loss of wash-effective substances. In addition, the mechanical processing of the textiles to be washed is reduced. For this reason, additional foam regulators are used. However, only a few of the existing regulators described in the patent literature have succeeded (e.g. soap).

Corrosion inhibitors:

The drum washing machines on the market today almost exclusively contain drums and laundry tubs made of corrosion-resistant stainless steel. Nevertheless, the fact that various machine components, particularly in older machines, are made of aluminium, has to be taken into account. To prevent the corrosive effect of the alkaline wash liquor on this metal, corrosion inhibitors in the form of sodium silicates are often mixed into washing powders.

Fluorescent whitening agents (FWAs):

FWAs are organic substances (fluorescent dyes) that are able to convert part of the invisible ultraviolet light contained in daylight into longer-wave blue light. The yellowish tone of a washed and bleached material is due to some of the blue wavelengths being absorbed from the white light falling on it, so that the reflected light lacks these blue wavelengths. The missing blue is added by the wavelengths emitted by the FWA, so that there is a higher level of whiteness and greater brightness. FWAs however have no actual washing action and are only effective when the laundry is clean.

Perfumes:

These substances are present not only to give the washing powder a pleasant smell, but more to cover the smell of the alkaline wash liquor that arises during the washing process. Furthermore, the washed laundry should be given a fresh and pleasant odour. The perfumes themselves are very complex mixtures.

Colorants:

Certain ingredients of washing powders are dyed for better recognition, so that the finished powder contains coloured particles as well as the basic white powder. Monochrome washing powders are also now available commercially. In liquid detergents the use of colorants is a general practice. Favourite colours are pink, blue and green.

Stabilisers and formulation additives:

Inorganic salts, especially sodium sulphates, are used as stabilisers for washing powders. They are intended to give the detergents good trickling ability, dosability and solubility. Clumping or dusting is also prevented.

2.6 Household Detergents

The detergents found on the market can be divided according to type of use into universal (all-purpose) detergents, special detergents, laundry aids and aftertreatment aids.

Under the term universal detergent we class detergents that are suitable for all washing procedures, and usually also at all washing temperatures. Previously, they were used in a somewhat different composition for whites up to 95 °C. Due to better-performing tenside combinations and enzymes, however, washing characteristics at temperatures between 30 °C and 60 °C have been significantly improved.

For certain kinds of textiles, such as woollens or synthetic fibres, special detergents were recommended for sensitive colours and when hand washing. They usually contain sodium perborate and no FWA. They are used if light-sensitive colours are present, or if there is a danger of dyes in pastel-coloured textiles running. Detergents for wool also contain no FWA. Many of these special detergents can be used for hand or machine washing. Greying inhibitors are added to detergents for net curtains, to prevent greying or yellowing. Strongly foaming hand-wash detergents are intended for occasional handwashing in basins.

Laundry aids of the pre-wash type assist in the removal of problematic stains, often only local, and are used in addition to detergents. Besides the classic fabric softeners, stain removers have also gained in significance. Fabric softeners are mostly strongly alkaline additives that loosen particularly tough soil, although pre-soaking plays a special role. Stain removers are tenside-rich products which are used to advantage on strong, localised fat stains. Water softeners are laundry aids that should also be mentioned. They usually contain sequestering agents and ion exchangers.

After the actual washing process, in which the main goal is the removal of dirt, the textiles are often submitted to after-treatment. This attempts to restore certain characteristics to the laundry. Thus the textiles should be given elastic stiffness, a good sit, fullness, fatness, shine, softness, antistatic properties and so on as required. The main active substances of the soft rinses are cation tensides. Stiffness and fullness are obtained, on the other hand, with the use of stiffeners based on copolymers of vinyl acetate. The textiles treated with them are made firmer.

2.7 Industrial Detergents

Although the detergents used in households and those used in commercial laundries act basically in the same way, the detergents used in laundries differ in that they must be tailored to the special conditions of institutional washing. In contrast to households, soft water is often available in well-run laundries. It is usually obtained from softening plants by means of ion exchangers.

The development of continuously operating large-scale laundry plants, with efficient use of water and energy, has made particular combinations of detergents necessary. This need has also arisen because

of the desire to modify the procedure according to the type of washing and soiling, or to strengthen the effect of the detergent in a particular direction. Various partly tailored combinations are therefore used. The trend however is towards the simpler use of a single product, like the universal household detergents.

According to application, industrial laundries also work with special products. These include, for example, whitener-free detergents for coloureds and delicates, detergents for work clothes, and enzyme-containing products for particular protein-containing stains, or detergents with a disinfecting action.

3 Applied Harmonisation Rules

As already mentioned, this report aims to transform all data listed in the EMPA-report "Life Cycle Inventories for the Production of Detergent Ingredients" (Dall'Acqua et al. (1999)) to the quality standards of ecoinvent (Frischknecht et al. (2007b)). This transformation comprised different working steps:

1. Definition of suitable processes: goal was to define single process steps. In some cases, sequencing processes had to be linked due to missing process data.
2. Evaluation of single process data: the EMPA-report Dall'Acqua et al. (1999) lists only fully aggregated inventory data. Therefore the data sources that have been used for the EMPA report had to be consulted.
3. Linking to ecoinvent-processes: the evaluated single process inventories had to be linked to the respective ecoinvent processes.
4. Transformation to relevant units: All data on single process level had to be transformed to the respective physical units as described in Frischknecht et al. (2007b).
5. Adding infrastructure data: data on infrastructure consumption was missing in the EMPA-report and had to be added.

Finally, all washing ingredients of the EMPA-report Dall'Acqua et al. (1999) have been successfully transformed in the ecoinvent compatible form.

For those datasets that are taken – due to a lack of other possibilities – as cumulated datasets directly from the mentioned EMPA report (similar to e.g. plastics in Hischier (2007)), general rules have been established in order to ensure a harmonized integration of these datasets. The respective rules for the integration of the resources are summarized in Tab. 3.1. Concerning the emissions to air and water, the same rules are applied like for the plastics. They are shown in details in table 5.3 to 5.6 of the plastics report of ecoinvent (Hischier (2007)).

Tab. 3.1 Translation rules for integration of resource consumption in datasets from Dall'Acqua et al. (1999) into the database ecoinvent

Dall'Aqua (1999)	Ecoinvent database	Remarks / Calculation procedure
(i) Resources, commercial fuels		
1 kg Raw brown coal	1 kg coal, brown, in ground	-
1 Nm ³ Crude gas (natural gas)	1 Nm ³ gas, natural, in ground	No distinction between feedstock and fuel use
1 kg Crude oil from drilling well	1 kg oil, crude, in ground	No distinction between feedstock and fuel use
1 kg Raw hard coal	1 kg coal, hard, unspec., in ground	-
1 g Uranium from ore	1 g Uranium, in ground	-
1 kg Wood	1.65 * 10 ⁻³ m ³ wood, unspecified, standing	Based on d=607 kg/m ³ – i.e. u=20% for wood, unspecified, CH-Mix (No distinction between feedstock and fuel use)
1 MJ Pot. Energy water	1 MJ energy, potential, stock, in barrage water	-
(ii) Resources, feedstock		
1 Nm ³ Crude gas (feedstock)	1 Nm ³ gas, natural, in ground	No distinction between feedstock and fuel use
1 kg Crude oil f. drilling mill (feedstock)	1 kg oil, crude, in ground	No distinction between feedstock and fuel use
1 kg Barytes	1 kg Barite, 15% in crude ore, in ground	-
1 kg Bauxite	0.24 kg Aluminium, 24% in bauxite, 11% in crude ore, in ground	Calculation based on assumption, that 1 kg bauxite results in 0.24 kg aluminium (see name of resource)
1 kg Lead	1 kg Lead, 5%, in sulfide, Pb 2.97% and Zn 5.34% in crude ore, in ground	-
1 kg Boron	1 kg Borax, in ground	-
1 kg Dolomite	1 kg Dolomite, in ground	-
1 kg Iron ore	1 kg Iron, 46% in ore, 25% in crude ore, in round	-
1 kg Fluorspar	1 kg Fluorspar, 92%, in ground	-
1 kg Wood	1.65 * 10 ⁻³ m ³ wood, unspecified, standing	Based on d=607 kg/m ³ – i.e. u=20% for wood, unspecified, CH-Mix (No distinction between feedstock and fuel use)
1 kg Potassium	0.476 kg Sylvite, in ground	-
1 kg Limestone	1 kg Calcite, in ground	-
1 kg Magnesium	1 kg Magnesite, 60% in crude ore, in round	-
1 kg Phosphate rock	0.44 kg Phosphorus, 18% in apatite, 12% in crude ore, in ground	-
1 kg Sand, clay	1 kg Clay, unspecified, in ground	-
1 kg Sulphur	1 kg Sulfur, in ground	-
1 kg SO ₂ secondary	-	It is already included into the amount of raw oil !
1 kg Rock salt	1 kg Sodium chloride, in ground	-

4 Tensides

4.1 Precursors of Fatty Alcohols

4.1.1 Production of n-olefins

Olefins is a common name for all aliphatic hydrocarbons that contain at least one carbon-carbon double bond (Behr (2000)) – i.e. monoolefins with only one C=C double bond have the empirical formula C_nH_{2n} – e.g. ethene, propene or isopropene. The name olefin signifies oil-forming gas – i.e. these substances have the property to form oily liquids when reacting with halogens. n-Olefins in the context of the detergents production means higher (C_{10} to C_{16}) olefins, produced either by dehydrogenation of the corresponding n-paraffins or by ethylene oligomerization.

Production and use

As just mentioned, two different raw materials can be used for the production of n-olefins: n-paraffins or ethylene.

- *Dehydrogenation of n-paraffins*: Therefore a combination of two processes – Pacol (paraffin catalyst olefin) and Olex (olefin extraction) – is most often used. The first step, at a temperature of 450 to 510°C and a pressure of 0.3 Mpa, uses a hydrogen : carbon ratio of 9:1. The whole process uses a platinum catalyst (0.8 wt%) on alumina, activated by adding lithium, arsenic or germanium. The resulting paraffin-olefin mixture is then separated in the second process, using molecular sieves, absorbing olefins more strongly (Behr (2000)).
- *Ethylene oligomerization*: Two different methods are mainly used for the production of n-olefins out of ethylene: the SHOP (Shell's higher olefin process) or the Ziegler reaction.

The total European production according to Franke et al. (1995) is about 500 kt/a, including the amounts used for the direct production of LAB/LAS. Its demand is almost entirely limited to detergents, and there to the surfactants producers.

Process data

Main data source is the petrochemical intermediates report from the ECOSOL study of the European LCI Surfactant Study group (Franke et al. (1995)), representing a mix of the two above shown production ways. Its exact relation is not mentioned in Franke et al. (1995), but assuming that both ways have a similar yield, the relation n-paraffins to ethylene for the synthesis of n-olefins is of 35% to 65%. A split into two separate modules is not possible with the information in Franke et al. (1995).

The data from this source have been transformed into the ecoinvent format according to the rules described in the chapter about the n-paraffins production (see Althaus et al. (2007)). Thereby it has been taken into consideration that all data described in Franke et al. (1995) represent cradle to gate inventories and therefore – only the difference between the data for n-olefins and its raw materials (n-paraffins, ethylene) have been taken as basis for the transformation. Tab. 4.1 summarizes the resulting input and output data for the production of 1 kg of n-olefins. As can be seen there, the mass balance between input and output shows less input. According to Janzen (1995) there are several possible reasons therefore. Nevertheless, in order of not adding any additional uncertainty / source of error, the data from Franke et al. (1995) are used here.

Tab. 4.1 Input and Output data for the production of 1 kg n-olefins (data from Franke et al. (1995), transformed according to the description in the n-paraffins chapter in Althaus et al. (2007))

[per kg n-olefin produced]			[per kg n-olefin produced]			[per kg n-olefin produced]		
INPUTS			OUTPUTS					
<i>materials</i>			<i>emissions to air</i>			dissolved solids	kg	3.00E-03
n-paraffins	kg	0.346	waste heat	MJ	7.75E-01	suspended solids	kg	2.93E-05
ethylene	kg	0.633	PM, > 10 µm	kg	1.55E-05	BOD	kg	3.99E-05
<i>energy</i>			PM, > 2.5 µm & < 10 µm	kg	2.08E-05	COD	kg	1.66E-03
electricity	kWh	2.15E-01	PM, < 2.5 µm	kg	1.21E-05	sulphides	kg	1.71E-05
heat from heavy oil	MJ	6.07E-02	nitrogen oxides	kg	2.10E-04	oil	kg	2.78E-05
heat from natural gas	MJ	8.27E+00	NM VOC, unspecified	kg	7.13E-03	chromium	kg	4.37E-08
heat from coal	MJ	9.90E-01	sulfur dioxide	kg	4.07E-05	iron	kg	1.07E-06
<i>transports</i>			carbon monoxide, fossil	kg	8.63E-05	aluminium	kg	9.52E-09
lorry, 32t	tkm	0.091	aldehydes, unspecified	kg	3.00E-06	nickel	kg	4.27E-08
<i>waste</i>			ammonia	kg	1.50E-06	phosphates as P ₂ O ₅	kg	3.94E-06
solid waste, incineration	kg	0.00536	carbon dioxide, fossil	kg	4.86E-02	zinc	kg	6.79E-07
			<i>emissions to water</i>			ammonia	kg	7.14E-07
			acid	kg	3.06E-05	sulphates	kg	4.81E-04
			fluorides	kg	1.30E-06	chloride	kg	2.30E-04

The shown transport amount is the result of a calculation, based on the reported amount of transport energy as well as the assumption, that all transports are on the road and therefore use only oil as energy source. More details about the calculation can be found in the n-paraffins chapter in Althaus et al. (2007).

Besides these inputs, the following additional inputs are integrated into the dataset of olefins production:

- *Water, cooling water:* no information about cooling installations is given in the different sources. Nevertheless, as both processes work at high temperatures it is assumed that a cooling system exists. According to the methodology used within the different chemicals, a cooling water amount of 0.024 m³/kg product is assumed based on Gendorf (2000) (see e.g. trimethylamine in) Althaus et al. (2007).
- *Water, process water:* no information about process water amounts is given in the different sources. Nevertheless, the emissions to water in Franke et al. (1995) shows that most of them are from the process – therefore, a process water amount of 25% of the cooling water amount is assumed.
- *Infrastructure:* As no information is given about the infrastructure, the general module for the chemicals production “chemical plant, organics” is used as a first approximation.

4.1.2 Production of crude palm oil and crude palm kernel oil

In the framework of the work on ecoinvent v2.0, new data for the palm oil chain have been established and they replace the former data out of Hirsinger et al. (1995b). These new data are described in ecoinvent report No. 17 (Jungbluth et al. (2007)).

4.1.3 Production of crude coconut oil

Coconut oil is – similar to palm oil and palm kernel oil – an oleochemical raw material used in the detergent surfactant production. They are all three triglycerides, consisting of fatty acids with different chain lengths – C₁₂ to C₁₄ in the case of coconut and palm kernel oil, C₁₆ to C₁₈ for palm oil – and of

glycerol (Hirsinger et al. (1995b)). Although about 75% of these oils are used in the food industry, they are also important raw materials for the surfactant industry – especially for the fatty alcohol production.

Production and use

The production starts with harvesting and husking of coconuts, followed by the copra production and the coconut oil mill producing crude coconut oil. The main characteristics of these three steps are described in the following:

- *Coconut harvesting and husking:* Coconut palm trees start bearing fruits at an age of about 5 to 7 years and have than a productive life of about 60 to 70 years. Within this productive life, every 45 to 60 days the coconuts can be harvested from the tree. Mature coconuts are cut from the trees with a sharp knife. Then the thick, fibrous husk is removed and the remaining part of coconuts is transported from there to a central copra drying area.
- *Copra production:* In a first step the coconuts are cut into two halves, and the coconut water is taken away. This step is followed by a first drying step, after which a separation of meat and shell is made manually. Another drying step follows resulting in the production of copra (coconut meat with a moisture content of 3 to 12%). Finally copra is transported – either in bulk or in 60 kg cloth bags – to the copra or coconut oil mill.

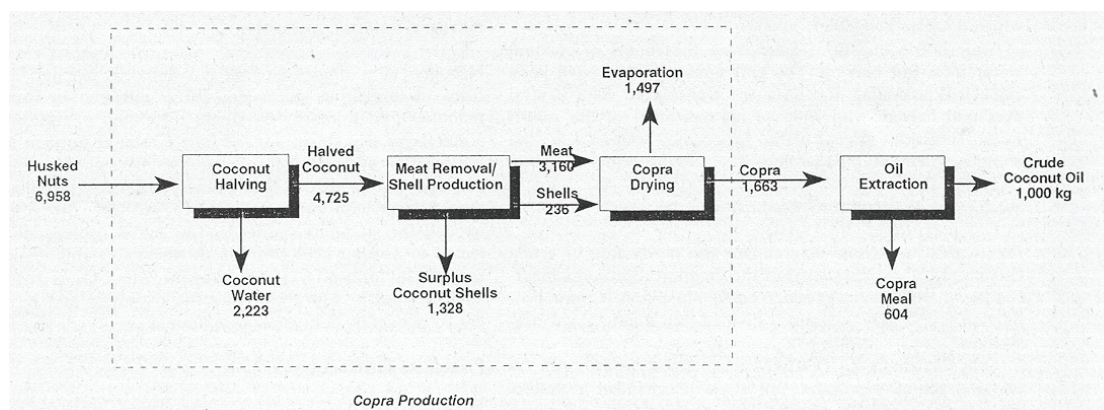


Fig. 4.1 flow diagramm of the production of crude coconut oil (fig.4 out of Hirsinger et al. (1995b))

- *Copra mill (Coconut oil mill):* two different extraction processes are possible for the production of crude coconut oil – solvent extraction and full mechanical pressing process. On the phillipines – the main producer of coconut oil – about 70% are mechanical pressing processes. There, high-pressure screw presses are used, while in the solvent extraction much lower pressure in screw presses is used. The remaining copra cake is in both cases further treated for animal feed.

The World production according to Hirsinger et al. (1995b) in 1993 was about 2'900 kt of coconut oil. Thereof about 1'800 kt are used in the food industry and only about 1'000 kt are used in the industry (i.e. surfactants production).

Process data

Main source for the data is the oleochemical raw material report from the ECOSOL study of the European LCI Surfactant Study group (Hirsinger et al. (1995b)). There, the production of crude coconut oil is described in details, including all co-products that got allocated parts of the environmental load of

these processes. For the study of Hirsinger et al. (1995b) it is assumed that 55% of the copra is dried by the sun, while the remaining 45% are tapahans dried. Furthermore, it is assumed in that study, that 90% of the crude oil is produced by means of mechanical extraction. A split into the above described different production steps is not possible with the information in Hirsinger et al. (1995b), and therefore only a summary dataset for the production of coconut oil can be established here.

The data from Hirsinger et al. (1995b) have been transformed into the ecoinvent format according to the rules described in the chapter about the n-paraffins production (see Althaus et al. (2007)). Instead of the UCTE mix, the philippinean one (according to the information in Dall'Acqua et al. (1999)) is used for the calculation of the amount of electricity within the total energy consumption, while for the dataset itself, due to a lack of other electricity mixes, the UCTE mix is used as a first approximation. Tab. 4.2 summarizes the resulting input and output data for the production of 1 kg of crude coconut oil.

Tab. 4.2 Input and Output data for the production of 1 kg coconut oil (data from Hirsinger et al. (1995b), transformed according to the description in the n-paraffins chapter in Althaus et al. (2007))

[per kg coconut oil produced]			[per kg coconut oil produced]			[per kg coconut oil produced]		
INPUTS			OUTPUTS					
<i>materials</i>			<i>emissions to air</i>			<i>emissions to water</i>		
husked coconuts	kg	2.966	PM, > 10 um	kg	3.39E-04	acids	kg	4.79E-03
<i>energy</i>			PM, > 2.5 um & < 10 um	kg	4.56E-04	dissolved solids	kg	4.16E-02
electricity, UCTE-mix	kWh	0.172	PM, < 2.5 um	kg	2.65E-04	BOD	kg	1.38E-02
heat from heavy oil	MJ	2.97E-01	nitrogen oxides	kg	8.80E-05	COD	kg	1.38E-02
<i>transports</i>			NMVOC, unspecified	kg	2.00E-04	oil	kg	4.42E-03
lorry, 32t	tkm	1.34E-01	sulfur dioxide	kg	5.80E-05	nitrogen	kg	4.42E-03
<i>waste</i>			carbon dioxide, biogen	kg	2.68E-01			
ash from burning	kg	0.00104						

The shown transport amount is the result of a calculation, based on the reported amount of transport energy as well as the assumption, that all transports are on the road and therefore use only oil as energy source. More details about the calculation can be found in the n-paraffins chapter in Althaus et al. (2007).

Besides these inputs, the following additional inputs are integrated into the dataset of coconut oil production:

- *Water, cooling water:* no information about cooling installations is given in the different sources, but it can be assumed that some kind of cooling system exists. According to the methodology used within the different chemicals, a cooling water amount of 0.024 m³/kg product is assumed based on Gendorf (2000) (see e.g. trimethylamine in Althaus et al. (2007)).
- *Water, process water:* no information about process water amounts is given in the different sources. On the other hand side, process emissions to water are indicated and it is mentioned that steam is used for deactivating enzymes – therefore, a process water amount of 25% of the cooling water amount is assumed.
- *Energy content of Biomass:* Due to the fact that the used data source (i.e. Hirsinger et al. (1995b)) is not 100% transparent, it was not possible to add the energy content to the coconuts. Hence, the energy content has been added to the oil dataset. The respective gross calorific value for coconut oil is taken from Mühlbauer et al. (1998).
- *Infrastructure:* As no information is given about the infrastructure, the general module for the chemicals production “chemical plant, organics” is used as a first approximation.

Also for the first process step, the harvesting and husking of coconuts, information out of the ECOSOL (Hirsinger et al. (1995b)) study are used. Besides the rules described in the chapter about n-paraffins production (Althaus et al. (2007)), the following assumptions are used:

- *Fertilizer consumption*: according to the information in Hirsinger et al. (1995b), no fertilizers are used within a coconut plantation. Therefore, in this study no input of fertilizer occurs.
- *Waste*: according to information in Hirsinger et al. (1995b), the waste produced in this process step is pure organic. It is either left on the ground below the trees where it degrades rapidly or – in case of husking in a central location outside the plantations – this waste is piled and therefore will degrade less rapidly. Nevertheless, for this study this waste is seen as organic waste and therefore is not included into the data.
- *Transports*: it is assumed that the next step occurs next to the plantations and therefore no transport is included into this dataset here.
- *Process energy*: the reported amount of process energy is represented within this study by the dataset “diesel, in construction machine” – representing the different activities during the year on a coconut plantation. All according emissions are already included into this module.
- *Land use*: According to Hirsinger et al. (1995b) the ECOSOL study bases on the situation on the Philippines, occupying 3 Mio ha of land. On this area 11291 mio coconuts had been produced in 1991. A coconut tree has a productive life time of about 60 to 70 years and a total life time of 65 to 77 years. For this study, it is assumed that a coconut tree bares coconuts during 65 years and has accordingly a total life time of 71 years. The total weight of a coconut is in the range of 1 to 2 kg, whereof around 0.8 kg remains after removing the husks (Foodmarket Exchange.com (2002)). This remaining “husked” coconut is used for the copra production.

Tab. 4.3 summarizes the resulting input and output data for the production of 1 kg of harvested and husked coconuts on the Philippines, assuming an occupation of 100 years time.

Tab. 4.3 Input and Output data for the production of 1 kg harvested and husked coconuts (data out of Hirsinger et al. (1995b))

[per kg harvested & husked coconuts]		
INPUTS		
<i>land use</i>		
Transformation, from forest, extensive	m2	5.11E-02
Transformation, to permanent crop, fruit, intensive	m2	5.11E-02
Occupation, permanent crop, vine, intensive	m2a	3.63E+00
<i>energy</i>		
diesel, in construction machine	MJ	6.68E-04

4.1.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Main source used here is the European Surfactant LCI group's ECOSOL study. Although it is not possible to see from the different reports of this study, how many companies behind the different datasets are, the uncertainty is in general quite low. Higher values are found especially for the water consumption and the infrastructure due to the fact, that these values are based on one chemical production area in Germany (Gendorf (2000)). All other differences are due to the different basic uncertainties used for the different emissions and inputs within this study.

The resulting input and output data as well as the uncertainty scores are summarized for the different products examined within this chapter in Tab. 4.4 to Tab. 4.6.

Tab. 4.4 In-/Outputs and uncertainty informations for the dataset "n-olefins, at plant"

Explanation	Name	Location	Unit	n-olefins, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resources	Water, cooling, unspecified natural origin		m3	2.40E-02	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
Input from Technosphere	Water, unspecified natural origin		m3	6.00E-03	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
	paraffin, at plant	RER	kg	3.46E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	ethylene, average, at plant	RER	kg	6.33E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	chemical plant, organics	RER	unit	4.00E-10	1	3.56	(5,5,1,5,4,5); estimated from a large chem. plant
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	2.15E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	6.08E-02	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, natural gas, at industrial furnace >100kW	RER	MJ	8.27E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	9.90E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
Waste	transport, lorry 32t	RER	tkm	9.10E-02	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	kg	5.36E-03	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Output	n-olefins, at plant	RER	kg	1			
Air emission	Heat, waste		MJ	7.75E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Particulates, > 10 um		kg	1.55E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Particulates, > 2.5 um, and < 10um		kg	2.08E-05	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Particulates, < 2.5 um		kg	1.21E-05	1	3.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Nitrogen oxides		kg	2.10E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	7.13E-03	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Sulfur dioxide		kg	4.07E-05	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Carbon monoxide, fossil		kg	8.63E-05	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Aldehydes, unspecified		kg	3.00E-06	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Ammonia		kg	1.50E-06	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Carbon dioxide, fossil		kg	4.86E-02	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Acidity, unspecified		kg	3.06E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Fluoride		kg	1.30E-06	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Solved solids		kg	3.00E-03	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Suspended solids, unspecified		kg	2.93E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
water emission	BOD5, Biological Oxygen Demand		kg	3.99E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	COD, Chemical Oxygen Demand		kg	1.66E-03	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Sulfide		kg	1.71E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Oils, unspecified		kg	2.78E-05	1	3.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Chromium, ion		kg	4.37E-08	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Iron, ion		kg	1.07E-06	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Aluminum		kg	9.52E-09	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Nickel, ion		kg	4.27E-08	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Phosphate		kg	3.94E-06	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Zinc, ion		kg	6.79E-07	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Ammonium, ion		kg	7.14E-07	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Sulfate		kg	4.81E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Chloride		kg	2.30E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)

Tab. 4.5 In-/Outputs and uncertainty informations for the dataset "husked nuts harvesting, at farm"

Explanation	Name	Unit	husked nuts harvesting, at farm	uncertaintyType	standardDeviation95%	GeneralComment
	Location		PH			
	InfrastructureProcess		0			
	Unit		kg			
Resources	Transformation, from forest, extensive	m2	5.11E-02	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Input Tech'sph.	Transformation, to permanent crop, fruit, intensive	m2	5.11E-02	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Occupation, permanent crop, fruit, intensive	m2a	3.63E+00	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	diesel, burned in building machine	MJ	6.68E-04	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Output	husked nuts harvesting, at farm	kg	1			

Tab. 4.6 In-/Outputs and uncertainty informations for the dataset "crude coco nut oil, at plant"

Explanation	Name	Unit	crude coco nut oil, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location		PH			
	InfrastructureProcess		0			
	Unit		kg			
Resources	Water, cooling, unspecified natural origin	m3	2.40E-02	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
	Water, unspecified natural origin	m3	6.00E-03	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
	Energy, gross calorific value, in biomass	MJ	3.75E+01			
Input from Technosphere	husked nuts harvesting, at farm	kg	2.97E+00	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	chemical plant, organics	unit	4.00E-10	1	3.56	(5,5,1,5,4,5); estimated from a large chem. plant
	electricity, medium voltage, production UCTE, at grid	kWh	1.72E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, heavy fuel oil, at industrial furnace 1MW	MJ	2.97E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	transport, lorry 32t	t.km	1.34E-01	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Waste	disposal, wood ash mixture, pure, 0% water, to sanitary landfill	kg	1.04E-03	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Output	crude coco nut oil, at plant	kg	1			
Air emission	Heat, waste	MJ	6.19E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Particulates, > 10 um	kg	3.39E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Particulates, > 2.5 um, and < 10um	kg	4.56E-04	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Particulates, < 2.5 um	kg	2.65E-04	1	3.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Nitrogen oxides	kg	8.80E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	NM VOC, non-methane volatile organic compounds, unspecified origin	kg	2.00E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Sulfur dioxide	kg	5.80E-05	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Carbon dioxide, biogenic	kg	2.68E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Acidity, unspecified	kg	4.79E-03	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Solved solids	kg	4.16E-02	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
water emission	BOD5, Biological Oxygen Demand	kg	1.38E-02	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	COD, Chemical Oxygen Demand	kg	1.38E-02	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Oils, unspecified	kg	4.42E-03	1	3.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Nitrogen	kg	4.42E-03	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)

4.2 Fatty Alcohols

4.2.1 Introduction

Fatty alcohols are aliphatic alcohols with chain lengths between C₆ and C₂₂:



They are predominantly linear and monohydric and can be saturated or have one or more double bonds. Alcohols with a carbon chain length above C₂₂ are referred to as wax alcohols. Diols with chain lengths exceeding C₈ are regarded as substituted fatty alcohols. The character of fatty alcohols (primary or secondary, linear or branched, saturated or unsaturated) is determined by the manufacturing process and the raw material used. Fatty alcohols and their derivatives are used in polymers, surfactants, oil additives and cosmetics and have many specialty uses.

While saturated fatty alcohols up to dodecanol are clear colorless liquids, the next higher homologues are soft materials. Saturated alcohols crystallize in a nearly orthorhombic lattice and all of them have a lower specific density than water. While lower members have a characteristic odor, higher fatty alcohols are odorless. The boiling as well as the melting points increase uniformly with the chain length and are both significantly higher compared with hydrocarbons having the same number of carbon atoms. Suitable solvents therefore are common organic solvents – i.e. petroleum ether, lower alcohols, diethyl ether. Concerning the solubility in water, only hexanol and octanol show some solubility – all higher fatty alcohols are immiscible with water (Noweck (2001)).

4.2.2 Production and Use

Production processes

Up to 1930, the manufacturing of fatty alcohols was based almost exclusively on the splitting of sperm oil from whales. The production of fatty alcohols there consists of a simple hydrolysis or a reduction with sodium (Noweck (2001)). However, nowadays the depletion of whale populations is banned in

most countries and therefore other sources are used for production. By 1962, the world production capacity from natural raw materials – besides sperm oil also fats, oils from plants (e.g. coconut, palm) – had grown to approx. 200.000 mt/a. New processes utilizing petrochemical raw materials, e.g. the Ziegler alcohol process, the SHOP-process, the Oxo-process, and the construction of additional high-pressure hydrogenation plants for natural raw materials, allowed a further increase.

In 1999, the worldwide production capacity of fatty alcohols was estimated to be 2 Mio mt/year, being nearly evenly distributed to natural and petrochemical feedstocks. Production and consumption were estimated to be 80-90% of this capacity.

Concerning the production out of petrochemical resources, two main routes can be distinguished:

- *Ziegler process*: According to Franke et al. (1995) this process is a catalytic process involving as catalyst triethylaluminium. Fig. 4.2 shows the flow diagram of this process. After a hydrogenation of the catalyst, ethylene is inserted into the aluminium-carbon bond of this molecule to build up the carbon chain. This step is then followed by a growth reaction of the aluminium alkyls to aluminium alcoholates by oxidation with air. These are then subsequently hydrolysed to alcohols, as shown in the process steps listed.

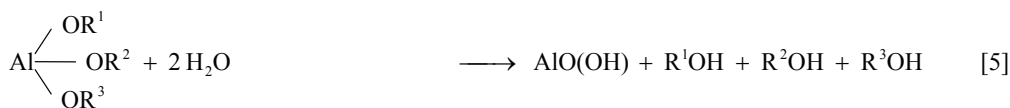
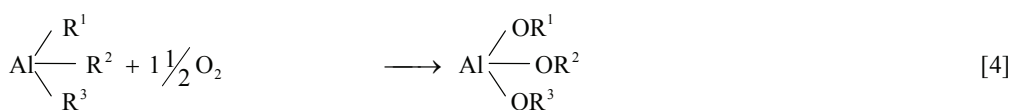
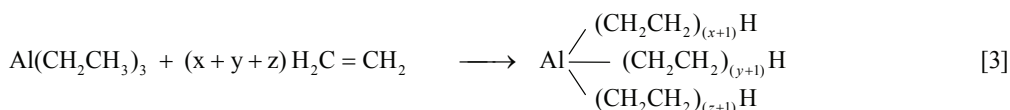
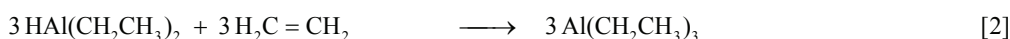
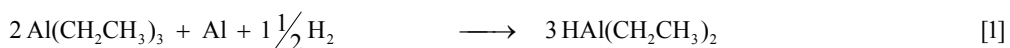
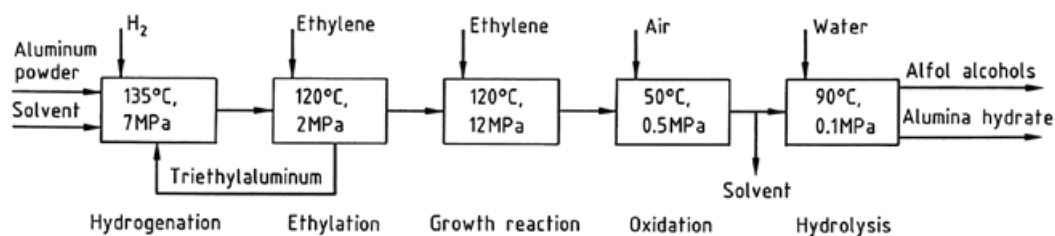


Fig. 4.2 Flow diagram (top) of the Ziegler process for the production of petrochemical fatty alcohols (Fig. 5 of Noweck (2001)) and the different process reactions (bottom): Hydrogenation [1], ethylation [2], growth reaction [3], oxidation [4] and final hydrolysis [5]

- *Oxo process*: Again according to Franke et al. (1995), this process constitutes a hydroformulation in which long chain olefins are reacted with hydrogen and carbon monoxide. An intermediate alde-

hyde is formed, which is subsequently reduced by hydrogen. Fig. 4.3 shows the flow diagram of this process.

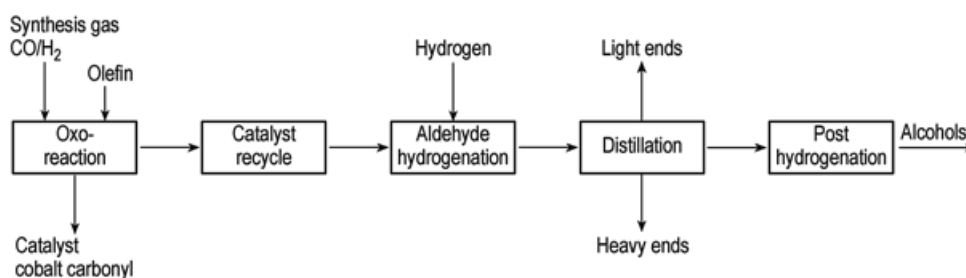


Fig. 4.3 Flow diagram of the oxo process for the production of petrochemical fatty alcohols (Fig. 7 of Noweck (2001))

For the production out of palm or coconut oil, again two main routes exist that are shown in Fig. 4.4 and described in the following:

- *Fatty acid splitting and hydrogenation:* In this case, according to Franke et al. (1995), triglycerides are split, producing fatty acids that are transformed into alcohols by hydrogenation. Raw material therefore is the crude vegetable oil and no additional refining processes are needed herefore. This crude oil is contacted with high pressure/high temperature steam in the splitting step, resulting in the production of an aqueous phase containing glycerol (accounted as co-product within ECOSOL) and a gaseous phase containing crude fatty acids. In a second step, these fatty acids are transformed into fatty alcohols by the so-called “Lurgi direct hydrogenation”.
- *Methyl ester production and hydrogenation:* According to Franke et al. (1995), in a first step, transesterification of crude or refined vegetable oils together with methanol is made and in a second step these methyl ethers are hydrogenated. In fact, methyl ethers from oleochemical raw materials are important intermediates not only for the production of fatty acids, but also in the production of i.e. methyl ester sulphonates. In the alcohol production, thus it is possible to recover methanol and to return it to the first step - the transesterification.

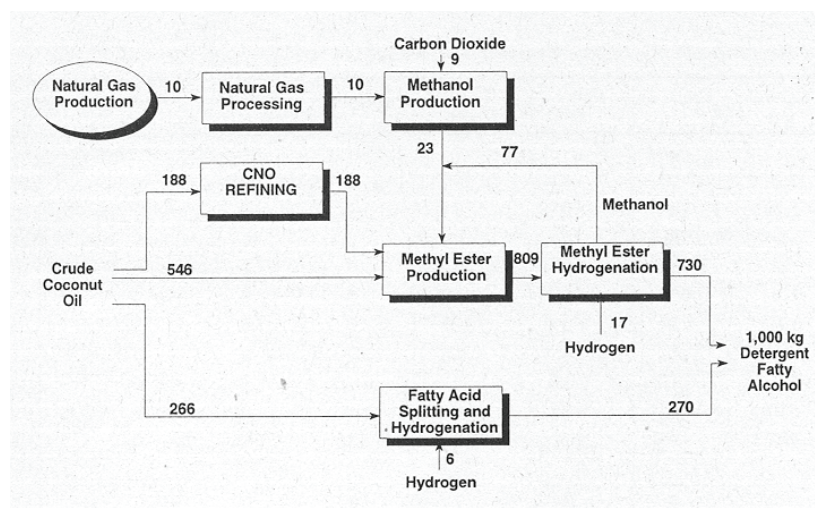


Fig. 4.4 Flow diagram of the different production ways for the production of fatty alcohols out of oleochemical raw materials (here for the example of coconut oil) (Fig.5 out of Hirsinger et al. (1995a))

In both cases, the hydrogenation process needs hydrogen as further input. In Franke et al. (1995) it is argued that the energy demand and the emissions for the hydrogen production can be neglected due to the fact that hydrogen is often produced as co-product in another production process.

Use

Fatty alcohols are mainly used as derivatives. In Western Europe only 5% of the production is used directly and 95% is used as intermediates. Major property of fatty alcohols is the amphiphilic character. Therefore, surfactants account for 70-75% of total fatty alcohol production Knaut & Richtler (1985). They can be used in detergents, emulsions, cosmetic creams, lotions, or as lubricants. The most important surfactants derived from fatty alcohols are:

- *Alcohol ethoxylates*: the first nonionic surfactants produced on industrial scale. These weakly-foaming surfactants are produced by condensing ethylene oxide to fatty alcohols (see chapter 4.4).
- *Alkyl sulfates*: fatty alcohol sulfates that belong to the group of anionic surfactants and that are in use as detergents since the 1930s. They are synthesized by the reaction of fatty alcohols with sulfur intermediates and subsequent neutralization with sodium hydroxide (see chapter 4.3).
- *Alkyl polyglycol ether sulfates*: fatty alcohol ether sulfates are obtained by the reaction of fatty alcohols with sulfur intermediates and subsequent neutralization with caustic soda or ammonia.

4.2.3 Systems characterization

Fig. 4.5 shows the processes for the production of the different types of fatty alcohols included into this study.

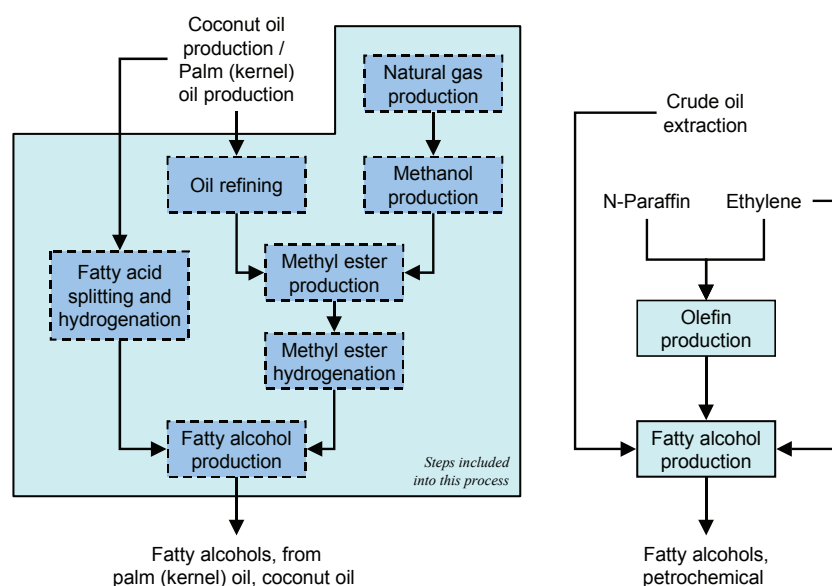


Fig. 4.5 Production systems of the different types of fatty alcohols (the process steps described within this chapter are slightly coloured)

As can be seen there, four different raw materials are used as starting material – palm oil, palm kernel oil, coconut oil and petrochemical resources (n-paraffin and ethylene). All of them are described elsewhere within this study (precursor substances in chapter 4.1 of this report – n-paraffin in Althaus et al. (2007) and ethylene in Hischier (2007)). In each step examined here, the material and energy input

(incl. cooling and process water), the infrastructure and transport of the raw materials to the production site as well as the emissions to air and water and the produced waste are included. Therefore, in the following chapters, the production of petrochemical fatty alcohols and of fatty alcohols from palms and coconuts (oleochemical fatty acids) are described in further details.

4.2.4 Production of fatty alcohols, petrochemical

Main source for the data is the detergent-grade alcohol report from the ECOSOL study of the European LCI Surfactant Study group (Hirsinger et al. (1995a)), representing a mix of 82% of fatty alcohols, produced with the Oxo process and the remaining 18% by the Ziegler process. A split into two separate modules is not possible with the information in Hirsinger et al. (1995a).

The data from this source have been transformed into the ecoinvent format according to the rules described in the chapter about the n-paraffins production (see Althaus et al. (2007)). Thereby it has been taken into consideration that all data described in Hirsinger et al. (1995a) represent cradle to gate inventories and therefore – only the difference between the data for fatty alcohols and its raw materials (n-olefins, ethylene) have been taken as basis for the transformation. Tab. 4.7 summarizes the resulting input and output data for the production of 1 kg of petrochemical fatty alcohols.

Tab. 4.7 Input and Output data for the production of 1 kg petrochemical fatty alcohols (data from Hirsinger et al. (1995a), transformed according to the description in the n-paraffins chapter in Althaus et al. (2007))

[per kg fatty alcohol]			[per kg fatty alcohol]			[per kg fatty alcohol]		
INPUTS			OUTPUTS					
<i>materials</i>			<i>emissions to air</i>			fluorides	kg	1.00E-06
crude oil (feedstock)	kg	0.012	waste heat	MJ	5.97E-01	suspended solids	kg	9.48E-05
natural gas (feedstock)	kg	0.061	PM, > 10 µm	kg	7.95E-06	BOD	kg	6.01E-05
n-olefin	kg	0.778	PM, > 2.5 µm & < 10 µm	kg	1.07E-05	oil	kg	1.82E-07
ethylene	kg	0.177	PM, < 2.5 µm	kg	6.21E-06	iron	kg	7.98E-07
<i>energy</i>			nitrogen oxides	kg	2.06E-04	phosphates as P ₂ O	kg	6.43E-07
electricity	kWh	1.66E-01	sulfur dioxide	kg	7.50E-04	zinc	kg	1.34E-07
heat from heavy oil	MJ	1.99E+00	carbon monoxide, fossil	kg	1.41E-04	ammonia	kg	8.42E-06
heat from natural gas	MJ	3.23E+00	ammonia	kg	1.68E-05	hydrocarbons	kg	1.20E-09
heat from coal	MJ	5.86E-01	<i>emissions to water</i>			sulphates	kg	3.57E-04
<i>transports</i>			acid	kg	2.99E-06	chloride	kg	2.01E-04
lorry, 32t	tkm	0.183						
<i>waste</i>								
solid waste, incineration	kg	0.00339						

The shown transport amount is the result of a calculation, based on the reported amount of transport energy as well as the assumption, that all transports are on the road and therefore use only oil as energy source. More details about the calculation can be found in the n-paraffins chapter in Althaus et al. (2007).

Besides these inputs, the following additional inputs are integrated into the dataset of petrochemical fatty acid production:

- *Water, cooling water:* no quantitative information about cooling installations are given in the different sources, but in Noweck (2001) it is mentioned that especially in the beginning of the Ziegler process cooling is necessary. According to the methodology used within the different chemicals, a cooling water amount of 0.024 m³/kg product is assumed based on Gendorf (2000) (see e.g. trimethylamine in Althaus et al. (2007)).

- *Water, process water*: no information about process water amounts is given in the different sources. Nevertheless, the emissions to water in Hirsinger et al. (1995a) shows that most of them are from the process – therefore, a process water amount of 25% of the cooling water amount is assumed.
- *Infrastructure*: As no information is given about the infrastructure, the general module for the chemicals production “chemical plant, organics” is used as a first approximation.

4.2.5 Production of fatty alcohols, oleochemical

Main source for the data is the detergent-grade alcohol report from the ECOSOL study of the European LCI Surfactant Study group (Hirsinger et al. (1995a)), representing a separate module for each of the different examined natural oil sources (palm oil, palm kernel oil, coconut oil). In each case, a mix of 27% produced from fatty acid splitting, 56% produced from methyl ester on the basis of crude vegetable oil and 17% from methyl ester out of refined oil is behind the final result. Therefore several production steps are included into the examined fatty alcohol production scheme (see Fig. 4.5). A split into these different process steps is not possible with the information given in Hirsinger et al. (1995a). Due to the fact that no inventory for methanol of the ECOSOL study is published, it is even not possible to use for methanol the module of the ecoinvent database – instead, its production is part of the dataset for the different fatty alcohols from oleochemical sources.

The data from this source have been transformed into the ecoinvent format according to the rules described in the chapter about the n-paraffins production (see Althaus et al. (2007)). Thereby it has been taken into consideration that all data described in Hirsinger et al. (1995a) represent cradle to gate inventories and therefore – only the difference between the data for fatty alcohols and its raw materials (palm oil resp. coconut oil resp. palm kernel oil) have been taken as basis for the transformation. Tab. 4.8 to Tab. 4.10 summarize the resulting input and output data for the production of 1 kg of different oleochemical fatty alcohols. As can be seen in Tab. 4.8, the mass balance between input and output shows less input for palm oil alcohol. According to Janzen (1995) there are several possible reasons therefore. In order of not adding any additional uncertainty / source of error, nevertheless the data from Hirsinger et al. (1995a) are used here.

Tab. 4.8 Input and Output data for the production of 1 kg palm oil alcohol (data from Hirsinger et al. (1995a), transformed according to the description in the n-paraffins chapter in Althaus et al. (2007)).

[per kg fatty alcohol]			[per kg fatty alcohol]		
INPUTS			OUTPUTS		
<i>raw material</i>			<i>emissions to air</i>		
palm oil	kg	0.927	waste heat	MJ	4.34E-01
natural gas	kg	0.010	NM VOC, unspecified	kg	2.01E-03
hydrogen, H ₂	kg	0.006	carbon dioxide, fossil	kg	5.00E-05
<i>energy</i>			<i>emissions to water</i>		
electricity	kWh	1.21E-01	dissolved solids	kg	7.80E-04
heat from heavy oil	MJ	1.175	suspended solids	kg	1.83E-04
heat from natural gas	MJ	5.303	BOD	kg	2.83E-04
heat from coal	MJ	0.914	COD	kg	1.42E-03
<i>transports</i>			oil	kg	2.35E-05
lorry, 32t	tkm	0.049			
ship, ocean	tkm	15.500			
<i>waste</i>					
solid waste, incineration	kg	0.021424			

Tab. 4.9 Input and Output data for the production of 1 kg palm kernel alcohol (data from Hirsinger et al. (1995a), transformed according to the description in the n-paraffins chapter in Althaus et al. (2007)).

[per kg fatty alcohol]			[per kg fatty alcohol]		
INPUTS			OUTPUTS		
<i>materials</i>			<i>emissions to air</i>		
palm kernel oil	kg	1.004	waste heat	MJ	5.79E-01
natural gas	kg	0.010	NM VOC, unspecified	kg	2.05E-03
hydrogen, H ₂	kg	0.006	carbon dioxide, fossil	kg	6.10E-05
<i>energy</i>			<i>emissions to water</i>		
electricity	kWh	1.61E-01	dissolved solids	kg	4.28E-04
heat from heavy oil	MJ	3.715	BOD	kg	2.79E-04
heat from natural gas	MJ	6.728	COD	kg	1.33E-03
heat from coal	MJ	1.395	oil	kg	2.08E-05
<i>transports</i>					
lorry, 32t	tkm	0.058			
ship, ocean	tkm	15.500			
<i>waste</i>					
solid waste, incineration	kg	0.019472			

Tab. 4.10 Input and Output data for the production of 1 kg coconut oil alcohol (data from Hirsinger et al. (1995a), transformed according to the description in the n-paraffins chapter in Althaus et al. (2007)).

[per kg fatty alcohol]			[per kg fatty alcohol]		
INPUTS			OUTPUTS		
<i>materials</i>			<i>emissions to air</i>		
husked coconuts	kg	1.001	waste heat	MJ	4.34E-01
natural gas	kg	0.01	NM VOC, unspecified	kg	2.75E-03
hydrogen, H ₂	kg	0.006	carbon monoxide	kg	3.10E-06
<i>energy</i>			methane	kg	1.40E-05
electricity	kWh	1.21E-01	carbon dioxide, fossil	kg	3.30E-04
heat from heavy oil	MJ	2.538	<i>emissions to water</i>		
heat from natural gas	MJ	6.672	suspended solids	kg	3.60E-05
heat from coal	MJ	1.356	oil	kg	1.20E-05
<i>transports</i>					
lorry, 32t	tkm	0.291			
ship, ocean	tkm	15.500			
<i>waste</i>					
solid waste, incineration	kg	0.00336			

As the production of the different fatty alcohols is situated in Europe, but the production of the used oleochemical input materials takes place in Asia, the transports take place by ocean ship and by lorry and are calculated the following way:

- *Transport by ship*: According to Dataloy (2002) the distance from Manila (Philippines) to Europe (Rotterdam) by ship is about 15'500 km (9650 miles). This leads to an additional input of 15.5 tkm of "transport, transoceanic freight ship" for each of the three datasets.
- *Transport by lorry*: The shown transport amount by lorry results from the difference of the reported transport energy in Hirsinger et al. (1995a) and the transport energy for the 15.5 tkm by oceanic ship. The latter one is calculated from the data documented in Frischknecht et al. (1996). For the remaining amount of transport energy, the method described in the n-paraffin chapter of Althaus et al. (2007) is used for the calculation of the respective transport distance in tkm.

Besides the shown inputs, the following additional inputs are integrated into the dataset of oleochemical fatty acid production:

- *Water, cooling water:* no quantitative information about cooling installations are given in the different sources, but in Noweck (2001) it is mentioned that especially in the beginning of the Ziegler process cooling is necessary. According to the methodology used within the different chemicals, a cooling water amount of 0.024 m³/kg product is assumed based on Gendorf (2000) (see e.g. trimethylamine in Althaus et al. (2007)).
- *Water, process water:* no information about process water amounts is given in the different sources. Nevertheless, the emissions to water in Hirsinger et al. (1995a) shows that most of them are from the process – therefore, a process water amount of 25% of the cooling water amount is assumed.
- *Infrastructure:* As no information is given about the infrastructure, the general module for the chemicals production “chemical plant, organics” is used as a first approximation.

4.2.6 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Main source used here is the European Surfactant LCI group's ECOSOL study. Although it is not possible to see from the different reports of this study, how many companies behind the different datasets are, the uncertainty is in general quite low. Higher values are found especially for the water consumption and the infrastructure due to the fact, that these values are based on one chemical production area in Germany (Gendorf (2000)). All other differences are due to the different basic uncertainties used for the different emissions and inputs within this study.

The resulting input and output data as well as the uncertainty scores are summarized for the different products examined within this chapter in Tab. 4.11 to Tab. 4.14.

Tab. 4.11 In-/Outputs and uncertainty informations for the dataset "fatty alcohol, petrochemical, at plant"

Explanation	Name	Location	Unit	fatty alcohol, petrochemical , at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location						
	InfrastructureProcess						
	Unit						
Resources	Water, cooling, unspecified natural origin		m3	2.40E-02	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
Resources	Water, unspecified natural origin		m3	6.00E-03	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
Input from Technosphere	Oil, crude, in ground		kg	1.20E-02	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Gas, natural, in ground		Nm3	7.63E-02	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	n-olefins, at plant	RER	kg	7.78E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	ethylene, average, at plant	RER	kg	1.77E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	chemical plant, organics	RER	unit	4.00E-10	1	3.56	(5,5,1,5,4,5); estimated from a large chem. plant
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.66E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.99E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, natural gas, at industrial furnace >100kW	RER	MJ	3.23E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	5.86E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	transport, lorry 32t	RER	tkm	1.83E-01	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Waste	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	kg	3.39E-03	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Output	fatty alcohol, petrochemical, at plant	RER	kg	1			
Air emission	Heat, waste		MJ	5.97E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Particulates, > 10 um		kg	7.95E-06	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Particulates, > 2.5 um, and < 10um		kg	1.07E-05	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Particulates, < 2.5 um		kg	6.21E-06	1	3.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Nitrogen oxides		kg	2.06E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Sulfur dioxide		kg	7.50E-04	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Carbon monoxide, fossil		kg	1.41E-04	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Ammonia		kg	1.68E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Acidity, unspecified		kg	2.99E-06	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Fluoride		kg	1.00E-06	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Suspended solids, unspecified		kg	9.48E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	BOD5, Biological Oxygen Demand		kg	6.01E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Oils, unspecified		kg	1.82E-07	1	3.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Iron, ion		kg	7.98E-07	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Phosphate		kg	6.43E-07	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Zinc, ion		kg	1.34E-07	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Ammonium, ion		kg	8.42E-06	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Hydrocarbons, unspecified		kg	1.20E-09	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Sulfate		kg	3.57E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Chloride		kg	2.01E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)

Tab. 4.12 In-/Outputs and uncertainty informations for the dataset "fatty alcohol, from palm oil, at plant"

Explanation	Name	Location	Unit	fatty alcohol, from palm oil, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location						
	InfrastructureProcess						
	Unit						
Resources	Water, cooling, unspecified natural origin		m3	2.40E-02	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
	Water, unspecified natural origin		m3	6.00E-03	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
Input from Technosphere	Gas, natural, in ground		Nm3	1.19E-02	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	hydrogen, liquid, at plant	RER	kg	6.00E-03	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	crude palm oil, at plant	MY	kg	9.27E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	chemical plant, organics	RER	unit	4.00E-10	1	3.56	(5,5,1,5,4,5); estimated from a large chem. plant
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.21E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.18E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, natural gas, at industrial furnace >100kW	RER	MJ	5.30E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	9.14E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	transport, lorry 32t	RER	tkm	4.90E-02	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	transport, transoceanic freight ship	OCE	tkm	1.55E+01	1	2.11	(4,4,2,1,1,5); estim., based on km Manila - Rotterdam
Waste	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	kg	2.14E-02	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Output	fatty alcohol, from palm oil, at plant	RER	kg	1			
Air emission	Heat, waste		MJ	4.34E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	2.01E-03	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
water emission	Carbon dioxide, fossil		kg	5.00E-05	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Solved solids		kg	7.80E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Suspended solids, unspecified		kg	1.83E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	BOD5, Biological Oxygen Demand		kg	2.83E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	COD, Chemical Oxygen Demand		kg	1.42E-03	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Oils, unspecified		kg	2.35E-05	1	3.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)

Tab. 4.13 In-/Outputs and uncertainty informations for the dataset "fatty alcohol, from palm kernel oil, at plant"

Explanation	Name	Location	Unit	fatty alcohol, from palm kernel oil, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resources	Water, cooling, unspecified natural origin		m3	2.40E-02	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
	Water, unspecified natural origin		m3	6.00E-03	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
	Gas, natural, in ground		Nm3	1.25E-02	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Input from Technosphere	hydrogen, liquid, at plant	RER	kg	6.00E-03	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	crude palm kernel oil, at plant	MY	kg	1.00E+00	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	chemical plant, organics	RER	unit	4.00E-10	1	3.56	(5,5,1,5,4,5); estimated from a large chem. plant
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.61E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	3.72E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, natural gas, at industrial furnace >100kW	RER	MJ	6.73E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.40E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	transport, lorry 32t	RER	tkm	5.80E-02	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	transport, transoceanic freight ship	OCE	tkm	1.55E+01	1	2.11	(4,4,2,1,1,5); estim., based on km Manila - Rotterdam
Waste	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	kg	1.95E-02	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Output	fatty alcohol, from palm kernel oil, at plant	RER	kg	1			
Air emission	Heat, waste		MJ	5.79E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	2.05E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Carbon dioxide, fossil		kg	6.10E-05	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
water emission	Solved solids		kg	4.28E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	BOD5, Biological Oxygen Demand		kg	2.79E-04	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	COD, Chemical Oxygen Demand		kg	1.33E-03	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Oils, unspecified		kg	2.08E-05	1	3.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)

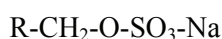
Tab. 4.14 In-/Outputs and uncertainty informations for the dataset "fatty alcohol, from coconut oil, at plant"

Explanation	Name	Location	Unit	fatty alcohol, from coconut oil, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resources	Water, cooling, unspecified natural origin		m3	2.40E-02	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
	Water, unspecified natural origin		m3	6.00E-03	1	1.89	(5,5,1,5,4,5); estimated from a large chem. plant
	Gas, natural, in ground		Nm3	1.25E-02	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Input from Technosphere	hydrogen, liquid, at plant	RER	kg	6.00E-03	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	crude coco nut oil, at plant	PH	kg	1.00E+00	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	chemical plant, organics	RER	unit	4.00E-10	1	3.56	(5,5,1,5,4,5); estimated from a large chem. plant
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.21E-01	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.54E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, natural gas, at industrial furnace >100kW	RER	MJ	6.67E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.36E+00	1	1.38	(4,5,2,1,1,5); estimation, based on ECOSOL data
	transport, lorry 32t	RER	tkm	2.91E-01	1	2.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	transport, transoceanic freight ship	OCE	tkm	1.55E+01	1	2.11	(4,4,2,1,1,5); estim., based on km Manila - Rotterdam
Waste	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	kg	3.36E-03	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
Output	fatty alcohol, from coconut oil, at plant	RER	kg	1			
Air emission	Heat, waste		MJ	4.34E-01	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	2.75E-03	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Methane, fossil		kg	1.40E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Carbon monoxide, fossil		kg	3.10E-06	1	5.11	(2,5,2,1,1,5); data from ECOSOL study (Literature)
water emission	Carbon dioxide, fossil		kg	3.30E-04	1	1.31	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Suspended solids, unspecified		kg	3.60E-05	1	1.62	(2,5,2,1,1,5); data from ECOSOL study (Literature)
	Oils, unspecified		kg	1.20E-05	1	3.1	(2,5,2,1,1,5); data from ECOSOL study (Literature)

4.3 Fatty Alcohol Sulfates (FAS)

4.3.1 Introduction

Fatty alcohol sulfates (FAS) belong to the group of anionic surfactants and have been used as detergents already since the 1930s. They are synthesized by sulfation to an acid form followed by a neutralization with sodium hydroxide. The generic chemical structure of FAS is:



FAS generally exhibit good detergency, wetting and foaming properties. However, the exact properties of FAS are determined by molecular structure and chain length of the alcohol and the type of

cation. Products with a chain length of C_{12} - C_{15} , that are derived from coconut oil, palm kernel oil and petrochemical sources, have good foaming power and show good low temperature detergency. At higher temperatures, chain lengths of C_{16} - C_{18} , derived from palm oil, are more effective (Hirsinger & Schick (1995)).

4.3.2 Production and use

Production processes

Fatty alcohols are the starting product for manufacturing FAS. They can be produced oleochemically or petrochemically. FAS are obtained by sulphonation of the alcohols. A thin film of alcohol is brought together with a mixture of SO_3 and dry air inside a vertical tube. The heat produced by the reaction is removed using coolants outside the tube. SO_3 itself is produced through a catalytic oxidation of SO_2 , which derives from the combustion of sulphur or as a secondary product. After sulphonation, sodium hydroxide is applied to neutralize the acid that has been formed (Hirsinger & Schick (1995)).

Amount of production

It is estimated that approx. 40% of total fatty alcohol production is transformed to FAS.

Storage and transport

No specific data on transport distances of alcohol ethoxylates were available. Therefore, an assessment of average transport distances for organic chemicals on rail and lorry has been applied according to Frischknecht et al. (2007b).

Use

FAS are mainly used as surfactants in washing powders, especially in gentle washing powders. FAS are also used in toiletry and cosmetic formulations such as shampoos, toothpastes and cosmetic lotions.

4.3.3 Systems characterization

This study provides life-cycle data of FAS with short chain lengths of C_{12} - C_{15} , that can be derived from petrochemical, palm kernel, or coconut oil. The chemical and physical properties of the final product are similar for all three pathways, providing that their carbon chainlength and ethoxylate distribution are similar (Stalmans et al. (1995)). Additionally this study also includes a LCA of FAS based on palm oil with chain lengths of C_{16} - C_{18} .

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

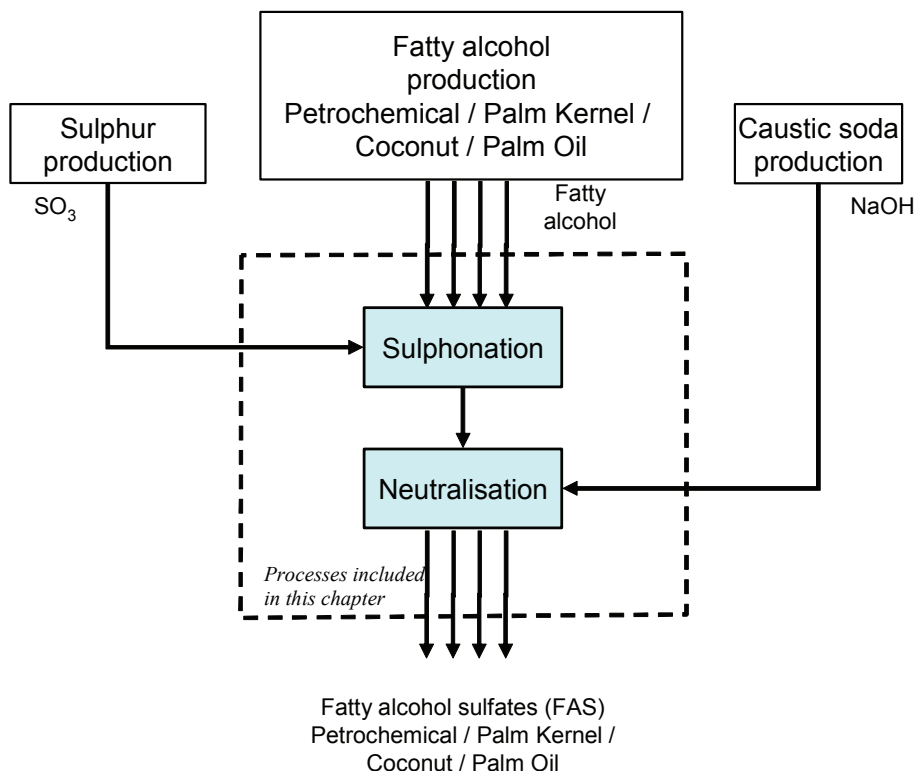


Fig. 4.6 Flow diagram for the production of four different types of FAS

4.3.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the EMPA-study on tensides (Dall'Acqua et al. (1999)). Masses may not balance exactly due to conventions and boundary conditions (Janzen (1995)). Process-related emissions have been assessed by comparing the aggregated emissions of the four different types of FAS (Hirsinger & Schick (1995)) with the aggregated emissions of the respective types of fatty alcohols (Hirsinger et al. (1995a)) and of sulphur and caustic soda (Postlethwaite et al. (1995)). Although it is not possible to see from the different used reports, how many companies behind the different datasets are, the uncertainty is in general quite low. The differences are due to the different basic uncertainties used for the different emissions and inputs within this study.

Additionally, an average FAS dataset, based on the four different types of FAS reported, is established. Due to a lack of information about the market distribution of the four types of FAS, an equal distribution to these four datasets is used here. For the uncertainty, a similar characterization like for the four specific FAS is used.

The resulting input and output data as well as the uncertainty scores are summarized in the following tables.

Tab. 4.15 Input/Output data and data uncertainty of the dataset “fatty alcohol sulfate, petrochemical, at plant (RER)”

Explanation	Name	Unit	fatty alcohol sulfate, petrochemical, at plant RER 0 kg	Uncertainty Type	Standard Deviation 95 %	General Comment
	Location					
	Infrastructure					
	Process					
Input from Technosphere	Unit					
	fatty alcohol, petrochemical, at plant	kg	6.98E-01	1	1.25	(2,3,3,2,1,4);
	secondary sulphur, at refinery	kg	1.13E-01	1	1.25	(2,3,3,2,1,4);
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	kg	1.44E-01	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	kWh	1.49E-01	1	1.25	(2,3,3,2,1,4);
	heat, unspecific, in chemical plant	MJ	1.87E+00	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	unit	4.00E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	tkm	9.55E-02	1	2.07	(2,3,3,2,1,4);
	transport, freight, rail	tkm	5.73E-01	1	2.07	(2,3,3,2,1,4);
Output	disposal, inert waste, 5% water, to inert material landfill	kg	6.20E-04	1	1.25	(2,3,3,2,1,4);
	fatty alcohol sulfate, petrochemical, at plant	kg	1			
	Particulates, > 10 um	kg	1.13E-04	1	1.58	(2,3,3,2,1,4);
	Nitrogen oxides	kg	7.41E-04	1	1.58	(2,3,3,2,1,4);
	NM VOC, non-methane volatile organic compounds, unspecified	kg	1.04E-03	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide	kg	1.64E-03	1	1.25	(2,3,3,2,1,4);
	Carbon monoxide, fossil	kg	5.68E-05	1	5.07	(2,3,3,2,1,4);
	Aldehydes, unspecified	kg	3.34E-07	1	2.07	(2,3,3,2,1,4);
	Methane, fossil	kg	0.00E+00	1	1.33	(2,3,3,2,1,4);
	Ammonia	kg	6.44E-06	1	1.58	(2,3,3,2,1,4);
	Chlorine	kg	9.60E-09	1	1.58	(2,3,3,2,1,4);
	Carbon dioxide, biogenic	kg	0.00E+00	1	1.25	(2,3,3,2,1,4);
	Acidity, unspecified	kg	3.59E-06	1	1.58	(2,3,3,2,1,4);
	Fluoride	kg	1.10E-07	1	1.58	(2,3,3,2,1,4);
	Solved solids	kg	3.12E-03	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand	kg	4.62E-05	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand	kg	6.25E-04	1	1.58	(2,3,3,2,1,4);
	Phenol	kg	4.25E-09	1	3.06	(2,3,3,2,1,4);
	Sulfide	kg	1.52E-06	1	1.58	(2,3,3,2,1,4);
	Oils, unspecified	kg	2.25E-06	1	3.06	(2,3,3,2,1,4);
	Chromium, ion	kg	1.57E-06	1	5.07	(2,3,3,2,1,4);
	Iron, ion	kg	4.36E-08	1	5.07	(2,3,3,2,1,4);
	Nitrogen	kg	0.00E+00	1	5.07	(2,3,3,2,1,4);
	Zinc, ion	kg	1.92E-07	1	5.07	(2,3,3,2,1,4);
	Hydrocarbons, unspecified	kg	2.83E-07	1	3.06	(2,3,3,2,1,4);
	Sulfate	kg	4.25E-03	1	1.58	(2,3,3,2,1,4);
	Phosphate	kg	1.99E-07	1	1.58	(2,3,3,2,1,4);
	Ammonium, ion	kg	2.24E-06	1	1.58	(2,3,3,2,1,4);
	Chloride	kg	1.48E-05	1	1.58	(2,3,3,2,1,4);

Tab. 4.16 Input/Output data and data uncertainty of the dataset “fatty alcohol sulfate, palm kernel oil, at plant (RER)”

Explanation	Name	Unit	fatty alcohol sulfate, palm kernel oil, at plant	Uncertainty Type	Standard Deviation 95 %	General Comment
	Location		RER			
	Infrastructure	Process	0			
	Unit		kg			
Input from Technosphere	fatty alcohol, from palm kernel oil, at plant	kg	6.98E-01	1	1.25	(2,3,3,2,1,4);
	secondary sulphur, at refinery	kg	1.13E-01	1	1.25	(2,3,3,2,1,4);
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	kg	1.44E-01	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	kWh	1.49E-01	1	1.25	(2,3,3,2,1,4);
	heat, unspecific, in chemical plant	MJ	1.87E+00	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	unit	4.00E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	tkm	9.55E-02	1	2.07	(2,3,3,2,1,4);
	transport, freight, rail	tkm	5.73E-01	1	2.07	(2,3,3,2,1,4);
Output	disposal, inert waste, 5% water, to inert material landfill	kg	6.20E-04	1	1.25	(2,3,3,2,1,4);
	fatty alcohol sulfate, palm kernel oil, at plant	kg	1			
	Particulates, > 10 um	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Nitrogen oxides	kg	1.14E-04	1	1.58	(2,3,3,2,1,4);
	NM VOC, non-methane volatile organic compounds, unspecified	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide	kg	6.07E-05	1	1.25	(2,3,3,2,1,4);
	Carbon monoxide, fossil	kg	0.00E+00	1	5.07	(2,3,3,2,1,4);
	Aldehydes, unspecified	kg	0.00E+00	1	2.07	(2,3,3,2,1,4);
	Methane, fossil	kg	0.00E+00	1	1.33	(2,3,3,2,1,4);
	Ammonia	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Chlorine	kg	8.60E-09	1	1.58	(2,3,3,2,1,4);
	Carbon dioxide, biogenic	kg	0.00E+00	1	1.25	(2,3,3,2,1,4);
	Acidity, unspecified	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Fluoride	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Solved solids	kg	2.65E-03	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand	kg	7.08E-04	1	1.58	(2,3,3,2,1,4);
	Phenol	kg	5.80E-09	1	3.06	(2,3,3,2,1,4);
	Sulfide	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Oils, unspecified	kg	0.00E+00	1	3.06	(2,3,3,2,1,4);
	Chromium, ion	kg	2.43E-06	1	5.07	(2,3,3,2,1,4);
	Iron, ion	kg	0.00E+00	1	5.07	(2,3,3,2,1,4);
	Nitrogen	kg	4.16E-08	1	5.07	(2,3,3,2,1,4);
	Zinc, ion	kg	1.30E-07	1	5.07	(2,3,3,2,1,4);
	Hydrocarbons, unspecified	kg	1.90E-07	1	3.06	(2,3,3,2,1,4);
	Sulfate	kg	1.20E-03	1	1.58	(2,3,3,2,1,4);
	Phosphate	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Ammonium, ion	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Chloride	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);

Tab. 4.17 Input/Output data and data uncertainty of the dataset “fatty alcohol sulfate, coconut oil, at plant (RER)”

Explanation	Name	Unit	fatty alcohol sulfate, coconut oil, at plant RER 0 kg	Uncertainty Type	Standard Deviation 95 %	General Comment
	Location					
	Infrastructure					
	Process					
Input from Technosphere	Unit					
	fatty alcohol, from coconut oil, at plant	kg	6.98E-01	1	1.25	(2,3,3,2,1,4);
	secondary sulphur, at refinery	kg	1.13E-01	1	1.25	(2,3,3,2,1,4);
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	kg	1.44E-01	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	kWh	1.49E-01	1	1.25	(2,3,3,2,1,4);
	heat, unspecific, in chemical plant	MJ	1.87E+00	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	unit	4.00E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	tkm	9.55E-02	1	2.07	(2,3,3,2,1,4);
	transport, freight, rail	tkm	5.73E-01	1	2.07	(2,3,3,2,1,4);
Output	disposal, inert waste, 5% water, to inert material landfill	kg	6.20E-04	1	1.25	(2,3,3,2,1,4);
	fatty alcohol sulfate, coconut oil, at plant	kg	1			
	Particulates, > 10 um	kg	6.98E-05	1	1.58	(2,3,3,2,1,4);
	Nitrogen oxides	kg	1.10E-04	1	1.58	(2,3,3,2,1,4);
	NM VOC, non-methane volatile organic compounds, unspecified	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide	kg	9.24E-05	1	1.25	(2,3,3,2,1,4);
	Carbon monoxide, fossil	kg	7.36E-07	1	5.07	(2,3,3,2,1,4);
	Aldehydes, unspecified	kg	0.00E+00	1	2.07	(2,3,3,2,1,4);
	Methane, fossil	kg	0.00E+00	1	1.33	(2,3,3,2,1,4);
	Ammonia	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Chlorine	kg	8.60E-09	1	1.58	(2,3,3,2,1,4);
	Carbon dioxide, biogenic	kg	1.24E-02	1	1.25	(2,3,3,2,1,4);
	Acidity, unspecified	kg	2.24E-04	1	1.58	(2,3,3,2,1,4);
	Fluoride	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Solved solids	kg	4.91E-03	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand	kg	7.85E-04	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand	kg	2.19E-03	1	1.58	(2,3,3,2,1,4);
	Phenol	kg	5.80E-09	1	3.06	(2,3,3,2,1,4);
	Sulfide	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Oils, unspecified	kg	0.00E+00	1	3.06	(2,3,3,2,1,4);
	Chromium, ion	kg	2.43E-06	1	5.07	(2,3,3,2,1,4);
	Iron, ion	kg	2.11E-04	1	5.07	(2,3,3,2,1,4);
	Nitrogen	kg	2.11E-04	1	5.07	(2,3,3,2,1,4);
	Zinc, ion	kg	1.30E-07	1	5.07	(2,3,3,2,1,4);
	Hydrocarbons, unspecified	kg	1.90E-07	1	3.06	(2,3,3,2,1,4);
	Sulfate	kg	1.20E-03	1	1.58	(2,3,3,2,1,4);
	Phosphate	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Ammonium, ion	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Chloride	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);

Tab. 4.18 Input/Output data and data uncertainty of the dataset “fatty alcohol sulfate, palm oil, at plant (RER)”

Explanation	Name	Unit	fatty alcohol sulfate, palm oil, at plant RER 0 kg	UncertaintyType	StandardDeviation95 %	GeneralComment
	Location					
	InfrastructureProcess					
	Unit					
Input from Technosphere	fatty alcohol, from palm oil, at plant	kg	6.98E-01	1	1.25	(2,3,3,2,1,4);
	secondary sulphur, at refinery	kg	1.13E-01	1	1.25	(2,3,3,2,1,4);
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	kg	1.44E-01	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	kWh	1.49E-01	1	1.25	(2,3,3,2,1,4);
	heat, unspecific, in chemical plant	MJ	1.87E+00	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	unit	4.00E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	tkm	9.55E-02	1	2.07	(2,3,3,2,1,4);
	transport, freight, rail	tkm	5.73E-01	1	2.07	(2,3,3,2,1,4);
	disposal, inert waste, 5% water, to inert material landfill	kg	6.20E-04	1	1.25	(2,3,3,2,1,4);
Output air emission water emission	fatty alcohol sulfate, palm oil, at plant	kg	1			
	Particulates, > 10 um	kg	1.56E-04	1	1.58	(2,3,3,2,1,4);
	Nitrogen oxides	kg	1.08E-04	1	1.58	(2,3,3,2,1,4);
	NM VOC, non-methane volatile organic compounds, unspecified	kg	3.62E-07	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide	kg	1.10E-04	1	1.25	(2,3,3,2,1,4);
	Carbon monoxide, fossil	kg	6.24E-06	1	5.07	(2,3,3,2,1,4);
	Aldehydes, unspecified	kg	0.00E+00	1	2.07	(2,3,3,2,1,4);
	Methane, fossil	kg	1.05E-03	1	1.33	(2,3,3,2,1,4);
	Ammonia	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Chlorine	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Carbon dioxide, biogenic	kg	3.11E-02	1	1.25	(2,3,3,2,1,4);
	Acidity, unspecified	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Fluoride	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Solved solids	kg	3.02E-03	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand	kg	1.59E-05	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand	kg	8.86E-04	1	1.58	(2,3,3,2,1,4);
	Phenol	kg	0.00E+00	1	3.06	(2,3,3,2,1,4);
	Sulfide	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Oils, unspecified	kg	5.36E-07	1	3.06	(2,3,3,2,1,4);
	Chromium, ion	kg	2.43E-06	1	5.07	(2,3,3,2,1,4);
	Iron, ion	kg	0.00E+00	1	5.07	(2,3,3,2,1,4);
	Nitrogen	kg	4.14E-06	1	5.07	(2,3,3,2,1,4);
	Zinc, ion	kg	1.10E-07	1	5.07	(2,3,3,2,1,4);
	Hydrocarbons, unspecified	kg	0.00E+00	1	3.06	(2,3,3,2,1,4);
	Sulfate	kg	1.06E-03	1	1.58	(2,3,3,2,1,4);
	Phosphate	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Ammonium, ion	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);
	Chloride	kg	0.00E+00	1	1.58	(2,3,3,2,1,4);

Tab. 4.19 Input/Output data and data uncertainty of the dataset “fatty alcohol sulfate, mix, at plant (RER)”

Explanation	Name	Unit	fatty alcohol sulfate, mix, at plant RER 0 kg	UncertaintyType	StandardDeviation95 %	GeneralComment
	Location					
	InfrastructureProcess					
	Unit					
Input from Technosphere	fatty alcohol sulfate, petrochemical, at plant	kg	2.50E-01	1	1.25	(2,3,3,2,1,4);
	fatty alcohol sulfate, palm kernel oil, at plant	kg	2.50E-01	1	1.25	(2,3,3,2,1,4);
	fatty alcohol sulfate, coconut oil, at plant	kg	2.50E-01	1	1.25	(2,3,3,2,1,4);
	fatty alcohol sulfate, palm oil, at plant	kg	2.50E-01	1	1.25	(2,3,3,2,1,4);
Output	fatty alcohol sulfate, mix, at plant	kg	1			

4.4 Linear Alkylbenzene Sulphonate (LAS)

4.4.1 Introduction

Since its introduction to the market in the early 1960's, linear alkylbenzene sulphonate (LAS) is the predominant anionic surfactant in commercial detergents. Compatibility and synergistic interaction with other ingredients lead to the enhanced performance of LAS. European LAS is a mixture of different homologues (C_{10} - C_{13} alkyl chains) including all positional isomers with the exception of 1-alkyl substituted ones. The typical homologue distribution is 15%, 30%, 30% and 25% for C_{10} , C_{11} , C_{12} and C_{13} with an average alkyl chain of $C_{11.6}$ and an average molecular weight of 342 as the sodium salt (Berna et al. (1995)).

4.4.2 Production and use

Production processes

LAS is produced by sulphonation of linear alkylbenzene (LAB) followed by neutralization of the corresponding sulphonic acid with caustic soda. The overall flow diagram is shown in Fig. 4.7.

LAB is produced by catalytic alkylation of benzene using two different processes: the aluminium chloride ($AlCl_3$) process or the hydrofluoric acid (HF) process. In the $AlCl_3$ process the benzene is alkylated with chloro-paraffins and/or n-olefins with the help of an $AlCl_3$ catalyst. In this process, part of the n-paraffins are chlorinated, the remainder is used to produce n-olefins. In the HF process, benzene is alkylated directly with n-olefins, using HF as a catalyst. The n-olefins themselves are produced by catalytic dehydrogenation of n-paraffins. In both processes the n-paraffins are responsible for the linearity of the alkyl chain. Sulphonation of LAB is then usually carried out by adding an SO_3 group to the aromatic ring of the LAB molecule. The reaction is exothermic and requires careful control of the temperature. Finally, sodium hydroxide is used to neutralize the acid that has been produced (Berna et al. (1995)).

Amount of production

The world LAS production in 1993 was nearly 2 million tonnes and 510,000 tonnes in Western Europe. LAS production in both cases represents approximately 30% of total surfactant production, excluding soap. Due to its poor anaerobic biodegradability, consumption of LAS is currently decreasing in environmentally concerned countries. In Sweden, the use of LAS has decreased over the last ten years by 95% (KTF (2000)). In countries like Poland, however, LAS is still the most important tenside.

Use

LAS is widely used in powders and liquids for textile, dishwashers and industrial cleaners.

4.4.3 Systems characterization

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

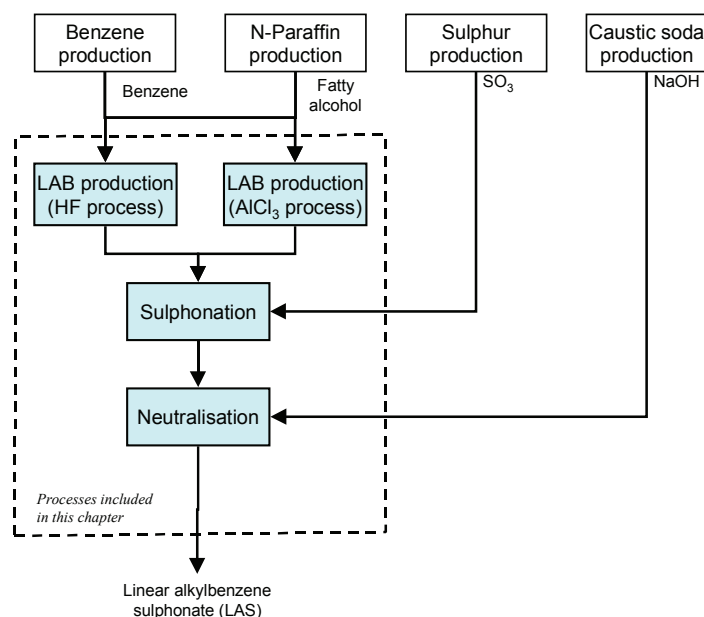


Fig. 4.7 Flow diagram of the production of LAS.

4.4.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the EMPA-study on tensides (Dall'Acqua et al. (1999)). Masses may not balance exactly due to conventions and boundary conditions (Janzen (1995)). Process-related emissions have been assessed by comparing the aggregated emissions of LAS out of Berna et al. (1995) with the aggregated emissions of its most important precursory substances, namely benzene and n-paraffins out of Franke et al. (1995).

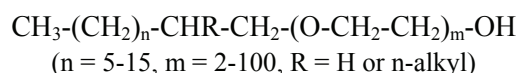
Tab. 4.20 In-/Output data and uncertainty information of the process “alkylbenzene sulfonate, linear, petrochemical, at plant RER)”

Explanation	Name	Location	Unit	alkylbenzene sulfonate, linear, petrochemical, at plant RER 0 kg	Uncertainty Type	Standard Deviation 95%	General Comment
	Location						
	Infrastructure Process Unit						
Input from Technosphere	benzene, at plant	RER	kg	2.51E-01	1	1.25	(2,3,3,2,1,4);
	paraffin, at plant	RER	kg	5.16E-01	1	1.25	(2,3,3,2,1,4);
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	RER	kg	1.27E-01	1	1.25	(2,3,3,2,1,4);
	secondary sulphur, at refinery	RER	kg	1.00E-01	1	1.25	(2,3,3,2,1,4);
	hydrogen fluoride, at plant	GLO	kg	1.00E-02	1	1.25	(2,3,3,2,1,4);
	aluminium, production mix, at plant	RER	kg	1.00E-02	1	1.25	(2,3,3,2,1,4);
	heat, unspecific, in chemical plant	RER	MJ	3.78E+00	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4.00E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	1.00E-01	1	2.07	(2,3,3,2,1,4);
	transport, freight, rail	RER	tkm	6.00E-01	1	2.07	(2,3,3,2,1,4);
Output air emission water emission	alkylbenzene sulfonate, linear, petrochemical, at plant	RER	kg	1			
	Chlorine	RER	kg	7.80E-08	1	1.58	(2,3,3,2,1,4);
	Mercury	RER	kg	3.20E-07	1	1.58	(2,3,3,2,1,4);
	Carbon monoxide, fossil	RER	kg	4.31E-06	1	1.58	(2,3,3,2,1,4);
	Nitrogen oxides	RER	kg	1.00E-04	1	1.58	(2,3,3,2,1,4);
	Hydrocarbons, aliphatic, alkanes, unspecified	RER	kg	2.31E-04	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide	RER	kg	2.93E-04	1	1.25	(2,3,3,2,1,4);
	Carbon dioxide, fossil	RER	kg	9.07E-03	1	1.25	(2,3,3,2,1,4);
	Particulates, > 10 um	RER	kg	1.64E-05	1	1.58	(2,3,3,2,1,4);
	Lead	RER	kg	1.30E-08	1	5.07	(2,3,3,2,1,4);
	Mercury	RER	kg	3.00E-08	1	5.07	(2,3,3,2,1,4);
	Hydrocarbons, unspecified	RER	kg	1.488E-07	1	3.06	(2,3,3,2,1,4);
	Phenol	RER	kg	5.22E-07	1	3.06	(2,3,3,2,1,4);
	Zinc, ion	RER	kg	7.70E-07	1	5.07	(2,3,3,2,1,4);
	Phosphate	RER	kg	9.81E-07	1	1.58	(2,3,3,2,1,4);
	Aluminum	RER	kg	1.06E-06	1	5.07	(2,3,3,2,1,4);
	Nickel, ion	RER	kg	2.08E-06	1	5.07	(2,3,3,2,1,4);
	Iron, ion	RER	kg	2.40E-06	1	5.07	(2,3,3,2,1,4);
	Chromium, ion	RER	kg	4.28E-06	1	5.07	(2,3,3,2,1,4);
	Fluoride	RER	kg	7.58E-06	1	1.58	(2,3,3,2,1,4);
	Ammonium, ion	RER	kg	1.59E-05	1	1.58	(2,3,3,2,1,4);
	Sulfate	RER	kg	6.68E-05	1	1.58	(2,3,3,2,1,4);
	Sulfide	RER	kg	1.10E-04	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand	RER	kg	1.26E-03	1	1.58	(2,3,3,2,1,4);
	Chloride	RER	kg	1.32E-03	1	1.58	(2,3,3,2,1,4);
	Solved solids	RER	kg	2.78E-03	1	1.58	(2,3,3,2,1,4);
	Suspended solids, unspecified	RER	kg	2.40E-04	1	1.58	(2,3,3,2,1,4);

4.5 Alcohol Ethoxylates (AE)

4.5.1 Introduction

Alcohol Ethoxylates (AE) are the largest group within the class of nonionic tensides. The hydrophobic part is formed by fatty alcohols with chain lengths from C₉-C₁₈. The hydrophilic part consists of a polyether chain formed by condensation of ethylene oxide to the fatty alcohol. Several hundred different AE are commercially available, differing by alcohol structure and number of polyether units. The generic chemical structure of AE is:



Because of their structural variety, AE have very different physical and chemical properties. This study focusses on AE primarily used in detergents exhibiting alcohol chain lengths of C₁₂-C₁₅. Two types of alcohol ethoxylates are principally used in the detergents industry, one with an average value of three polyether units (AE₃), the other with an average value of seven (AE₇). In comparison with other tensides like anionic surfactants, alcohol ethoxylates exhibit a better fat soil removal capacity, especially at low temperatures.

4.5.2 Production and use

Production processes

AE are produced by the reaction of fatty alcohols with ethylene oxide leading to a condensation of polyethylene glycolether groups on the alcohol chains. The fatty alcohols can be derived from petrochemical, palm kernel, or coconut oil. However the chemical and physical properties of the final product are similar for all three pathways, provided their carbon chainlength and ethoxylate distribution is similar (Stalmans et al. (1995)). The reaction takes place at temperatures ranging from 150°C to 200°C. This ethoxylation reaction can be catalyzed by alkaline catalysts like potassium hydroxide or by acidic catalysts like boron trifluoride or zinc chloride. For detergent range alcohol ethoxylates, the alkaline catalysis is normally used (Schul (1995)). The grade of ethoxylation, e.g. the specific number of polyether units condensed to the alcohol chains, is controlled by the concentration of ethylene oxid in the ethoxylation process.

Amount of production

In 1994, about 400,000 tonnes of alcohol ethoxylates are used in Western Europe in various industries. In granular and liquid detergent formulations, about 60% are based on petrochemical alcohols and about 40% on oleochemical alcohols (Schul (1995)). For this study, we assumed that oleochemical alcohols are based by 50% on palm kernel oil and by 50% on coconut oil.

Storage and transport

No specific data on transport distances of alcohol ethoxylates were available. Therefore, an assessment of average transport distances for organic chemicals on rail and lorry has been applied according to Frischknecht et al. (2007b).

4.5.3 Systems characterization

Fig. 4.8 shows the major process steps for the production of alcohol ethoxylates based on palm kernel oil, coconut oil and petrochemic oil. This inventory only comprises the process step of ethoxylation. Inventories of intermediates are shown in the respective report (Althaus et al. (2007)). The inventory has been calculated for seven different typical alcohol ethoxylates: AE with 3 polyether groups (AE₃) and AE with 7 polyether groups (AE₇), each derived from petrochemical alcohols, from palm kernel oil and from coconut oil, and additionally AE with 11 polyether groups (AE₁₁) derived from palm oil.

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

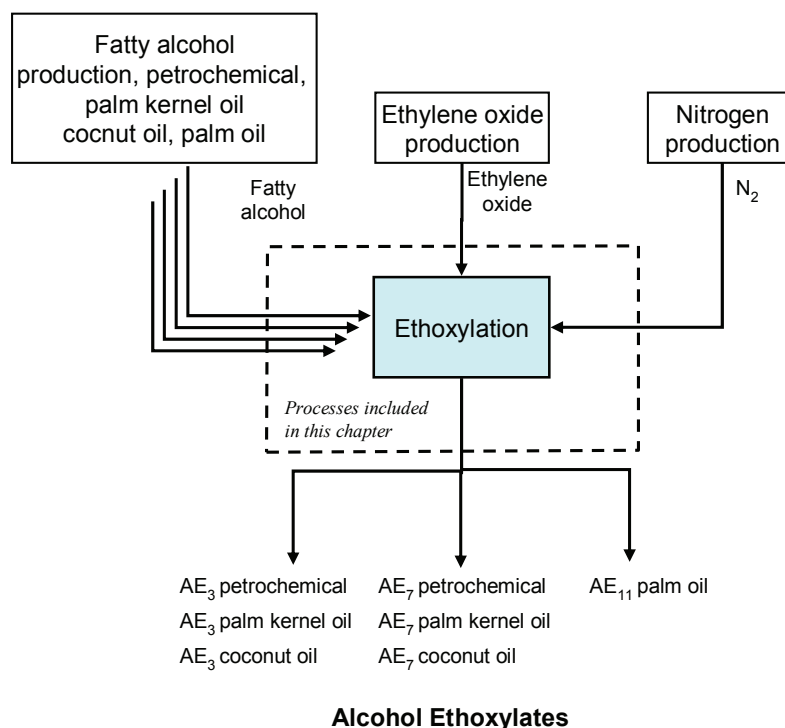


Fig. 4.8 Flow diagram of the production of alcohol ethoxylates.

4.5.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the study of Schul (1995) that examined the alcohol ethoxylate production in Western Europe. Manufacturing processes of the major producers have been considered. The original publication neither included data on infrastructure and land use nor on transport distances on road and rail. This information has been added using basic assumptions for generic organic chemicals (Frischknecht et al. (2007b)). The output of solid waste of the ethoxylation process is zero, and the energy consumption of this last process step is low compared to some of the precedent steps. Process-related emissions have been assessed by comparing the aggregated emissions of the seven different types of alcohol ethoxylates (Schul (1995)) with the aggregated emissions of the respective types of fatty alcohols (Hirsinger et al. (1995a)) and of ethylene oxide (Franke et al. (1995)). Although it is not possible to see from the used report, how many companies behind the different datasets are, the uncertainty is in general assumed to be quite low. The differences are thus due to the different basic uncertainties used for the different emissions and inputs within this study.

Additionally, an average AE dataset, based on all specific types of AE reported, is established. Due to a lack of information about the market distribution of the various AE forms, an equal distribution to all these datasets is used here. For the uncertainty, a similar characterization like for the four specific AE is used.

The resulting input and output data as well as the uncertainty scores are summarized in the following tables.

Tab. 4.21 Input/Output data and data uncertainty of the dataset “ethoxylated alcohols (AE3), petrochemical, at plant (RER)”

Explanation	Name	Location	Unit	ethoxylated alcohols (AE3), petrochemical, at plant	Uncertainty Type	Standard Deviation 95%	General Comment
	Location			RER			
	Infrastructure			0			
	Unit			kg			
Input from Technosphere	ethylene oxide, at plant	RER	kg	3.88E-01	1	1.25	(2,3,3,2,1,4);
	fatty alcohol, petrochemical, at plant	RER	kg	6.15E-01	1	1.25	(2,3,3,2,1,4);
	nitrogen, liquid, at plant	RER	kg	1.80E-02	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	6.24E-02	1	1.25	(2,3,3,2,1,4);
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.80E-01	1	1.25	(2,3,3,2,1,4);
	heat, natural gas, at industrial furnace >100kW	RER	MJ	2.40E-01	1	1.25	(2,3,3,2,1,4);
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.80E-01	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	0.7752	1	2.07	(2,3,3,2,1,4);
Output	ethoxylated alcohols (AE3), petrochemical, at plant	RER	kg	1			
	Particulates, > 10 um		kg	6.14E-06	1	1.58	(2,3,3,2,1,4);
	Nitrogen oxides		kg	7.68E-05	1	1.58	(2,3,3,2,1,4);
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	2.00E-03	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide		kg	1.89E-04	1	1.25	(2,3,3,2,1,4);
	Carbon monoxide, fossil		kg	5.86E-05	1	5.07	(2,3,3,2,1,4);
	Ammonia		kg	6.93E-06	1	1.58	(2,3,3,2,1,4);
	Carbon dioxide, fossil		kg	4.44E-03	1	1.25	(2,3,3,2,1,4);
	Suspended solids, unspecified		kg	5.16E-06	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand		kg	8.99E-04	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand		kg	1.21E-03	1	1.58	(2,3,3,2,1,4);
	Phenol		kg	6.42E-10	1	3.06	(2,3,3,2,1,4);
	Chromium, ion		kg	4.01E-07	1	5.07	(2,3,3,2,1,4);
	Zinc, ion		kg	2.07E-08	1	5.07	(2,3,3,2,1,4);

Tab. 4.22 Input/Output data and data uncertainty of the dataset “ethoxylated alcohols (AE3), palm kernel oil, at plant (RER)”

Explanation	Name	Location	Unit	ethoxylated alcohols (AE3), palm kernel oil, at plant	Uncertainty Type	Standard Deviation 95%	General Comment
	Location			RER			
	Infrastructure Process			0			
	Unit			kg			
Input from Technosphere	ethylene oxide, at plant	RER	kg	3.98E-01	1	1.25	(2,3,3,2,1,4);
	fatty alcohol, from palm kernel oil, at plant	RER	kg	5.90E-01	1	1.25	(2,3,3,2,1,4);
	nitrogen, liquid, at plant	RER	kg	1.76E-02	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	6.24E-02	1	1.25	(2,3,3,2,1,4);
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.80E-01	1	1.25	(2,3,3,2,1,4);
	heat, natural gas, at industrial furnace >100kW	RER	MJ	2.40E-01	1	1.25	(2,3,3,2,1,4);
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.80E-01	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	0.7752	1	2.07	(2,3,3,2,1,4);
Output	ethoxylated alcohols (AE3), palm kernel oil, at plant	RER	kg	1			
	air emission		kg	6.40E-07	1	1.58	(2,3,3,2,1,4);
	Particulates, > 10 um		kg	1.39E-05	1	1.58	(2,3,3,2,1,4);
	Nitrogen oxides		kg	1.72E-03	1	1.58	(2,3,3,2,1,4);
	NMVOC, non-methane volatile organic compounds, unspecified origin		kg	1.31E-05	1	1.25	(2,3,3,2,1,4);
	Sulfur dioxide		kg	6.93E-06	1	2.07	(2,3,3,2,1,4);
	Aldehydes, unspecified		kg	3.51E-05	1	1.33	(2,3,3,2,1,4);
	Methane, fossil		kg	8.36E-03	1	1.25	(2,3,3,2,1,4);
	Carbon dioxide, fossil		kg	6.00E-08	1	1.58	(2,3,3,2,1,4);
	Acidity, unspecified		kg	8.97E-04	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand		kg	1.24E-03	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand		kg	5.33E-10	1	3.06	(2,3,3,2,1,4);
	Phenol		kg	2.54E-07	1	1.58	(2,3,3,2,1,4);
	Sulfide		kg	4.26E-07	1	5.07	(2,3,3,2,1,4);
	Chromium, ion		kg	3.71E-09	1	5.07	(2,3,3,2,1,4);
	Iron, ion		kg	2.00E-06	1	5.07	(2,3,3,2,1,4);
	Nitrogen		kg	6.62E-08	1	5.07	(2,3,3,2,1,4);
	Zinc, ion		kg	1.32E-08	1	3.06	(2,3,3,2,1,4);
	Hydrocarbons, unspecified		kg	9.24E-10	1	1.58	(2,3,3,2,1,4);
	Ammonium, ion		kg				
	water emission						

Tab. 4.23 Input/Output data and data uncertainty of the dataset “ethoxylated alcohols (AE3), coconut oil, at plant (RER)”

Explanation	Name	Location	Unit	ethoxylated alcohols (AE3), coconut oil, at plant	Uncertainty	Standard Deviation	General Comment
	Location			RER			
	Infrastructure			0			
	Process			kg			
	Unit						
Input from Technosphere	ethylene oxide, at plant	RER	kg	3.98E-01	1	1.25	(2,3,3,2,1,4);
	fatty alcohol, from coconut oil, at plant	RER	kg	5.90E-01	1	1.25	(2,3,3,2,1,4);
	nitrogen, liquid, at plant	RER	kg	1.76E-02	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	6.24E-02	1	1.25	(2,3,3,2,1,4);
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.80E-01	1	1.25	(2,3,3,2,1,4);
	heat, natural gas, at industrial furnace >100kW	RER	MJ	2.40E-01	1	1.25	(2,3,3,2,1,4);
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.80E-01	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	0.7752	1	2.07	(2,3,3,2,1,4);
Output	ethoxylated alcohols (AE3), coconut oil, at plant	RER	kg	1			
	Particulates, > 10 um		kg	5.61E-05	1	1.58	(2,3,3,2,1,4);
air emission	Nitrogen oxides		kg	7.78E-06	1	1.58	(2,3,3,2,1,4);
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	1.33E-03	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide		kg	1.85E-05	1	1.25	(2,3,3,2,1,4);
	Aldehydes, unspecified		kg	6.93E-06	1	2.07	(2,3,3,2,1,4);
	Carbon dioxide, fossil		kg	8.20E-03	1	1.25	(2,3,3,2,1,4);
	Carbon dioxide, biogenic		kg	1.46E-02	1	1.25	(2,3,3,2,1,4);
	Acidity, unspecified		kg	2.81E-03	1	1.58	(2,3,3,2,1,4);
	Solved solids		kg	2.40E-03	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand		kg	1.76E-03	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand		kg	2.61E-03	1	1.58	(2,3,3,2,1,4);
	Phenol		kg	5.33E-10	1	3.06	(2,3,3,2,1,4);
	Sulfide		kg	2.54E-07	1	1.58	(2,3,3,2,1,4);
	Chromium, ion		kg	4.26E-07	1	5.07	(2,3,3,2,1,4);
	Iron, ion		kg	2.34E-04	1	5.07	(2,3,3,2,1,4);
	Nitrogen		kg	2.34E-04	1	5.07	(2,3,3,2,1,4);
	Zinc, ion		kg	6.62E-08	1	5.07	(2,3,3,2,1,4);
	Hydrocarbons, unspecified		kg	1.32E-08	1	3.06	(2,3,3,2,1,4);
	Ammonium, ion		kg	9.24E-10	1	1.58	(2,3,3,2,1,4);
water emission							

Tab. 4.24 Input/Output data and data uncertainty of the dataset “ethoxylated alcohols (AE7), petrochemical, at plant (RER)”

Explanation	Name	Location	Unit	ethoxylated alcohols (AE7), petrochemical, at plant	Uncertainty Type	Standard Deviation 95%	General Comment
	Location						
	Infrastructure Process						
	Unit			RER 0 kg			
Input from Technosphere	ethylene oxide, at plant	RER	kg	5.56E-01	1	1.25	(2,3,3,2,1,4);
	fatty alcohol, petrochemical, at plant	RER	kg	4.16E-01	1	1.25	(2,3,3,2,1,4);
	nitrogen, liquid, at plant	RER	kg	1.90E-02	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	6.24E-02	1	1.25	(2,3,3,2,1,4);
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.80E-01	1	1.25	(2,3,3,2,1,4);
	heat, natural gas, at industrial furnace >100kW	RER	MJ	2.40E-01	1	1.25	(2,3,3,2,1,4);
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.80E-01	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	0.7752	1	2.07	(2,3,3,2,1,4);
Output	ethoxylated alcohols (AE7), petrochemical, at plant	RER	kg	1			
	Particulates, > 10 um		kg	8.44E-06	1	1.58	(2,3,3,2,1,4);
	Nitrogen oxides		kg	9.93E-05	1	1.58	(2,3,3,2,1,4);
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	2.52E-03	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide		kg	1.50E-04	1	1.25	(2,3,3,2,1,4);
	Carbon monoxide, fossil		kg	5.18E-05	1	5.07	(2,3,3,2,1,4);
	Aldehydes, unspecified		kg	1.01E-05	1	2.07	(2,3,3,2,1,4);
	Ammonia		kg	4.51E-06	1	1.58	(2,3,3,2,1,4);
	Carbon dioxide, fossil		kg	2.13E-02	1	1.25	(2,3,3,2,1,4);
	Acidity, unspecified		kg	6.00E-07	1	1.58	(2,3,3,2,1,4);
	Fluoride		kg	2.00E-08	1	1.58	(2,3,3,2,1,4);
	Solved solids		kg	2.56E-05	1	1.58	(2,3,3,2,1,4);
	Suspended solids, unspecified		kg	5.60E-06	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand		kg	8.79E-04	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand		kg	1.24E-03	1	1.58	(2,3,3,2,1,4);
	Phenol		kg	3.17E-09	1	3.06	(2,3,3,2,1,4);
	Sulfide		kg	2.60E-07	1	1.58	(2,3,3,2,1,4);
	Oils, unspecified		kg	7.93E-07	1	3.06	(2,3,3,2,1,4);
	Chromium, ion		kg	3.50E-08	1	5.07	(2,3,3,2,1,4);
	Iron, ion		kg	4.29E-08	1	5.07	(2,3,3,2,1,4);
	Aluminum		kg	1.84E-10	1	1.58	(2,3,3,2,1,4);
	Nickel, ion		kg	1.39E-09	1	5.07	(2,3,3,2,1,4);
	Zinc, ion		kg	1.88E-08	1	5.07	(2,3,3,2,1,4);
	Hydrocarbons, unspecified		kg	4.48E-08	1	3.06	(2,3,3,2,1,4);
	Sulfate		kg	8.88E-06	1	1.58	(2,3,3,2,1,4);
	Chloride		kg	1.92E-06	1	1.58	(2,3,3,2,1,4);

Tab. 4.25 Input/Output data and data uncertainty of the dataset “ethoxylated alcohols (AE7), palm kernel oil, at plant (RER)”

Explanation	Name	Location	Unit	ethoxylated alcohols (AE7), palm kernel oil, at plant	Uncertainty Type	Standard Deviation 95%	General Comment
	Location			RER			
	Infrastructure Process			0			
	Unit			kg			
Input from Technosphere	ethylene oxide, at plant	RER	kg	5.89E-01	1	1.25	(2,3,3,2,1,4);
	fatty alcohol, from palm kernel oil, at plant	RER	kg	4.13E-01	1	1.25	(2,3,3,2,1,4);
	nitrogen, liquid, at plant	RER	kg	1.90E-02	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	6.24E-02	1	1.25	(2,3,3,2,1,4);
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.80E-01	1	1.25	(2,3,3,2,1,4);
	heat, natural gas, at industrial furnace >100kW	RER	MJ	2.40E-01	1	1.25	(2,3,3,2,1,4);
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.80E-01	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	0.7752	1	2.07	(2,3,3,2,1,4);
Output	ethoxylated alcohols (AE7), palm kernel oil, at plant	RER	kg	1			
water emission	Nitrogen oxides		kg	1.33E-05	1	1.58	(2,3,3,2,1,4);
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	2.62E-03	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide		kg	1.50E-05	1	1.25	(2,3,3,2,1,4);
	Aldehydes, unspecified		kg	1.09E-05	1	2.07	(2,3,3,2,1,4);
	Carbon dioxide, fossil		kg	1.12E-02	1	1.25	(2,3,3,2,1,4);
	Acidity, unspecified		kg	3.30E-07	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand		kg	9.24E-04	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand		kg	1.27E-03	1	1.58	(2,3,3,2,1,4);
	Phenol		kg	7.70E-10	1	3.06	(2,3,3,2,1,4);
	Sulfide		kg	9.70E-08	1	1.58	(2,3,3,2,1,4);
	Chromium, ion		kg	8.43E-07	1	5.07	(2,3,3,2,1,4);
	Iron, ion		kg	5.32E-09	1	5.07	(2,3,3,2,1,4);
	Nitrogen		kg	1.70E-06	1	5.07	(2,3,3,2,1,4);
	Zinc, ion		kg	1.74E-07	1	5.07	(2,3,3,2,1,4);
	Hydrocarbons, unspecified		kg	1.56E-08	1	3.06	(2,3,3,2,1,4);
	Ammonium, ion		kg	1.31E-09	1	1.58	(2,3,3,2,1,4);

Explanation	Name	Location	Unit	ethoxylated alcohols (AE7), coconut oil, at plant	Uncertainty	Type	Standard Deviation	General Comment	
	Location								
	Infrastructure								Process
	Unit								
Input from Technosphere	ethylene oxide, at plant	RER	kg	5.89E-01	1	1.25	(2,3,3,2,1,4);		
	fatty alcohol, from coconut oil, at plant	RER	kg	4.13E-01	1	1.25	(2,3,3,2,1,4);		
	nitrogen, liquid, at plant	RER	kg	1.90E-02	1	1.25	(2,3,3,2,1,4);		
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	6.24E-02	1	1.25	(2,3,3,2,1,4);		
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.80E-01	1	1.25	(2,3,3,2,1,4);		
	heat, natural gas, at industrial furnace >100kW	RER	MJ	2.40E-01	1	1.25	(2,3,3,2,1,4);		
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.80E-01	1	1.25	(2,3,3,2,1,4);		
	chemical plant, organics	RER	unit	4E-10	1	3.06	(2,3,3,2,1,4);		
transport, lorry 32t	RER	tkm	0.7752	1	2.07	(2,3,3,2,1,4);			
Output air emission	ethoxylated alcohols (AE7), coconut oil, at plant	RER	kg	1					
	Particulates, > 10 um		kg	4.45E-05	1	1.58	(2,3,3,2,1,4);		
water emission	Nitrogen oxides		kg	9.99E-06	1	1.58	(2,3,3,2,1,4);		
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	2.29E-03	1	1.58	(2,3,3,2,1,4);		
	Sulfur dioxide		kg	1.98E-05	1	1.25	(2,3,3,2,1,4);		
	Aldehydes, unspecified		kg	1.09E-05	1	2.07	(2,3,3,2,1,4);		
	Carbon dioxide, fossil		kg	1.11E-02	1	1.25	(2,3,3,2,1,4);		
	Carbon dioxide, biogenic		kg	9.64E-03	1	1.25	(2,3,3,2,1,4);		
	Acidity, unspecified		kg	1.76E-04	1	1.58	(2,3,3,2,1,4);		
	Solved solids		kg	1.59E-03	1	1.58	(2,3,3,2,1,4);		
	BOD5, Biological Oxygen Demand		kg	1.53E-03	1	1.58	(2,3,3,2,1,4);		
	COD, Chemical Oxygen Demand		kg	2.23E-03	1	1.58	(2,3,3,2,1,4);		
	Phenol		kg	7.70E-10	1	3.06	(2,3,3,2,1,4);		
	Sulfide		kg	9.70E-08	1	1.58	(2,3,3,2,1,4);		
	Chromium, ion		kg	8.43E-07	1	5.07	(2,3,3,2,1,4);		
	Iron, ion		kg	1.64E-04	1	5.07	(2,3,3,2,1,4);		
	Nitrogen		kg	1.64E-04	1	5.07	(2,3,3,2,1,4);		
	Zinc, ion		kg	1.74E-07	1	5.07	(2,3,3,2,1,4);		
Hydrocarbons, unspecified		kg	1.56E-08	1	3.06	(2,3,3,2,1,4);			
Ammonium, ion		kg	1.34E-09	1	1.58	(2,3,3,2,1,4);			

Tab. 4.27 Input/Output data and data uncertainty of the dataset “ethoxylated alcohols (AE11), palm oil, at plant (RER)”

Explanation	Name	Location	Unit	ethoxylated alcohols (AE11), palm oil, at plant	UncertaintyType	StandardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Input from Technosphere	ethylene oxide, at plant	RER	kg	6.52E-01	1	1.25	(2,3,3,2,1,4);
	fatty alcohol, from palm oil, at plant	RER	kg	3.42E-01	1	1.25	(2,3,3,2,1,4);
	nitrogen, liquid, at plant	RER	kg	1.90E-02	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	6.24E-02	1	1.25	(2,3,3,2,1,4);
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.80E-01	1	1.25	(2,3,3,2,1,4);
	heat, natural gas, at industrial furnace >100kW	RER	MJ	2.40E-01	1	1.25	(2,3,3,2,1,4);
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.80E-01	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	0.7752	1	2.07	(2,3,3,2,1,4);
Output	ethoxylated alcohols (AE11), palm oil, at plant	RER	kg	1			
	Particulates, > 10 um		kg	1.08E-03	1	1.58	(2,3,3,2,1,4);
air emission	Nitrogen oxides		kg	1.15E-04	1	1.58	(2,3,3,2,1,4);
	NMVOC, non-methane volatile organic compounds, unspecified origin		kg	3.03E-03	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide		kg	3.74E-04	1	1.25	(2,3,3,2,1,4);
	Carbon monoxide, fossil		kg	3.25E-05	1	5.07	(2,3,3,2,1,4);
	Aldehydes, unspecified		kg	1.19E-05	1	2.07	(2,3,3,2,1,4);
	Methane, fossil		kg	7.82E-03	1	1.33	(2,3,3,2,1,4);
	Carbon dioxide, fossil		kg	1.08E-02	1	1.25	(2,3,3,2,1,4);
	Carbon dioxide, biogenic		kg	2.23E-01	1	1.25	(2,3,3,2,1,4);
	Acidity, unspecified		kg	4.40E-07	1	1.58	(2,3,3,2,1,4);
	Solved solids		kg	1.67E-03	1	1.58	(2,3,3,2,1,4);
	Suspended solids, unspecified		kg	2.14E-04	1	1.58	(2,3,3,2,1,4);
	BOD5, Biological Oxygen Demand		kg	9.56E-04	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand		kg	1.77E-03	1	1.58	(2,3,3,2,1,4);
	Phenol		kg	9.47E-10	1	3.06	(2,3,3,2,1,4);
	Oils, unspecified		kg	1.23E-05	1	3.06	(2,3,3,2,1,4);
	Chromium, ion		kg	7.24E-07	1	5.07	(2,3,3,2,1,4);
	Iron, ion		kg	6.15E-09	1	5.07	(2,3,3,2,1,4);
	Nitrogen		kg	3.28E-05	1	5.07	(2,3,3,2,1,4);
	Zinc, ion		kg	1.79E-07	1	5.07	(2,3,3,2,1,4);
	Hydrocarbons, unspecified		kg	5.68E-08	1	3.06	(2,3,3,2,1,4);
	Ammonium, ion		kg	1.52E-09	1	1.58	(2,3,3,2,1,4);

Tab. 4.28 Input/Output data and data uncertainty of the dataset “ethoxylated alcohols, unspecified, at plant (RER)”

Explanation	Name	Location	Unit	ethoxylated alcohols, unspecified, at plant	UncertaintyType	StandardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Input from Technosphere	ethoxylated alcohols (AE3), petrochemical, at plant	RER	kg	1.67E-01	1	1.25	(2,3,3,2,1,4);
	ethoxylated alcohols (AE3), palm kernel oil, at plant	RER	kg	1.67E-01	1	1.25	(2,3,3,2,1,4);
	ethoxylated alcohols (AE3), coconut oil, at plant	RER	kg	1.67E-01	1	1.25	(2,3,3,2,1,4);
	ethoxylated alcohols (AE7), petrochemical, at plant	RER	kg	1.67E-01	1	1.25	(2,3,3,2,1,4);
	ethoxylated alcohols (AE7), palm kernel oil, at plant	RER	kg	1.67E-01	1	1.25	(2,3,3,2,1,4);
	ethoxylated alcohols (AE7), coconut oil, at plant	RER	kg	1.67E-01	1	1.25	(2,3,3,2,1,4);
Output	ethoxylated alcohols, unspecified, at plant	RER	kg	1			

4.6 Soap

4.6.1 Introduction

Soap is one of the oldest surfactants having been widely used since ancient times. It essentially contains the soluble sodium salt of fatty acid



and is produced by neutralisation of the fatty acid, usually with caustic soda. The fatty acid in soap comprises a range of chain lengths stemming from the basic oleochemical feedstocks; principally animal (tallow) or vegetable (palm, palm kernel, coconut).

Unlike other surfactants, soap is rarely produced as a single material. Modern soap production largely entails manufacturing of a blend of materials in order to achieve the required performance. The majority of soaps (90%) are produced by a combined saponification and neutralisation process performed on a blend of oleochemical feedstocks comprised of short chain (C_{12-14} , derived from coconut or palm kernel oil) and long chain (C_{16-18} , from tallow or palm oil) esters. Thus it is necessary to consider all four combinations in this study.

Glycerine is an important co-product of soap manufacturing. It is generated at a rate of approx. 105 g per 1 kg of soap blend and is itself a valuable intermediate. Although, some toilet soap bars retain a small amount of glycerine, it has been assumed in this study that all glycerine is being removed and used elsewhere as a co-product.

4.6.2 Production and use

Production processes

A great variety of soap manufacturing processes are known.

Amount of production

Soap continues to be a major surfactant and its worldwide use is still increasing (Postlethwaite (1995)). In Western Europe, however, soap production has declined. This decline is largely due to the increasing use of synthetic surfactants that are less sensitive to hard water and more suitable for machine usage applications.

Use

Major use for soap in Western Europe is in toilet bars. In detergents, soap is often used as a foam regulator.

4.6.3 Systems characterization

The ratio of short to long fatty acid chains in the final soap can vary but predominantly lies in the 20:80 region. This ratio and a moisture content of 13% has been assumed in the present study.

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

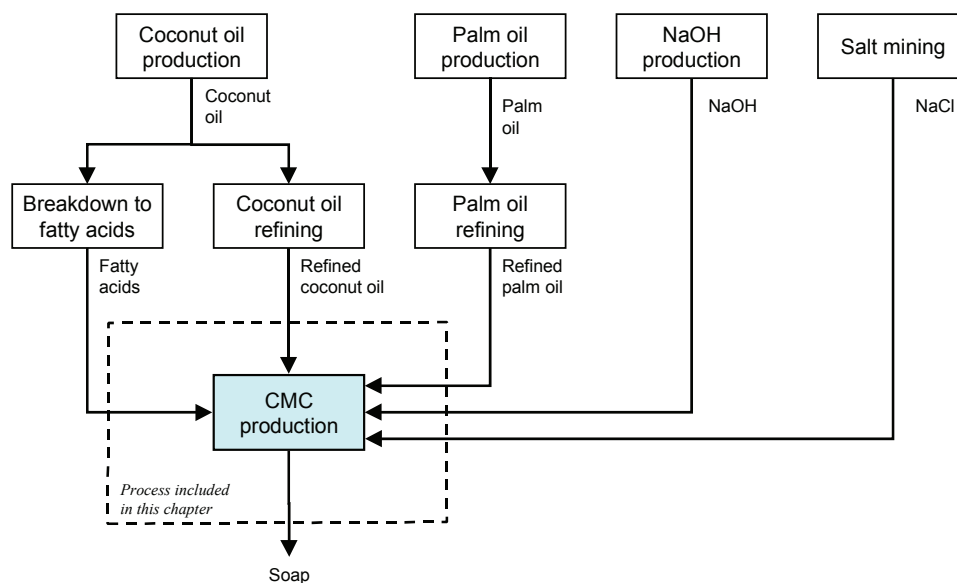


Fig. 4.9 Flow diagram of the production of soap.

4.6.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the EMPA-study on tensides. Dall'Acqua et al. (1999) Masses may not balance exactly due to conventions and boundary conditions. Janzen (1995) Process-related emissions have been assessed by comparing the aggregated emissions of Soap (Postlethwaite (1995)) with the aggregated emissions of palm oil and coconut oil (this study). Although it is not possible to see from the different used reports, how many companies behind the dataset are, the uncertainty is in general quite low. The differences are due to the different basic uncertainties used for the different emissions and inputs within this study.

The resulting input and output data as well as the uncertainty scores are summarized in the following table.

Tab. 4.29 Input/Output data and data uncertainty of the dataset “soap, at plant (RER)”

Explanation	Name	Location	Unit	soap, at plant	Uncertainty Type	Standard Deviation %	General Comment
	Location						
	Infrastructure Process						
	Unit						
Input	sodium hydroxide, 50% in H ₂ O, production mix, at plant	RER	kg	1.19E-01	1	1.25	(2,3,3,2,1,4);
	fatty acids, from vegetarian oil, at plant	RER	kg	1.60E-02	1	1.25	(2,3,3,2,1,4);
	crude coco nut oil, at plant	PH	kg	1.39E-01	1	1.25	(2,3,3,2,1,4);
	crude palm oil, at plant	MY	kg	6.14E-01	1	1.25	(2,3,3,2,1,4);
	sodium chloride, powder, at plant	RER	kg	5.00E-03	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.83E-02	1	1.25	(2,3,3,2,1,4);
	heat, unspecific, in chemical plant	RER	MJ	3.92E+00	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4.00E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	8.80E-02	1	2.07	(2,3,3,2,1,4);
	transport, freight, rail	RER	tkm	5.28E-01	1	2.07	(2,3,3,2,1,4);
	disposal, inert waste, 5% water, to inert material landfill	CH	kg	2.39E-03	1	1.25	(2,3,3,2,1,4);
Output	soap, at plant	RER	kg	1			
	Particulates, > 10 um		kg	1.51E-04	1	1.58	(2,3,3,2,1,4);
	NM VOC, non-methane volatile organic compounds, unspecified origin		kg	4.50E-06	1	1.58	(2,3,3,2,1,4);
	Sulfur dioxide		kg	1.11E-05	1	1.25	(2,3,3,2,1,4);
	Carbon monoxide, fossil		kg	3.32E-06	1	5.07	(2,3,3,2,1,4);
	Methane, fossil		kg	1.78E-04	1	1.33	(2,3,3,2,1,4);
	Mercury		kg	3.30E-07	1	5.07	(2,3,3,2,1,4);
	Chlorine		kg	6.11E-08	1	1.58	(2,3,3,2,1,4);
	Carbon dioxide, fossil		kg	3.50E-05	1	1.25	(2,3,3,2,1,4);
	Carbon dioxide, biogenic		kg	6.35E-03	1	1.25	(2,3,3,2,1,4);
	Acidity, unspecified		kg	4.19E-06	1	1.58	(2,3,3,2,1,4);
	Solved solids		kg	2.44E-03	1	1.58	(2,3,3,2,1,4);
	Suspended solids, unspecified		kg	1.44E-03	1	1.58	(2,3,3,2,1,4);
	BOD ₅ , Biological Oxygen Demand		kg	1.34E-03	1	1.58	(2,3,3,2,1,4);
	COD, Chemical Oxygen Demand		kg	6.38E-03	1	1.58	(2,3,3,2,1,4);
	Sulfide		kg	2.70E-06	1	1.58	(2,3,3,2,1,4);
	Chromium, ion		kg	2.01E-07	1	5.07	(2,3,3,2,1,4);
	Iron, ion		kg	6.10E-04	1	5.07	(2,3,3,2,1,4);
	Nickel, ion		kg	3.53E-08	1	5.07	(2,3,3,2,1,4);
	Mercury		kg	3.82E-08	1	5.07	(2,3,3,2,1,4);
	Lead		kg	1.61E-08	1	5.07	(2,3,3,2,1,4);
	Nitrogen		kg	6.06E-06	1	5.07	(2,3,3,2,1,4);
	Zinc, ion		kg	1.77E-07	1	5.07	(2,3,3,2,1,4);

4.7 Esterquat

4.7.1 Introduction

The most important cationic tensides are the quaternary nitrogen compounds, in particular tetraalkyl ammonium salts, N,N-dialkylimidazolines, and the N-alkylpyridium salts. The positive electrical charge of the hydrophilic moiety gives the cationic tensides properties that open up applications that are unsuitable or barely suitable for anionic or nonionic tensides. The cationic fabric softeners, for example, attach well to textiles thanks to their affinity for fibres. They give the washed textiles a soft feel and thus improve their wearability.

4.7.2 Production and use

Production processes

Quaternary ammonium compounds are obtained by alkylation of tertiary amines. At least one of the alkyl residues must be longer. In general, we start from tertiary amines with one or two long alkyl residues (fatty amines), reacted with shorter alkyl residues in the form of methyl chloride, ethyl chloride or bromide, or dimethyl sulphate. This then gives the relevant ammonium chloride or bromide or

methyl sulphate. The addition requires a polar solvent, e.g. alcohol for alkyl chlorides, which are primarily used for alkylation, several hours of reaction time at higher temperatures of 50°C to 100°C are required. The procedure may be simplified by carrying out the reaction in surplus alkylation solution, e.g. using methylene chloride as a solvent, under pressure, and obtaining the finished product after completion of the reaction through relaxation and evaporation of the unconverted alkyl chloride Dall'Acqua et al. (1999).

Quaternary ammonium compounds are not obtained from fatty amines. The compounds examined here are obtained by esterification of triethanolamines and subsequent quaternation with dimethylsulphate. Isopropanol is used as a solvent. Fatty acids can be obtained from vegetable oils as well as from tallow. In addition to quaternary ammonium compounds, imidazoline derivatives are also significant as a starting product for the manufacture of cationic tensides.

Use

Important applications for these cationic tensides are as microbicides, herbicides, inhibitors of corrosion or of oxidation and so on. In the detergents industry, esterquat is mainly used as intermediate for the production of fabric softeners.

4.7.3 Systems characterization

Two products are considered in this study: Esterquat based on tallow and Esterquat based on vegetable oils.

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

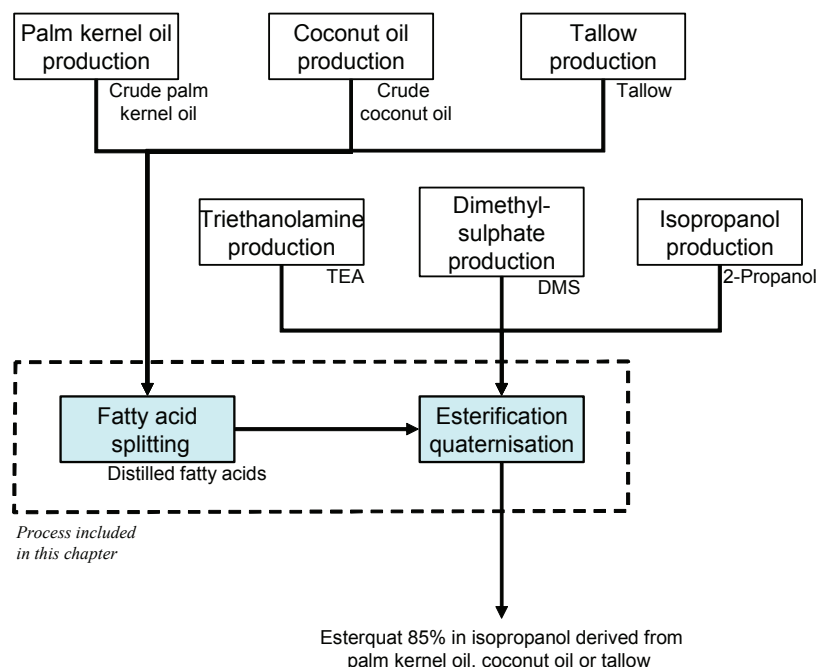


Fig. 4.10 Flow diagram of the production of esterquat.

4.7.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the EMPA-study on tensides (Dall'Acqua et al. (1999)). Masses may not balance exactly due to conventions and boundary conditions (Janzen (1995)). Process-related emissions were not available for esterquat and have been therefore omitted. Although according to the used data source, there is only one company behind the used numbers, the uncertainty is assumed to be in general low. The differences are due to the different basic uncertainties used within this study.

The resulting input and output data as well as the uncertainty scores are summarized in the following tables.

Tab. 4.30 Input/Output data and data uncertainty of the dataset “esterquat, coconut oil and palm kernel oil, at plant (RER)”

Explanation	Name	Location	Unit	esterquat, coconut oil and palm kernel oil, at plant RER 0 kg	UncertaintyType	StandardDeviation95 %	GeneralComm ent
	Location						
	InfrastructureProcess						
	Unit						
Input from Technosphere	crude coco nut oil, at plant	PH	kg	3.00E-01	1	1.25	(2,3,3,2,1,4);
	crude palm kernel oil, at plant	MY	kg	3.00E-01	1	1.25	(2,3,3,2,1,4);
	isopropanol, at plant	RER	kg	1.50E-01	1	1.25	(2,3,3,2,1,4);
	dimethyl sulphate, at plant	RER	kg	1.30E-01	1	1.25	(2,3,3,2,1,4);
	triethanolamine, at plant	RER	kg	1.70E-01	1	1.25	(2,3,3,2,1,4);
	nitrogen, liquid, at plant	RER	kg	2.80E-03	1	1.25	(2,3,3,2,1,4);
	heat, unspecific, in chemical plant	RER	MJ	2.11E+00	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4.00E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	1.05E-01	1	2.07	(2,3,3,2,1,4);
	transport, freight, rail	RER	tkm	6.32E-01	1	2.07	(2,3,3,2,1,4);
	treatment, sewage, unpolluted, to wastewater treatment, class 3	CH	m3	2.60E-04	1	1.25	(2,3,3,2,1,4);
Output	esterquat, coconut oil and palm kernel oil, at plant	RER	kg	1			

Tab. 4.31 Input/Output data and data uncertainty of the dataset “esterquat, tallow, at plant (RER)”

Explanation	Name	Location	Unit	esterquat, tallow, at plant RER 0 kg	uncertaintyType	standardDeviation95 %	GeneralComm ent
	Location						
	InfrastructureProcess						
	Unit						
Input from Technosphere	isopropanol, at plant	RER	kg	1.50E-01	1	1.25	(2,3,3,2,1,4);
	dimethyl sulphate, at plant	RER	kg	1.30E-01	1	1.25	(2,3,3,2,1,4);
	triethanolamine, at plant	RER	kg	1.70E-01	1	1.25	(2,3,3,2,1,4);
	nitrogen, liquid, at plant	RER	kg	2.80E-03	1	1.25	(2,3,3,2,1,4);
	heat, unspecific, in chemical plant	RER	MJ	2.11E+00	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4.00E-10	1	3.06	(2,3,3,2,1,4);
	transport, lorry 32t	RER	tkm	1.05E-01	1	2.07	(2,3,3,2,1,4);
	transport, freight, rail	RER	tkm	6.32E-01	1	2.07	(2,3,3,2,1,4);
	tallow, at plant	CH	kg	6.00E-01	1	1.25	(2,3,3,2,1,4);
	treatment, sewage, unpolluted, to wastewater treatment, class 3	CH	m3	2.60E-04	1	1.25	(2,3,3,2,1,4);
Output	esterquat, tallow, at plant	RER	kg	1			

4.8 EcoSpold Meta Information

Type	Field name				
ReferenceFunction	Name	n-olefins, at plant	husked nuts harvesting, at farm	crude coco nut oil, at plant	fatty alcohol, petrochemical, at plant
Geography	Location	RER	PH	PH	RER
ReferenceFunction	InfrastructureProcess	0	0	0	0
ReferenceFunction	Unit	kg	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input, production of waste and emissions for the production of n-olefins out of n-paraffins and ethylene. Water consumption and infrastructure have been estimated.	Land use for the plantation of coconut palm trees and diesel used during the year. No transports are included into this module.	This module contains material and energy input, production of waste and emissions for the steps from the harvested FFB to crude palm kernel oil (i.e. the extraction of kernel in the palm oil mill and the palm kernel oil mill). Water consumption and infrastructure have been estimated.	This module contains material and energy input, production of waste and emissions for the production of fatty alcohol out of n-olefins and ethylene. Water consumption and infrastructure have been estimated.
ReferenceFunction	Amount	1	1	1	1
ReferenceFunction	LocalName	n-Olefine, ab Werk	Kokosnuss Ernte, ab Hof	Palmkernöl, roh, ab Werk	Fettalkohol, petrochemisch, ab Werk
ReferenceFunction	Synonyms				
ReferenceFunction	GeneralComment	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.
ReferenceFunction	CASNumber				
TimePeriod	StartDate	1995	1995	1995	1995
TimePeriod	EndDate	1995	1995	1995	1995
TimePeriod	DataValidForEntirePeriod	1	1	1	1
TimePeriod	OtherPeriodText	data of published literature	data of published literature	data of published literature	data of published literature
Geography	Text	Data based on the European n-olefins production	Data based on the Philippines coconut production amount	Data based on palm oil and palm kernel oil mills in Malaysia	Data based on the European n-olefins production
Technology	Text	Average technology, typical for European production conditions in the mid 90s	average technology, typical for Philippines conditions in the mid 90s	Average technology, typical for Malaysian conditions in the mid 90s	Average technology, , representing a mix of 82% of fatty alcohols, produced with the Oxo process and the remaining 18% by the Ziegler process - typical for European production conditions in the mid 90s
Representativeness	Percent				
Representativeness	ProductionVolume	500 kt	unknown	worldwide 2'900 kt in	worldwide in 1999 2 Mio
Representativeness	SamplingProcedure				
Representativeness	Extrapolations	data out of ECOSOL	data out of ECOSOL	data out of ECOSOL	data out of ECOSOL
Representativeness	UncertaintyAdjustments				

4. Tensides

Type	Field name				
ReferenceFunction	Name	fatty alcohol, from palm oil, at plant	fatty alcohol, from palm kernel oil, at plant	fatty alcohol, from coconut oil, at plant	fatty alcohol sulfonate, petrochemical, at plant
Geography	Location	RER	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0	0
ReferenceFunction	Unit	kg	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input, production of waste and emissions for the production of fatty alcohol out of crude palm oil and natural gas. The production of the used methanol in the methyl ester production chain is included into this dataset here (therefore the input of natural	This module contains material and energy input, production of waste and emissions for the production of fatty alcohol out of crude palm kernel oil and natural gas. The production of the used methanol in the methyl ester production chain is included into this dataset here (therefore the input of natural	This module contains material and energy input, production of waste and emissions for the production of fatty alcohol out of crude coconut oil and natural gas. The production of the used methanol in the methyl ester production chain is included into this dataset here (therefore the input of natural	This module contains material and energy input, production of waste and emissions for the production of fatty alcohol sulfonate out of petrochemical resources. Transports and infrastructure have been estimated. No water consumption included.
ReferenceFunction	Amount	1	1	1	1
ReferenceFunction	LocalName	Fettalkohol, aus Palmöl, ab Werk	Fettalkohol, aus Palmkernöl, ab Werk	Fettalkohol, aus Kokosöl, ab Werk	Fettalkoholsulfonate, petrochemisch, ab Werk
ReferenceFunction	Synonyms				FAS, petrochemisch
ReferenceFunction	GeneralComment	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.
ReferenceFunction	CASNumber				-
TimePeriod	StartDate	1995	1995	1995	1992
TimePeriod	EndDate	1995	1995	1995	1995
TimePeriod	DataValidForEntirePeriod	1	1	1	1
TimePeriod	OtherPeriodText	data of published literature	data of published literature	data of published literature	data of published literature
Geography	Text	Data based on the European n-olefins production	Data based on the European n-olefins production	Data based on the European n-olefins production	Data based on the European fatty alcohol sulfonate production
Technology	Text	Average technology, representing a mix of 27% produced from fatty acid splitting, 56% produced from methyl ester on the basis of crude vegetable oil and 17% from methyl ester out of refined oil - typical for European production conditions in the mid 90s	Average technology, representing a mix of 27% produced from fatty acid splitting, 56% produced from methyl ester on the basis of crude vegetable oil and 17% from methyl ester out of refined oil - typical for European production conditions in the mid 90s	Average technology, representing a mix of 27% produced from fatty acid splitting, 56% produced from methyl ester on the basis of crude vegetable oil and 17% from methyl ester out of refined oil - typical for European production conditions in the mid 90s	Average technology for the production of FAS with short chain length (C12-C15), representing typical European production mix in the mid 90s
Representativeness	Percent				60
Representativeness	ProductionVolume	worldwide in 1999 2 Mio	worldwide in 1999 2 Mio	worldwide in 1999 2 Mio	unknown
Representativeness	SamplingProcedure				data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).
Representativeness	Extrapolations	data out of ECOSOL	data out of ECOSOL	data out of ECOSOL	see technology
Representativeness	UncertaintyAdjustments				none

4. Tensides

Type	Field name				
ReferenceFunction	Name	fatty alcohol sulfonate, palm kernel oil, at plant	fatty alcohol sulfonate, coconut oil, at plant	fatty alcohol sulfonate, palm oil, at plant	fatty alcohol sulfonate, mix, at plant
Geography	Location	RER	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0	0
ReferenceFunction	Unit	kg	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input, production of waste and emissions for the production of fatty alcohol sulfonate out of palm kernel oil. Transports and infrastructure have been estimated. No water consumption included.	This module contains material and energy input, production of waste and emissions for the production of fatty alcohol sulfonate out of coconut oil. Transports and infrastructure have been estimated. No water consumption included.	This module contains material and energy input, production of waste and emissions for the production of fatty alcohol sulfonate out of palm oil. Transports and infrastructure have been estimated. No water consumption included.	This module represents an unspecified mix of the four different FAS datasets that are included into this database
ReferenceFunction	Amount	1	1	1	1
ReferenceFunction	LocalName	Fettalkoholsulfonate, Palmkernöl, ab Werk	Fettalkoholsulfonate, Kokosnussöl, ab Werk	Fettalkoholsulfonate, Palmöl, ab Werk	Fettalkoholsulfonate, Mix, ab Werk
ReferenceFunction	Synonyms	FAS aus Palmkernöl	FAS aus Kokosnussöl	FAS aus Palmöl	
ReferenceFunction	GeneralComment	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data represent unweighted average of four different FAS types
ReferenceFunction	CASNumber	-	-	-	-
TimePeriod	StartDate	1992	1992	1992	1992
TimePeriod	EndDate	1995	1995	1995	1995
TimePeriod	DataValidForEntirePeriod	1	1	1	1
TimePeriod	OtherPeriodText	data of published literature	data of published literature	data of published literature	data of used literature sources
Geography	Text	Data based on the European fatty alcohol sulfonate production	Data based on the European fatty alcohol sulfonate production	Data based on the European fatty alcohol sulfonate production	Data based on the different FAS production
Technology	Text	Average technology for the production of FAS with short chain length (C12-C15), representing typical European production mix in the mid 90s	Average technology for the production of FAS with short chain length (C12-C15), representing typical European production mix in the mid 90s	Average technology for the production of FAS with medium chain length (C16-C18), representing typical European production mix in the mid 90s	unweighted average of FAS petrochemical, from palm kernel oil, from coconut oil and from palm oil
Representativeness	Percent	60	60	60	
Representativeness	ProductionVolume	unknown	unknown	unknown	unknown
Representativeness	SamplingProcedure	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	see technology
Representativeness	Extrapolations	see technology	see technology	see technology	see technology
Representativeness	UncertaintyAdjustments	none	none	none	none

4. Tensides

Type	Field name				
ReferenceFunction	Name	alkylbenzene sulfonate, linear, petrochemical, at plant	ethoxylated alcohols (AE3), petrochemical, at plant	ethoxylated alcohols (AE3), palm kernel oil, at plant	ethoxylated alcohols (AE3), coconut oil, at plant
Geography	Location	RER	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0	0
ReferenceFunction	Unit	kg	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input, production of waste and emissions for the production of linear alkylbenzene sulfonate out of benzene and paraffins. Transports and infrastructure have been estimated. No water consumption included.	This module contains material and energy input, transport efforts, production of waste and emissions for the production of alcohol etoxylate out of fatty alcohols and ethylene oxide. Infrastructure has been estimated. No water consumption included.	This module contains material and energy input, transport efforts, production of waste and emissions for the production of alcohol etoxylate out of fatty alcohols and ethylene oxide. Infrastructure has been estimated. No water consumption included.	This module contains material and energy input, transport efforts, production of waste and emissions for the production of alcohol etoxylate out of fatty alcohols and ethylene oxide. Infrastructure has been estimated. No water consumption included.
ReferenceFunction	Amount	1	1	1	1
ReferenceFunction	LocalName	Alkylbenzolsulfonate, lineare, petrochemisch, ab Mitte 90er	Alkoholethoxylate (AE3), petrochemisch, ab Mitte 90er	Alkoholethoxylate (AE3), Palmkernöl, ab Mitte 90er	Alkoholethoxylate (AE3), Kokosnussöl, ab Mitte 90er
ReferenceFunction	Synonyms				
ReferenceFunction	GeneralComment	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.
ReferenceFunction	CASNumber	42615-29-2			
TimePeriod	StartDate	1992	1992	1992	1992
TimePeriod	EndDate	1995	1995	1995	1995
TimePeriod	DataValidForEntirePeriod	1	1	1	1
TimePeriod	OtherPeriodText	data of published literature	data of published literature	data of published literature	data of published literature
Geography	Text	Data based on the European linear alkylbenzene sulfonate production	Data based on the European ethoxylated alcohols production	Data based on the European ethoxylated alcohols production	Data based on the European ethoxylated alcohols production
Technology	Text	Average technology, representing a mix of 50% aluminium chloride (AlCl3) process and 50% hydrofluoric acid (HF) process - typical for European production conditions in the mid 90s.	Average technology for the production of AE with 3 polyether units, representing typical European production mix in the mid 90s	Average technology for the production of AE with 3 polyether units, representing typical European production mix in the mid 90s	Average technology for the production of AE with 3 polyether units, representing typical European production mix in the mid 90s
Representativeness	Percent	40	40	40	40
Representativeness	ProductionVolume	510000000	150000000	150000000	150000000
Representativeness	SamplingProcedure	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).
Representativeness	Extrapolations	see technology	see technology	see technology	see technology
Representativeness	UncertaintyAdjustments	none	none	none	none

4. Tensides

Type	Field name				
ReferenceFunction	Name	ethoxylated alcohols (AE7), petrochemical, at plant	ethoxylated alcohols (AE7), palm kernel oil, at plant	ethoxylated alcohols (AE7), coconut oil, at plant	ethoxylated alcohols (AE11), palm oil, at plant
Geography	Location	RER	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0	0
ReferenceFunction	Unit	kg	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input, transport efforts, production of waste and emissions for the production of alcohol etoxylate out of fatty alcohols and ethylene oxide. Infrastructure has been estimated. No water consumption included.	This module contains material and energy input, transport efforts, production of waste and emissions for the production of alcohol etoxylate out of fatty alcohols and ethylene oxide. Infrastructure has been estimated. No water consumption included.	This module contains material and energy input, transport efforts, production of waste and emissions for the production of alcohol etoxylate out of fatty alcohols and ethylene oxide. Infrastructure has been estimated. No water consumption included.	This module contains material and energy input, transport efforts, production of waste and emissions for the production of alcohol etoxylate out of fatty alcohols and ethylene oxide. Infrastructure has been estimated. No water consumption included.
ReferenceFunction	Amount	1	1	1	1
ReferenceFunction	LocalName	Alkoholethoxylate (AE7), petrochemisch, ab Modell	Alkoholethoxylate (AE7), Palmkernöl, ab Modell	Alkoholethoxylate (AE7), Kokosnussöl, ab Modell	Alkoholethoxylate (AE11), Palmöl, ab Modell
ReferenceFunction	Synonyms				
ReferenceFunction	GeneralComment	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.
ReferenceFunction	CASNumber				
TimePeriod	StartDate	1992	1992	1992	1992
TimePeriod	EndDate	1995	1995	1995	1995
TimePeriod	DataValidForEntirePeriod	1	1	1	1
TimePeriod	OtherPeriodText	data of published literature	data of published literature	data of published literature	data of published literature
Geography	Text	Data based on the European ethoxylated alcohols production	Data based on the European ethoxylated alcohols production	Data based on the European ethoxylated alcohols production	Data based on the European ethoxylated alcohols production
Technology	Text	Average technology for the production of AE with 7 polyether units, representing typical European production mix in the mid 90s	Average technology for the production of AE with 7 polyether units, representing typical European production mix in the mid 90s	Average technology for the production of AE with 7 polyether units, representing typical European production mix in the mid 90s	Average technology for the production of AE with 11 polyether units, representing typical European production mix in the mid 90s
Representativeness	Percent	40	40	40	40
Representativeness	ProductionVolume	150000000	150000000	150000000	150000000
Representativeness	SamplingProcedure	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).
Representativeness	Extrapolations	see technology	see technology	see technology	see technology
Representativeness	UncertaintyAdjustments	none	none	none	none

4. Tensides

Type	Field name				
ReferenceFunction	Name	ethoxylated alcohols, unspecified, at plant	soap, at plant	esterquat, coconut oil and palm kernel oil, at plant	esterquat, tallow, at plant
Geography	Location	RER	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0	0
ReferenceFunction	Unit	kg	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module represents an unspecified mix of the six different AE3 and AE7 datasets that are included into this database	This module contains material and energy input, production of waste and emissions for the production of soap out of fatty acids from palm and coconut oil. Transports and infrastructure have been estimated. No water consumption included.	This module contains material and energy input, production of waste and emissions for the production of esterquat out of palm and coconut oil. Transports and infrastructure have been estimated. No water consumption and no process emissions are included.	This module contains material and energy input, production of waste and emissions for the production of esterquat out of tallow. Transports and infrastructure have been estimated. No water consumption and no process emissions are included.
ReferenceFunction	Amount	1	1	1	1
ReferenceFunction	LocalName	Alkoholethoxylate, unspezifisch, ab Werk	Seife, ab Werk	Esterquat, Kokos- und Palmöl, ab Werk	Esterquat, Rindertalg, ab Werk
ReferenceFunction	Synonyms				
ReferenceFunction	GeneralComment	data represent unweighted average of six different AE types	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.	data based on the ECOSOL study of the European surfactant industry. Allocations in multioutput processes were made, using the relative mass outputs of products.
ReferenceFunction	CASNumber		68952-95-4	-	-
TimePeriod	StartDate	1992	1992	1992	1992
TimePeriod	EndDate	1995	1995	1998	1998
TimePeriod	DataValidForEntirePeriod	1	1	1	1
TimePeriod	OtherPeriodText	data of used literature sources	data of published literature	data of published literature	data of published literature
Geography	Text	Data based on the different AE production	Data based on the European fatty alcohol sulfonate production	Data based on the European esterquat production	Data based on the European esterquat production
Technology	Text	unweighted average of AE3 and AE7 (petrochemical, from palm kernel oil, from coconut oil)	Average technology for the production of soap out of a blend of fatty acids from palm and coconut oil, representing typical European production mix in the mid 90s. The ratio of short to long fatty acid chains is 20:80, the moisture 13%.	Average technology for the production of esterquat out of palm kernel oil and coconut oil by esterification with triethanolamine and quaternication with DMS. Data representing typical European production mix in the mid 90s.	Average technology for the production of esterquat out of tallow by esterification with triethanolamine and quaternication with DMS. Data representing typical European production mix in the mid 90s.
Representativeness	Percent		60		
Representativeness	ProductionVolume	unknown	unknown	unknown	unknown
Representativeness	SamplingProcedure	see technology	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).	data out of ECOSOL study of European Surfactant Industry. Common translation rules used, reported in Chemical report (Althaus et al. 2003).
Representativeness	Extrapolations	see technology	unspecified metal	see technology	see technology
Representativeness	UncertaintyAdjustments	none	none	none	none

5 Builders

5.1 Sodium Silicates (water glass)

5.1.1 Introduction

The following paragraph is mainly based on Fawer (1997).

The solid portions of the surface of the Earth consist predominantly of silicon and oxygen atoms in the form of silica and silicates. The use of Earth's silica and silicate resources is thus the oldest technology on the planet – first antropogenic silicates were introduced with the domestication of the fire. The preparation of silicates by chemical reaction began late in the industrial revolution with the production of Portland cement in the 1830s and shortly after, soluble alkali silicates were produced on a commercial scale. About 100 years later, soluble silicates were used in aqueous effluent treatments and in the 1940s started the development of aqueous silica sols and of crystalline silicates by hydrothermal synthesis. Nowadays, the range of application of silica and silicates is mainly in the following three fields: detergents, chemicals, adhesives.

As a detergent ingredient or wash alkali, sodium silicates raise the pH value and dirt and textile fibres become more strongly negatively charged increasing their mutual electrostatic repulsion. Apart from this, the hardness components of the water are eliminated by complex formation.

Soluble silicate glasses, powders and liquids are among the largest volume synthetic chemicals with a world-wide volume for sodium silicates estimated as about 3×10^6 tonnes. They are essentially combinations of alkali metal oxide and silica, usually with some water. Here – the study is focusing on sodium as alkali metal and thus on sodium silicate (CAS-No. 1344-09-8).

5.1.2 Production and use

Production processes

Two main processes, furnace and hydrothermal, are used to produce sodium silicate:

- In the **furnace process**, water glass is produced directly by melting pure silicon sand and soda. Temperatures of approx. 1400 °C are required for this process. The sodium silicate produced can then be used directly in a solid form, but is usually in aqueous solution (37%).
- In the **hydrothermal process**, sand is dissolved in sodium hydroxide. This process is carried out under high temperature and pressure in an autoclave. After filtration sodium silicate is obtained in a 48% solution. Spray-drying this liquid reveals a powder of 80%. The sodium silicate solution can be reacted again with sodium hydroxide. Metasilicate crystallises out and can be used in its solid form.

A newer development is **crystalline layered sodium silicate (SKS-6)**. The hydrothermal 48% sodium silicate solution is dehydrated and crystallized at high temperatures to the d-modification of the disilicate. SKS-6 can subsequently be milled and turned into granulate if desired. Layered sodium silicates also occur naturally as zeolites. In its stoichiometric composition SKS-6 is equivalent to $\text{Na}_2\text{Si}_2\text{O}_5$ and its crystalline layer structure is closest to the mineral natrosilite. The crystal structure is built in layers.

Use

There are currently three principal areas of use for sodium silicates: as a raw material in detergents, as a starting material for the chemical industry, and as an adhesive in various fields. In detergents it serves as a builder, corrosion inhibitor and additive. SKS-6 can remove the hardness of water by ion

exchange. Besides calcium ions, it also binds magnesium ions. SKS-6 also provides the alkalinity necessary for the washing process, and stabilises the pH value over the whole procedure.

5.1.3 Systems characterization

Fig. 5.1 shows the processes for the production of the different types of sodium silicates included into this study. As can be seen there, furnace and hydrothermal process are the two starting points, producing each a form of sodium silicate that is used for the further production steps. In each step, the material and energy input (incl. cooling and process water), the infrastructure and transport of the raw materials to the production site as well as the emissions to air and water and the produced waste are included.

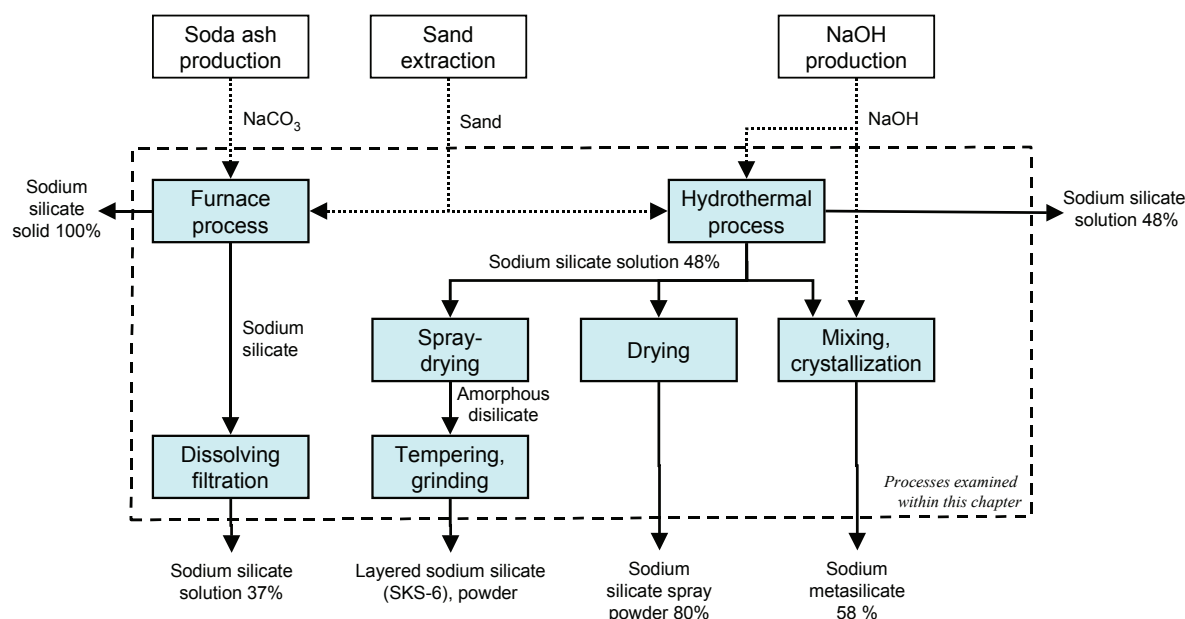


Fig. 5.1 Flow diagram of the production of the different types of sodium silicates

5.1.4 Sodium silicate from furnace process

Main data source for this process step is the sodium silicate study commissioned by the CEES¹ and established by EMPA in 1997 (Fawer (1997)). The data are average values from six different European producers for 1995, representing a production of 260 kt of solid sodium silicate by the furnace process. These data are shown in Tab. 5.1.

¹ CEES – Centre Européen d'Etude des Silicates

Tab. 5.1 Average data for the production of sodium silicate with the furnace process

Input		Na-silicate	Output		Na-silicate
<i>materials</i>			<i>waste</i>		
soda	g	399.60	inert chemicals	g	0.65
sand (100% dry)	g	772.10	<i>emissions to air</i>		
air	m3	0.015	water vapour (steam)	g	68.00
Additives	g	0.10	hydrofluoric acid, HF	g	4.00E-04
<i>energy</i>			fossil carbon dioxide, CO2	g	540.00
electricity	kWh	0.0332	carbon monoxide, CO	g	2.58E-01
heavy fuel oil	g	54.50	hydrochloric acid, HCl	g	2.40E-02
natural gas	m3	6.10E-02	sulphur oxides, as SO2	g	1.32E+00
steam	g	7.30	Dust, particles	g	2.12E-01
<i>water</i>			nitrogen oxides, NOx	g	2.03E+00
water (process)	m3	4.60E-04	<i>emissions to water</i>		
water (cooling)	m3	9.40E-04	waste water	m3	2.80E-04
Output			BOD	g	1.70E-04
<i>product</i>			COD	g	3.00E-06
Na-silicate, furnace, solid	g	1000.00	suspended solids	g	4.20E-03

For an integration of these data into the database ecoinvent, the following assumptions and transformations have been made:

- **Material input:** The amount of additives shown in Fawer (1997) is expressed as “chemical organics, at plant (GLO)” (instead as an energy consumption like in Fawer (1997)). The amount of compressed air consumed is not shown in the ecoinvent dataset according to Frischknecht et al. (2007b).
- **Energy input:** The input of the different fuels is taken directly from the information in Tab. 5.1. For all of them, the respective “heating” module is used, allowing to have a similar degree of details concerning the emissions to air. The amount of steam used is represented by the module “steam, at plant (RER)” – described in chapter 7.3 of this report here.
- **Transport and Infrastructure:** According to Fawer (1997), the authors included individual information for the transport distances of each involved company. As these informations are not any more available, for this study, standard distances and means of transport according to Frischknecht et al. (2007b) are used. For the infrastructure, due to a lack of data, the general module “chemical plant, organics (RER)” is used here.
- **Emissions to air and water:** The air emissions shown in Tab. 5.1 represent process and combustion emission. As the fuels are represented by their respective “heating” module, only the process emissions are inventorized within this study. A consumption of 400 g soda produces 165 g of carbon dioxide. Steam is a second output of the production process. All other emissions are supposed to be part of the combustion process and therefore are already included in the heating modules of the different fuels. In accordance with the method, steam and the electricity input are shown as waste heat output. For steam output, a heating value of 2.9 MJ/kg is used. The water emissions are inventorized according to the indications in Tab. 5.1.
- **Waste:** Small amounts of inert chemicals are reported as waste of this production process. As there are no further informations about the composition / origin of this waste, for this study, the dataset “disposal, average incineration residues, 0% water, to residual material landfill” is used according to the methodology chosen for the integration of the plastics data (see Hischier (2007)).

5.1.5 Sodium silicate from hydrothermal process

Main data source for this process step is again the sodium silicate study commissioned by the CEES and established by EMPA in 1997 (Fawer (1997)). The data are average values from six different European producers for 1995, representing a production of 91 kt of hydrothermal sodium silicate, liquid, as 48% solution. These data are shown in Tab. 5.2.

Tab. 5.2 Average data for the production of hydrothermal sodium silicate, liquor, 48%

Input		Na-silicate	Output		Na-silicate
<i>materials</i>			<i>waste</i>		
sodium hydroxide (100%)	g	209.00	inert chemicals	g	0.02
sand (100% dry)	g	325.00	mineral waste	g	8.70
air	m3	0.004	filter residues, disposal	g	1.20
Additives	g	0.70	filter residues, recycling	g	0.70
<i>energy</i>			<i>emissions to air</i>		
electricity	kWh	1.48E-02	water vapour (steam)	g	69.00
hard coal	g	0.90	<i>emissions to water</i>		
heavy fuel oil	g	0.10	waste water	m3	1.10E-04
natural gas	m3	9.70E-03	COD as g O2	g	6.30E-03
steam	g	143.00	Chlorides, Cl-	g	4.70E-02
<i>water</i>			Iron, Fe	g	4.00E-05
water (process)	m3	7.70E-04	Nitrates, NO3-	g	1.40E-04
water (cooling)	m3	1.00E-05	Phosphates, PO4 3-	g	1.00E-05
Output			Silicates	g	1.64E+00
<i>product</i>			Suspended solids	g	6.11E-01
Na-silicate, 48% liquor	g	1000.00			

For an integration of these data into the database ecoinvent, the following assumptions and transformations have been made:

- **Concentration factor:** The data reported in Fawer (1997) refer to a 48% solution of sodium silicate. As in this study here, all chemicals use a functional unit referring to 100% of sodium silicate, all used values have been multiplied with a factor of 2.08. The module inventorized is “sodium silicate, hydrothermal, 48% in water, at plant (RER)”.
- **Material input:** The amount of additives shown in Fawer (1997) is expressed as “chemical organics, at plant (GLO)” (instead as an energy consumption like in Fawer (1997)). The amount of compressed air consumed is not shown in the ecoinvent dataset according to Frischknecht et al. (2007b).
- **Energy input:** The input of the different fuels is taken directly from the information in Tab. 5.1. For all of them, the respective “heating” module is used, allowing to have a similar degree of details concerning the emissions to air. The amount of steam used is represented by the module “steam, at plant (RER)” – described in chapter 7.3 of this report here.
- **Transport and Infrastructure:** According to Fawer (1997), the authors included individual information for the transport distances of each involved company. As these informations are not any more available, for this study, standard distances and means of transport according to Frischknecht et al. (2007b) are used. For the infrastructure, due to a lack of data, the general module “chemical plant, organics (RER)” is used here.
- **Emissions to air and water:** No process emissions apart steam are reported and the combustion emissions are part of the respective heating modules used for the fuels. In accordance with the

method, steam as well as the electricity input are shown as waste heat output. For steam output, a heating value of 2.9 MJ/kg is used. The water emissions are inventorized according to the indications in Tab. 5.1.

- **Waste:** From the reported types of waste, no further information about the composition / origin of the respective waste is known. Therefore the different wastes are inventorized as following: inert chemicals as “disposal, average incineration residues, 0% water, to residual material landfill”, filter residues (only the amount for disposal) as “disposal, residues from cooling tower, 30% water, to sanitary landfill” and mineral waste as “disposal, inert waste, 5% water, to inert material landfill”.

5.1.6 Sodium silicates for detergents

Besides the direct use of the two above described forms of sodium silicate – they are also the starting point for the production of further forms of sodium silicates used within the detergent industry. Thereof the inventories of „sodium silicate, furnace liquor, 37% in water“, „sodium silicate, spray powder“ and „sodium metasilicate pentahydrate, 58% powder“ are based like the two already discussed forms on the EMPA study established for CEES (Fawer (1997)). For the integration into the database ecoinvent, all of them have been transformed accordingly to the descriptions in chapter 5.1.4 and 5.1.5 – using similar assumptions and transformation procedures.

The last form of sodium silicate – SKS-6 (layered sodium silicate, powder) – is based mainly on the information collected for the former EMPA detergent study (Dall'Acqua et al. (1999)). The data there are based on information from the main producer of this type of sodium silicate. For this study, these data are transformed and adapted according the methodology of this study and the assumptions used for the other forms of sodium silicate, described above. As these data are based on one single company, the unit process data are kept confidential and only a cumulated dataset is shown to the user of the database ecoinvent.

5.1.7 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. As the data for the five first forms of sodium silicate represent an average Western Europe production of 1995, including 93% of the European production volume, the respective uncertainty is quite low. In all cases the transports and the infrastructure have higher uncertainties due to the fact that they are not from Fawer (1997) but own assumptions from the authors of this study. The following tables summarize the input data as well as the uncertainties for the different forms of sodium silicate, with the exception of SKS-6 (confidential data).

Tab. 5.3 Input / Output and uncertainty values for the process "sodium silicate, hydrothermal liquor, 48%, at plant (RER)"

Explanation	Name	Location	Unit	sodium silicate, hydrothermal liquor, 48% in H ₂ O, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resource Input from Technosphere	Water, cooling, unspecified natural origin		m3	2.08E-05	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	Water, unspecified natural origin		m3	1.60E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	RER	kg	4.35E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	silica sand, at plant	CH	kg	6.76E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	chemicals organic, at plant	GLO	kg	1.46E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	3.08E-02	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	8.57E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, natural gas, at industrial furnace >100kW	RER	MJ	7.32E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	5.41E-02	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	steam, for chemical processes, at plant	RER	kg	2.97E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	chemical plant, organics	RER	unit	4.00E-10	1	3.77	(4,5,1,3,5,4); estimation
	transport, freight, rail	RER	tkm	9.28E-01	1	2.09	(4,5,na,na,na,na); standard distances
	transport, lorry 32t	RER	tkm	1.55E-01	1	2.09	(4,5,na,na,na,na); standard distances
	disposal, average incineration residue, 0% water, to residual material landfill	CH	kg	4.16E-05	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
Waste	disposal, residue from cooling tower, 30% water, to sanitary landfill	CH	kg	2.50E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	disposal, inert waste, 5% water, to inert material landfill	CH	kg	1.81E-02	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
Output	sodium silicate, hydrothermal liquor, 48% in H ₂ O, at plant	RER	kg	1			
Air emission Water emission	Heat, waste		MJ	3.22E+00	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	COD, Chemical Oxygen Demand		kg	1.31E-05	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers
	Suspended solids, unspecified		kg	1.27E-03	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers
	Nitrate		kg	2.91E-07	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers
	Phosphate		kg	2.08E-08	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers
	Chloride		kg	9.78E-05	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers
	Iron, ion		kg	8.32E-08	1	5.02	(2,2,2,1,1,4); Average data from 6 European producers
	Silicon		kg	3.41E-03	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers

Tab. 5.4 Input / Output and uncertainty values for process "sodium silicate, furnace process, pieces, at plant (RER)"

Explanation	Name	Location	Unit	sodium silicate, furnace process, pieces, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resource Input from Technosphere	Water, cooling, unspecified natural origin		m3	9.40E-04	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	Water, unspecified natural origin		m3	4.60E-04	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	soda, powder, at plant	RER	kg	4.00E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	silica sand, at plant	CH	kg	7.72E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	chemicals organic, at plant	GLO	kg	1.00E-04	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	3.32E-02	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.25E+00	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, natural gas, at industrial furnace >100kW	RER	MJ	2.21E+00	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	steam, for chemical processes, at plant	RER	kg	7.30E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	chemical plant, organics	RER	unit	4.00E-10	1	3.77	(4,5,1,3,5,4); estimation
	transport, freight, rail	RER	tkm	7.03E-01	1	2.09	(4,5,na,na,na,na); standard distances
	transport, lorry 32t	RER	tkm	1.17E-01	1	2.09	(4,5,na,na,na,na); standard distances
	disposal, average incineration residue, 0% water, to residual material landfill	CH	kg	6.50E-04	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	sodium silicate, furnace process, pieces, at plant	RER	kg	1			
Output	Heat, waste		MJ	1.96E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
Air emission Water emission	Carbon dioxide, fossil		kg	1.65E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	BOD5, Biological Oxygen Demand		kg	1.70E-07	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers
	COD, Chemical Oxygen Demand		kg	3.00E-09	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers
	Suspended solids, unspecified		kg	4.20E-06	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers

Tab. 5.5 Input / Output and uncertainty values for process “sodium silicate, furnace liquor, 37%, in H2O, at plant (RER)”

Explanation	Name	Location	Unit	sodium silicate, furnace liquor, 37% in H2O, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resource	Water, cooling, unspecified natural origin		m3	1.08E-06	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	Water, unspecified natural origin		m3	2.24E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
Input from Technosphere	sodium silicate, furnace process, pieces, at plant	RER	kg	1.01E+00	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	chemicals organic, at plant	GLO	kg	1.89E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	7.57E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.00E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, natural gas, at industrial furnace >100kW	RER	MJ	7.16E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	8.05E-02	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	steam, for chemical processes, at plant	RER	kg	3.03E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	chemical plant, organics	RER	unit	4.00E-10	1	3.77	(4,5,1,3,5,4); estimation
	transport, freight, rail	RER	tkm	6.04E-01	1	2.09	(4,5,na,na,na,na); standard distances
	transport, lorry 32t	RER	tkm	1.01E-01	1	2.09	(4,5,na,na,na,na); standard distances
Waste	disposal, residue from cooling tower, 30% water, to sanitary landfill	CH	kg	2.57E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	disposal, inert waste, 5% water, to inert material landfill	CH	kg	5.41E-05	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
Output	sodium silicate, furnace liquor, 37% in H2O, at plant	RER	kg	1			
Air emission	Heat, waste		MJ	9.05E-01	1	1.13	(2,2,2,1,1,4); calculated from electricity and steam input

Tab. 5.6 Input / Output and uncertainty values for process “sodium silicate, spray powder 80%, at plant (RER)”

Explanation	Name	Location	Unit	sodium silicate, spray powder 80%, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resource	Water, unspecified natural origin		m3	2.40E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	sodium silicate, hydrothermal liquor, 48% in H2O, at plant	RER	kg	1.04E+00	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
Input from Technosphere	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.87E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.03E-02	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, natural gas, at industrial furnace >100kW	RER	MJ	6.26E+00	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	5.22E-02	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	steam, for chemical processes, at plant	RER	kg	7.09E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	chemical plant, organics	RER	unit	4.00E-10	1	3.77	(4,5,1,3,5,4); estimation
	transport, freight, rail	RER	tkm	6.22E-01	1	2.09	(4,5,na,na,na,na); standard distances
	transport, lorry 32t	RER	tkm	1.04E-01	1	2.09	(4,5,na,na,na,na); standard distances
	disposal, inert waste, 5% water, to inert material landfill	CH	kg	2.71E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	sodium silicate, spray powder 80%, at plant	RER	kg	1			
Output	Heat, waste		MJ	2.73E+00	1	1.13	(2,2,2,1,1,4); calculated from electricity and steam input
Air emission	Silicon		kg	1.76E-02	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers

Tab. 5.7 Input / Output and uncertainty values for process “sodium metasilicate, pentahydrate, 58%, powder, at plant (RER)”

Explanation	Name	Location	Unit	sodium metasilicate pentahydrate, 58%, powder, at plant	uncertaintyType	standardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resource Input from Technosphere	Water, cooling, unspecified natural origin		m3	5.17E-05	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	Water, unspecified natural origin		m3	1.02E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	sodium silicate, hydrothermal liquor, 48% in H ₂ O, at plant	RER	kg	7.50E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	RER	kg	3.31E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	chemicals organic, at plant	GLO	kg	8.62E-05	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.35E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	heat, natural gas, at industrial furnace >100kW	RER	MJ	1.03E+00	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	steam, for chemical processes, at plant	RER	kg	3.84E-01	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
	chemical plant, organics	RER	unit	4.00E-10	1	3.77	(4,5,1,3,5,4); estimation
	transport, freight, rail	RER	tkm	6.49E-01	1	2.09	(4,5,na,na,na,na); standard distances
Waste	transport, lorry 32t	RER	tkm	1.08E-01	1	2.09	(4,5,na,na,na,na); standard distances
	disposal, inert waste, 5% water, to inert material landfill	CH	kg	1.12E-03	1	1.13	(2,2,2,1,1,4); Average data from 6 European producers
Output Air emission	sodium metasilicate pentahydrate, 58%, powder, at plant	RER	kg	1			
	Heat, waste		MJ	1.60E+00	1	1.13	(2,2,2,1,1,4); calculated from electricity and steam input
	Particulates, < 2.5 um		kg	5.86E-03	1	30.2	(2,2,2,1,1,4); Average data from 6 European producers
	Suspended solids, unspecified		kg	4.48E-03	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers
	Silicon		kg	6.12E-04	1	1.52	(2,2,2,1,1,4); Average data from 6 European producers

5.2 Sodium Tripolyphosphate (STPP)

5.2.1 Introduction

Sodium tripolyphosphate (Na₅P₃O₁₀, CAS-No. 7758-29-4) is a solid, inorganic compound present in the form of slightly hygroscopic granules. Its stable form is a hexahydrated salt. The anhydrous salt exists in two crystalline forms (HERA (2003)).

According to HERA (2003), STPP is used in a large variety of household cleaning products, mainly as builder – but also in human foodstuff, animal feed, industrial cleaning processes and ceramics manufacture. Within the cleaning products STPP can be one of the ingredients of regular and compact laundry detergents, of automatic dishwashing detergents, of toilet cleaners or of surface cleaners. As builder in these detergents, STPP provides various functions – among them e.g. the sequestration of the water hardness, enabling surfactants to function effectively. Further functions of STPP are pH buffering, dirt emulsification or hydrolysis of grease. For the year 2000 – according to HERA (2003) – about 300 kt of STPP have been used in detergents, representing 90-95% of the total use of STPP in Europe.

5.2.2 Production and use

The principle raw material for the production of STPP is phosphate rock. In a first step, the raw material is calcinated at temperatures of approx. 750 °C. It is subsequently reacted exothermically with sulphuric acid in the dihydrate process. The phosphoric acid produced is filtered and washed. It has a concentration of 29% and is concentrated to 55% by vaporisation. In the subsequent purification step, the phosphoric acid is dissolved in an organic solvent. The residues that occur are treated with a mixture of solvents and sulphuric acid. The produced waste is treated with calcium carbonate producing a stable material (raffinate solidification process) that can be deposited. The phosphoric acid solution itself is purified in a countercurrent with small quantities of the final product and separated again from the solvent which is reused. The purified phosphoric acid is neutralised with sodium carbonate, and the so called ortholiquor of dissolved sodium phosphate is produced. Finally, it is dried to obtain sodium tripolyphosphate.

5.2.3 Systems characterization

Fig. 5.2 shows the processes for the production of sodium tripolyphosphate included into this study. Starting point are soda and phosphoric acid, industrial grade – both described in Althaus et al. (2007). In a first step, the phosphoric acid is neutralized with soda, resulting in the production of sodium phosphate ortho-liquor. In the following spray-drying process this is transformed into sodium tripolyphosphate. In both steps, the material and energy input (incl. cooling and process water), the infrastructure and transport of the raw materials to the production site as well as the emissions to air and water and the produced waste are included.

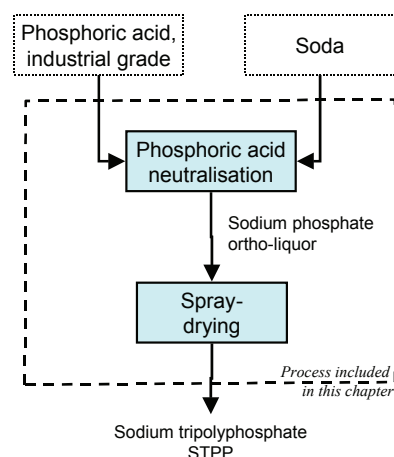


Fig. 5.2 Flow diagram of the production of STPP

5.2.4 Production of Sodium phosphate ortho-liquor

Main data source for this process step is the phosphate report (Landbank (1994)) – a life cycle study to evaluate the environmental impact of phosphates and zeolithe A-PCA as alternative builders in UK laundry detergent formulations, already used as main data source for the former detergent study from EMPA (Dall'Acqua et al. (1999)). The data represent the situation in UK in the mid 90s. These data are shown in Tab. 5.8.

Tab. 5.8 Input and output values for the phosphoric acid neutralization (data from Tab. 2.8 in Landbank (1994))

Input		ortho-liquor	Output		ortho-liquor
<i>materials</i>			<i>product</i>		
soda	g	412.00	Na-phosphate, ortho-liquor	g	1000.00
phosphoric acid, purified	g	553.00	<i>emissions to air</i>		
<i>energy</i>			fossil carbon dioxide, CO2	g	540.00
natural gas	m3	3.77E-03	<i>emissions to water</i>		
hard coal	g	2.50	waste water	m3	2.80E-04
heavy fuel oil	g	3.58			
<i>water</i>					
water (process)	m3	5.20E-04			

For an integration of these data into the database ecoinvent, the following assumptions and transformations have been made:

- **Material input:** According to Landbank (1994) the purified phosphoric acid has a concentration of 56%. As in this study here, all chemicals use a functional unit referring to 100% of their respective acting substance, here phosphoric acid, all used values have been multiplied with a factor of 1.79.
- **Energy input:** The input of the different fuels is taken directly from the information in Tab. 5.8. For all of them, the respective “heating” module is used, allowing having a similar degree of details concerning the emissions to air. No electricity input is reported in Landbank (1994) and therefore also no electricity input is assumed in this study.
- **Transport and Infrastructure:** As this kind of information in Landbank (1994) is very case specific for the UK situation, in this study, standard distances and means of transport according to Frischknecht et al. (2007b) are used. Therefore it has to be taken into account that the phosphoric acid, industrial grade has a concentration of 85%. For the infrastructure, due to a lack of data, the general module “chemical plant, organics (RER)” is used here.
- **Emissions to air and water:** The reported amount of carbon dioxide is part of the process and no combustion product. Thus, similar to the material inputs, its values are multiplied with a factor of 1.79. No further emissions (to air or to water) are reported and therefore no further emissions are inventorized.
- **Waste:** No waste is reported.

5.2.5 Production of Sodium tripolyphosphate

Sodium phosphate, ortho-liquor is the starting point for the production of sodium tripolyphosphate, used within the detergent industry. Thereof, the inventory of „sodium tripolyphosphate, at plant“ is based on the same study (Landbank (1994)) and has been established accordingly to the description in chapter 5.2.4 – using the same assumptions and transformation procedures.

5.2.6 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The data for the two examined forms of sodium phosphates represent industrial data provided to the authors of the used study (Landbank (1994)), resulting in a quite low uncertainty. Transports and the infrastructure have higher uncertainties due to the fact that they are own assumptions from the author of this study. The following tables summarize the input data as well as the uncertainties for sodium phosphate (ortho-liquor) and sodium tripolyphosphate.

Tab. 5.9 Input / Output and uncertainty data for the process "sodium phosphate, at plant (RER)"

Explanation	Name	Location	Unit	sodium phosphate, at plant	UncertaintyType	StandardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resource Input from Technosphere	Water, unspecified natural origin		m3	9.31E-04	1	1.33	(2,5,3,3,1,5); Data from UK study (Landbank report)
	phosphoric acid, industrial grade, 85% in H2O, at plant	RER	kg	9.90E-01	1	1.33	(2,5,3,3,1,5); Data from UK study (Landbank report)
	soda, powder, at plant	RER	kg	7.37E-01	1	1.33	(2,5,3,3,1,5); Data from UK study (Landbank report)
	heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.64E-01	1	1.33	(2,5,3,3,1,5); Data from UK study (Landbank report)
	heat, natural gas, at industrial furnace >100kW	RER	MJ	2.45E-01	1	1.33	(2,5,3,3,1,5); Data from UK study (Landbank report)
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.29E-01	1	1.33	(2,5,3,3,1,5); Data from UK study (Landbank report)
	chemical plant, organics	RER	unit	4.00E-10	1	3.77	(4,5,1,3,5,4); estimation
	transport, freight, rail	RER	tkm	1.14E+00	1	2.09	(4,5,na,na,na,na); standard distances
	transport, lorry 32t	RER	tkm	1.90E-01	1	2.09	(4,5,na,na,na,na); standard distances
	sodium phosphate, at plant	RER	kg	1			
Output Air emission	Carbon dioxide, fossil		kg	9.67E-01	1	1.33	(2,5,3,3,1,5); Data from UK study (Landbank report)

Tab. 5.10 Input / Output and uncertainty data for the process "sodium tripolyphosphate, at plant (RER)"

Explanation	Name	Location	Unit	sodium tripolyphosphate, at plant	UncertaintyType	StandardDeviation95%	General Comment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Resources Input from Technosphere	Water, unspecified natural origin		m3	3.60E-04	1	1.25	(2,3,3,2,1,4);
	sodium phosphate, at plant	RER	kg	1.84E+00	1	1.25	(2,3,3,2,1,4);
	heat, unspecific, in chemical plant	RER	MJ	4.88E+00	1	1.25	(2,3,3,2,1,4);
	electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.10E-01	1	1.25	(2,3,3,2,1,4);
	chemical plant, organics	RER	unit	4.00E-10	1	3.06	(2,3,3,2,1,4);
Output air emission	sodium tripolyphosphate, at plant	RER	kg	1			
	Particulates, > 10 um		kg	4.00E-05	1	1.25	(2,3,3,2,1,4);

5.3 Polycarboxylates

5.3.1 Introduction

Chemical structure: $[(-CH_2-CH)_x-COOH HOOC-(-CH-CH)_y-COOH]_n$

Polycarboxylates (PCA) have established themselves in connection with builders as water-soluble, linear polymers with numerous carboxy groups. In current detergents only homopolymers of acrylic acid and copolymers of acrylic and maleic acid are used. This study considers only these copolymers. They are used in detergents as dispersion agents.

5.3.2 Production and Use

Production processes

PCA are produced industrially by free radical polymerization of the monomeric acrylic acid alone or in a mixture with maleic anhydride to produce P(AA) and P(AA-MA). The overall patent claim describes polymers containing 40 to 90% acrylic acid and 10 to 60% maleic anhydride (Landbank (1994)). The manufacturing takes place in a batch reactor at approx. 100°C. The plants are usually multifunctional polymerisation plants, in which a multiplicity of products can be made (Dall'Acqua et al. (1999)). Mean molecular mass of the resulting PCA used for detergents is generally between 1000 and 100,000 amu.

Amount of Production

The amounts have increased over the last decades. In Germany, e.g., the amount was 15,000 tonnes in 1987 and 19,000 tonnes in 1990 when the marketshare for phosphate-free detergents was more than 95%. However, the amount of PCA used in the European detergent market has already peaked, due to new detergent formulations using less PCA (Folke (1996)).

Use

PCA have been used since the early 1980s in low-phosphate and phosphate-free detergents. Their effect is based on the dispersion of calcium carbonate or of calcium phosphate and the soil detached during washing. Between 2 and 5% of PCA were used in phosphate-free detergents based on zeolite/sodium carbonate, but recently, new formulations and development of new zeolites have reduced the amounts added (Folke (1996)). For example, Lever now has a formulation based on a new Zeolite P, which does not contain any PCA at all (Crosfield (1996)).

5.3.3 Systems characterization

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

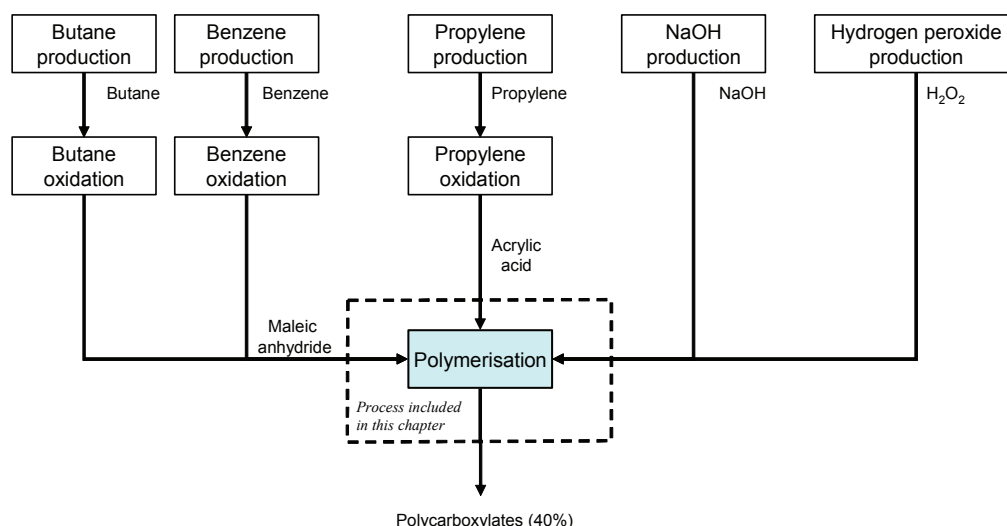


Fig. 5.3 Flow diagram of the production of polycarboxylates.

5.3.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the data used for the EMPA-study on detergents (Dall'Acqua et al. (1999)) and represent typical industrial data. Although from the data sources used it is not possible to see which part of the market is covered by the data, the uncertainty is assumed to be low. Differences are thus due to the different basic uncertainties used within this study.

Due to the fact that the data source used in Dall'Acqua et al. (1999) is not publicly available, the data are confidential and thus no unit process data of this process can be shown in this report here. Never-

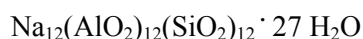
theless, the most important fields of the ecospol meta information from this dataset are listed in chapter 5.5.

5.4 Zeolite A

5.4.1 Introduction

The following paragraph is based on the former EMPA study about zeolites (Fawer (1996)). Zeolites are a well defined class of crystalline aluminosilicate minerals. In nature, they can be found in rocks of organic origin. They have been described for the first time in the 18th century by a Swedish mineralogist, observing that stones containing zeolites appeared to boil when rapidly heated. The name “zeolite” is therefore a combination of the greek words for “to boil” (zeo) and “stone” (lithos).

Nowadays about 40 different naturally occurring and another about 150 synthesised types of zeolites are known. Thereof, one specific type – called zeolite A – was developed as a detergent builder about 30 years ago and is in the meantime well established within the detergent industry (according to Fawer (1996), the production capacity in 1993 was in the order of 1.2 Mio t, whereof about 540 kt in Western Europe. This specific form of zeolite is insoluble in water and has the overall formula:



In the following only this specific type of zeolite is taken into account – all other types of zeolite are not further investigated within this study here.

5.4.2 Production and use

Production processes

Two commercially used production routes exist: the hydrogel process and the crystallisation of calcined clay. Worldwide, the **hydrogel process** has the greatest significance and is therefore described here. Solutions of aluminium hydroxide and sodium silicate are mixed. The amorphous sodium aluminium silicate produced is crystallised hydrothermally and filtered to form zeolite A. A portion (38%) is used directly as slurry. Slurry is a water suspension of zeolite A (approx. 50% water content) which is stabilised with surfactant substances (tensides) to prevent sedimentation. These surfactants are the same as those used normally in the final detergent. The other portion of zeolite A filter cake (62%) is dried (to approx. 20% water content) and used as a powder.

Use

Zeolite A was developed as a builder for washing powders. Furthermore, it is used in many other fields, e.g. as an additive in animal feeds, cleaning fluid, drying substances or as a catalyst.

5.4.3 Systems characterization

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

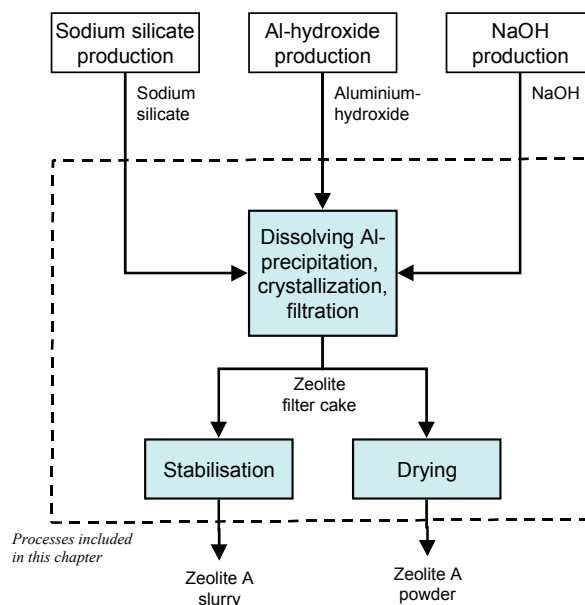


Fig. 5.4 Flow diagram of the production of zeolite A.

5.4.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the data used for the EMPA-study on detergents (Dall'Acqua et al. (1999)) and are from a former EMPA study (Fawer (1996)). That former study from EMPA is based on data from 5 different European producers. Although from the data sources used it is not possible to see which part of the market is covered by these data, the uncertainty can be assumed to be low. Differences are thus due to the different basic uncertainties used within this study.

Due to the fact that both data sources mentioned (Dall'Acqua et al. (1999) and Fawer (1996)) contain only aggregated data for these two forms of zeolithe, it can be deducted that the respective unit process data are confidential and thus no unit process data of those processes can be shown here. Nevertheless, the most important fields of the ecospold meta information from those datasets are listed in chapter 5.5.

5.5 EcoSpold Meta Information

ReferenceFunction	Name	sodium silicate, hydrothermal liquor, 48% in H2O, at plant	sodium silicate, furnace process, pieces, at plant	sodium silicate, furnace liquor, 37% in H2O, at plant	sodium silicate, spray powder 80%, at plant
Geography	Location	RER	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0	0
ReferenceFunction	Unit	kg	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input, production of waste and emissions for the production of sodium phosphate out of phosphoric acid. Transport and infrastructure have been estimated. The input of 0.015 m3 air is not reported in the data according to the methodology of the study.	This module contains material and energy input, production of waste and emissions for the production of sodium phosphate out of phosphoric acid. Transport and infrastructure have been estimated. The input of 0.01 m3 air is not reported in the data according to the methodology of the study.	This module contains material and energy input, production of waste and emissions for the production of sodium silicate, 37% furnace liquid. Transport and infrastructure have been estimated. The input of compressed air is not reported in the data according to the methodology of the study.	This module contains material and energy input, production of waste and emissions for the production of sodium silicate spray powder. Transport and infrastructure have been estimated. The input of compressed air is not reported in the data according to the methodology of the study.
ReferenceFunction	Amount	1	1	1	1
ReferenceFunction	LocalName	Natronwasserglas, hydrothermal, 48% in H2O, ab Werk	Natronwasserglas, Schmelzprozess, Stücke, ab Werk	Natronwasserglas, Schmelzprozess, 37% in H2O, ab Werk	Natriumsilikat, sprühetrocknetes Pulver 80%, ab Werk
ReferenceFunction	Synonyms				
ReferenceFunction	GeneralComment	data based on a EMPA study, commissioned by the CEES.	data based on a EMPA study, commissioned by the CEES.	data based on a EMPA study, commissioned by the CEES.	data based on a EMPA study, commissioned by the CEES.
ReferenceFunction	CASNumber	1344-09-8	1344-09-8	1344-09-8	1344-09-8
TimePeriod	StartDate	1995	1995	1995	1995
TimePeriod	EndDate	1995	1995	1995	1995
TimePeriod	DataValidForEntirePeriod	1	1	1	1
TimePeriod	OtherPeriodText	producer information for production year 1995	producer information for production year 1995	producer information for production year 1995	producer information for production year 1995
Geography	Text	average data from 6 European producers	average data from 6 European producers	average data from 6 European producers	average data from 6 European producers
Technology	Text	average technology used from these European producers	average technology used from these European producers	average technology used from these European producers	average technology used from these European producers
Representativeness	Percent				
Representativeness	ProductionVolume	260'000 t	91'000 t	unknown	unknown
Representativeness	SamplingProcedure			sampling of questionnaires	sampling of questionnaires
Representativeness	Extrapolations			see included processes	see included processes
Representativeness	UncertaintyAdjustments			none	none

5. Builders

ReferenceFunction	Name	sodium metasilicate pentahydrate, 58%, powder, at plant	layered sodium silicate, SKS-6, powder, at plant	sodium phosphate, at plant	sodium tripolyphosphate, at plant
Geography	Location	RER	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0	0
ReferenceFunction	Unit	kg	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input, production of waste and emissions for the production of sodium metasilicate pentahydrate powder. Transport and infrastructure have been estimated. The input of compressed air is not reported in the data according to the methodology of the study.	This module contains material and energy input, production of waste and emissions for the production of SKS-6 sodium silicate powder. Transport and infrastructure have been estimated. The input of compressed air is not reported in the data according to the methodology of the study.	This module contains material and energy input, production of waste and emissions for the production of sodium phosphate out of phosphoric acid. Transport and infrastructure have been estimated. No water emissions are accounted for.	This module contains material and energy input for the production of sodium tripolyphosphate out of phosphoric acid. Infrastructure has been estimated. No process emissions are accounted for.
ReferenceFunction	Amount	1	1	1	1
ReferenceFunction	LocalName	Natriummetasilikat-5-hydrat, 58%, Pulver, ab Werk	Natriumschichtsilikat, SKS-6, Pulver, ab Werk	Natriumphosphat, ab Werk	Natriumtripolyphosphat, sprühgetrocknetes Pulver, ab Werk
ReferenceFunction	Synonyms				
ReferenceFunction	GeneralComment	data based on a EMPA study, commissioned by the CEES.	data based on information from the major European producer	The functional unit represents a 56% solution of sodium phosphate, but all data refer to 100% of the acting substance. The data are based on Landbank report about phosphate in detergents (established for UK conditions)	The functional unit represents spray-dried powder of sodiumtripolyphosphate (STPP). The data are based on one single company and on literature (Landbank report about phosphate in detergents established for UK conditions).
ReferenceFunction	CASNumber	10213-79-3		7601-54-9	7758-29-4
TimePeriod	StartDate	1995	1997	1994	1986
TimePeriod	EndDate	1995	1997	1994	1994
TimePeriod	DataValidForEntirePeriod	1	1	1	1
TimePeriod	OtherPeriodText	producer information for production year 1995	producer information for production year 1997	date of published literature	
Geography	Text	data representing European conditions	data represent production of major producer	average data for UK consumption - here used as average values for the whole European area	data for UK conditions - here used as average values for the whole European area
Technology	Text	average technology used from these European producers	technology from major European producer	average technology for the examined area / time	dihydrate process
Representativeness	Percent				
Representativeness	ProductionVolume	unknown	unknown	unknown	unknown
Representativeness	SamplingProcedure	sampling of questionnaires	questionnaire		literature
Representativeness	Extrapolations	see included processes	none		see geography and IncludedProcesses
Representativeness	UncertaintyAdjustments	none	none		none

5. Builders

ReferenceFunction	Name	polycarboxylates, 40% active substance, at plant	zeolite, powder, at plant	zeolite, slurry, 50% in H2O, at plant
Geography	Location	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0
ReferenceFunction	Unit	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input for the production of polycarboxylates out of acrylic acid and maleic anhydride. Transport and infrastructure have been estimated. No process emissions are accounted for.	This module contains material and energy input, water consumption as well as emissions and waste amounts from the production of zeolithe out of aluminium hydrate, sodium silicate and sodium hydroxide. Transport and infrastructure have been estimated.	This module contains material and energy input, water consumption as well as emissions and waste amounts from the production of zeolithe out of aluminium hydrate, sodium silicate and sodium hydroxide. Transport and infrastructure have been estimated.
ReferenceFunction	Amount	1	1	1
ReferenceFunction	LocalName	Polycarboxylate, 40% aktive Substanz, ab Werk	Zeolith A, Pulver, ab Werk	Zeolith A, Aufschlämmung, 50% in H2O, ab Werk
ReferenceFunction	Synonyms			
ReferenceFunction	GeneralComment	The data are based on site specific data, used for a EMPA study.	The data represents average data from 5 European producers, established in a former EMPA study.	The data represents average data from 5 European producers, established in a former EMPA study.
ReferenceFunction	CASNumber		1318-02-1	1318-02-1
TimePeriod	StartDate	1993	1990	1990
TimePeriod	EndDate	1998	1995	1995
TimePeriod	DataValidForEntirePeriod	1	1	1
TimePeriod	OtherPeriodText	data of used literature		
Geography	Text	data represent European dataset	Average data for European production	Average data for European production
Technology	Text	Co-polymerization process of maleic anhydride and acrylic acid	hydrogel process with typical parameters representing production technolog of mid 90s	hydrogel process with typical parameters representing production technolog of mid 90s
Representativeness	Percent		76	76
Representativeness	ProductionVolume	unknown	unknown	unknown
Representativeness	SamplingProcedure	literature (data from one single producer)	company data, average	company data, average
Representativeness	Extrapolations	see IncludedProcesses	see IncludedProcesses	see IncludedProcesses
Representativeness	UncertaintyAdjustments	none	none	none

6 Bleachers

6.1 General

In chemical bleaching, stains that have an affinity to fibres and cannot be washed out are decoloured oxidatively through degradation of the chromophore system. The bleaching effect achieved is determined by the type of whitening agent, its concentration and the time it spends in the washing process, the washing temperature and the stains which are to be bleached. Bleachable stains are usually of vegetable origin.

Two processes of oxidative whitening have been successful: peroxide and hypochlorite bleaching. In peroxide bleaching, which is most common in Europe, perhydroxyl anions arise as an active transition substance in the alkaline medium of hydrogen peroxide. Hydrogen peroxide is delivered by inorganic peroxides and peroxyhydrates, the most important representative is sodium perborate. It hydrolyses in water with the formation of hydrogen peroxide, but persists in detergents. Sodium percarbonate must be additionally stabilised in order to be stored.

6.2 Sodium Perborates

6.2.1 Introduction

Chemical structure: $\text{NaBO}_3 \cdot 4\text{H}_2\text{O}$ (tetrahydrate), $\text{NaBO}_3 \cdot \text{H}_2\text{O}$ (monohydrate)

In the washing process, perborate serves as provider of hydrogen peroxide, which achieves the bleaching effect.

6.2.2 Production and use

Production processes

The main sources of the boron present in perborates are the minerals kernite and tincal. These boron-containing minerals are first dissolved in sodium hydroxide. The concentrated solution is filtered and thinned. The solution is then reacted with hydrogen peroxide, which causes the tetrahydrate to crystallise out. Monohydrate is then produced by drying in warm air.

Use

Perborates are used in detergents as oxidisers and bleachers. Perborates are also used in cleaning products, cosmetics, and pharmaceuticals.

6.2.3 Systems characterization

Two products have been considered: sodium perborate tetrahydrate and sodium perborate monohydrate.

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

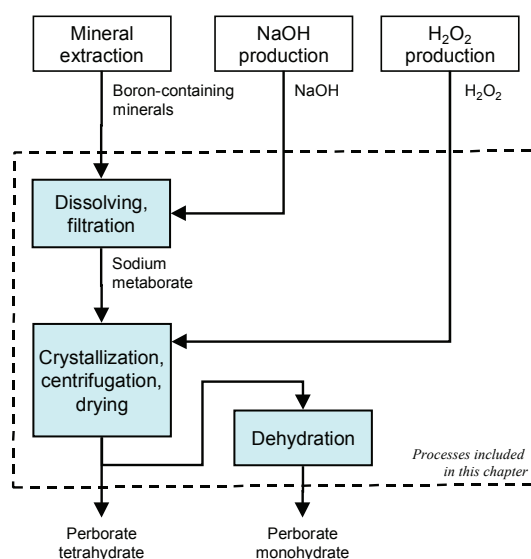


Fig. 6.1 Flow diagram of the production of sodium perborates.

6.2.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the data used for the EMPA-study on detergents Dall'Acqua et al. (1999) and are from a former study of EMPA and Boustead-Consulting for the “sector group peroxygen” of the European Chemical Industry Council (CEFIC) (Boustead & Fawer (1998)). That former study is based on data from 9 different European producers and are accordingly representative and of high quality. Thus, the uncertainty can be assumed to be low. Differences are then due to the different basic uncertainties used within this study.

Due to the fact that the data source used (Boustead & Fawer (1998)) shows only aggregated data from these two forms of perborate examined, it can be deduced that the respective unit process data are confidential and thus no unit process data of those processes can be shown here. Nevertheless, the most important fields of the ecospold meta information from those datasets are listed in chapter 6.4.

6.3 Sodium Percarbonate

6.3.1 Introduction

Chemical structure: $2\text{Na}_2\text{CO}_3 \cdot 3\text{H}_2\text{O}_2$

Sodium percarbonate is a term used unspecifically for sodium carbonate peroxyhydrate. Sodium percarbonate forms a white, water-soluble powder which readily decomposes to sodium carbonate and bleaching oxygen. It is therefore used in detergents as a whitening agent.

6.3.2 Production and use

Production processes

Sodium percarbonate is preferably produced from an aqueous solution (crystallisation process). Crystalline sodium percarbonate is obtained by the conversion of aqueous solutions of sodium carbonate and hydrogen peroxide at 10-20 °C in the presence of sodium chloride and possibly aluminium ions.

The centrifuged crystallised salt contains approx. 10% mother liquor and is dried in a fluidised-bed dryer at an air temperature of 90 °C. The centrifuged mother liquor is then reused. Stabilisation during the manufacturing procedure is not adequate for the required storage stability. Ideally, the percarbonate receives additional stabilisation, for example, by coating with sodium silicate.

6.3.3 Systems characterization

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

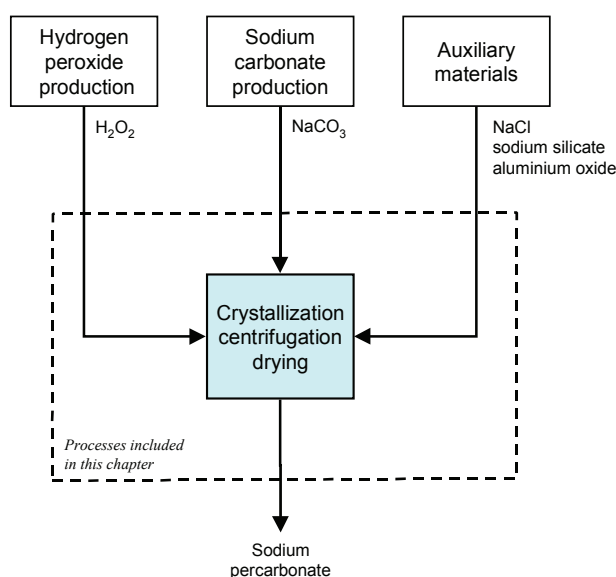


Fig. 6.2 Flow diagram of the production of sodium percarbonate.

6.3.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the data used in the EMPA-study on detergents (Dall'Acqua et al. (1999)) and are based on data from 4 different European producers. Although from the data sources used it is not possible to see which part of the market is covered by these data, the uncertainty can be assumed to be low. Differences in the uncertainty are then due to the different basic uncertainties used within this study.

Due to the fact that the mentioned former detergent study from EMPA (Dall'Acqua et al. (1999)) shows only aggregated data from percarbonate, it can be deduced that the respective unit process data are confidential and thus no unit process data of this process can be shown here. Nevertheless, the most important fields of the ecospold meta information from this dataset are listed in chapter 6.4.

6.4 EcoSpold Meta Information

ReferenceFunction	Name	sodium perborate, tetrahydrate, powder, at plant	sodium perborate, monohydrate, powder, at plant	sodium percarbonate, powder, at plant
Geography	Location	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0
ReferenceFunction	Unit	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input for the production of perborates out of borax, NaOH and hydrogen peroxide. Transport and infrastructure have been estimated.	This module contains material and energy input for the production of perborates out of borax, NaOH and hydrogen peroxide. Transport and infrastructure have been estimated.	This module contains material and energy input, production of waste and emissions for the production of sodium percarbonate. Transport and infrastructure have been estimated.
ReferenceFunction	Amount	1	1	1
ReferenceFunction	LocalName	Natriumperborat, Tetrahydrat, Pulver, ab Werk	Natriumperborat, Monohydrat, Pulver, ab Werk	Natriumpercarbonat, Pulver, ab Werk
ReferenceFunction	Synonyms			
ReferenceFunction	GeneralComment	data based on a study, performed by EMPA and Boustead Consulting	data based on a study, performed by EMPA and Boustead Consulting	data based on company information for a former detergent study of EMPA
ReferenceFunction	CASNumber	7632-04-4	7632-04-4	15630-89-4
TimePeriod	StartDate	1990	1990	1994
TimePeriod	EndDate	1995	1995	1994
TimePeriod	DataValidForEntirePeriod	1	1	1
TimePeriod	OtherPeriodText			year of reported data
Geography	Text	average data from 9 European producers	average data from 9 European producers	average data from 4 European producers, used as European average
Technology	Text	average technology used from these European producers	average technology used from these European producers	Crystallization out of a aqueous solution of soda and hydrogen peroxide, representing average technology for the mid 90s
Representativeness	Percent	85	85	
Representativeness	ProductionVolume	unknown	unknown	
Representativeness	SamplingProcedure	company data used for average	company data used for average	company data
Representativeness	Extrapolations	none	none	
Representativeness	UncertaintyAdjustments	none	none	

7 Auxiliaries

7.1 General

Tensides, builders and whitening agents are the main components of modern detergents in terms of quantity. Further additives are used only in small quantities. However, they should not be overlooked in today's detergent formulations. Here, two types of such auxiliaries are further investigated: Carboxymethyl cellulose (CMC) and fluorescent whitening agents (FWA).

7.2 Carboxymethyl Cellulose (CMC)

7.2.1 Introduction

Chemical structure: Cellulose-O-CH₂-COO⁻Na⁺

CMC is used in detergents as a soil carrier.

7.2.2 Production and use

Production processes

The starting product for its manufacture is cellulose sulphate or sulphite from pine wood. In a first step, the cellulose is activated using sodium salt. This may take place in the presence or absence of a solvent (e.g. isopropanol). Because of the production method, CMC contains sodium chloride and sodium glycolate. These salts are stepwise or continuously removed in a counter-current system. Residual alkali is neutralised with acid. Finally, the product is dried and ground. The final product is a granulate or a powder (Häussinger et al. (2000)).

Use

Main use of CMC is as soil carrier in detergents. In other fields it is employed, for example, as a thickener in salves in the food industry as an emulgator and stabiliser, in the paint industry as a stabiliser or thickener.

7.2.3 Systems characterization

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been applied.

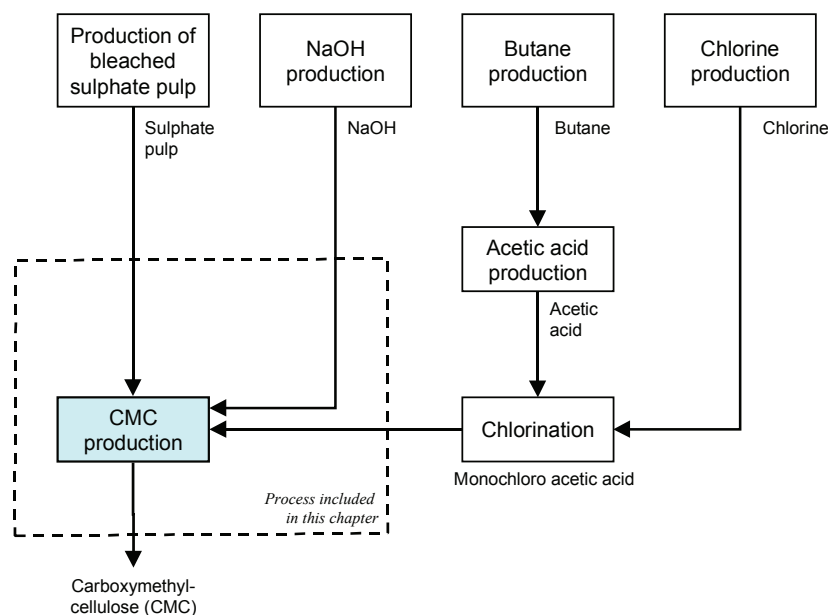


Fig. 7.1 Flow diagram of the production of carboxymethyl cellulose.

7.2.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the data used in the EMPA-study on detergents (Dall'Acqua et al. (1999)) and are based on data from one European producer. Although from the data sources used it is not possible to see which part of the market is covered by these data, the uncertainty is assumed to be low. Differences in the uncertainty are then due to the different basic uncertainties used within this study.

Due to the fact that only one company is behind these data and also due to the fact that Dall'Acqua et al. (1999) shows only aggregated data for CMC, it can be deduced that the respective unit process data are confidential and thus no unit process data of this process can be shown here. Nevertheless, the most important fields of the ecospold meta information from this dataset are listed in chapter 7.4.

7.3 Fluorescent Whitening Agents (FWA)

7.3.1 Introduction

Fluorescent whitening agents (FWA) are colorless organic substances that are able to absorb ultraviolet radiance and to re-emit the absorbed energy as blue fluorescent light (400 - 500 nm). FWAs are used as additives in detergents. By coating the textile fibres during the washing process, they compensate the natural (but continuously increasing) yellow touch of the textiles resulting in a high grade of whiteness (Häussinger et al. (2000)).

In this study, two important classes of FWA are examined. The widely used Triazinylaminostilbene (DAS-1) and the relatively new Distyrylbiphenyle (DSPB). The latter exhibit a more complex production path, but due to their higher efficiency, they can be applied in smaller quantities and they exhibit a nearly complete photolytical induced biodegradability in surface waters.

7.3.2 Fluorescent Whitening Agent Triazinylaminostilbene Type (DAS-1)

Production processes

The starting material for the triazinylaminostilbenes is 4,4'-Dinitrostilbene-2,2'-disulphonic acid, which is produced by oxidation of 4-nitrotoluene-2-sulphonic acid with sodium hypochlorite. A subsequent Béchamps reduction with iron fillings and HCl gives 4,4'-diaminostilbene-2,2'-disulphonic acid, that is reacted at both amino groups with cyanuric chloride. The two remaining Cl atoms in the cyanuric chloride rings react one after another with two amines, in the case of DAS-1 with aniline and morpholine. Subsequent steps produce the desired β -crystal modification and deliver the finished commercial product, that contains 67% active substance.

Systems characterization

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been multiplied by 3, due to the many aggregated processes.

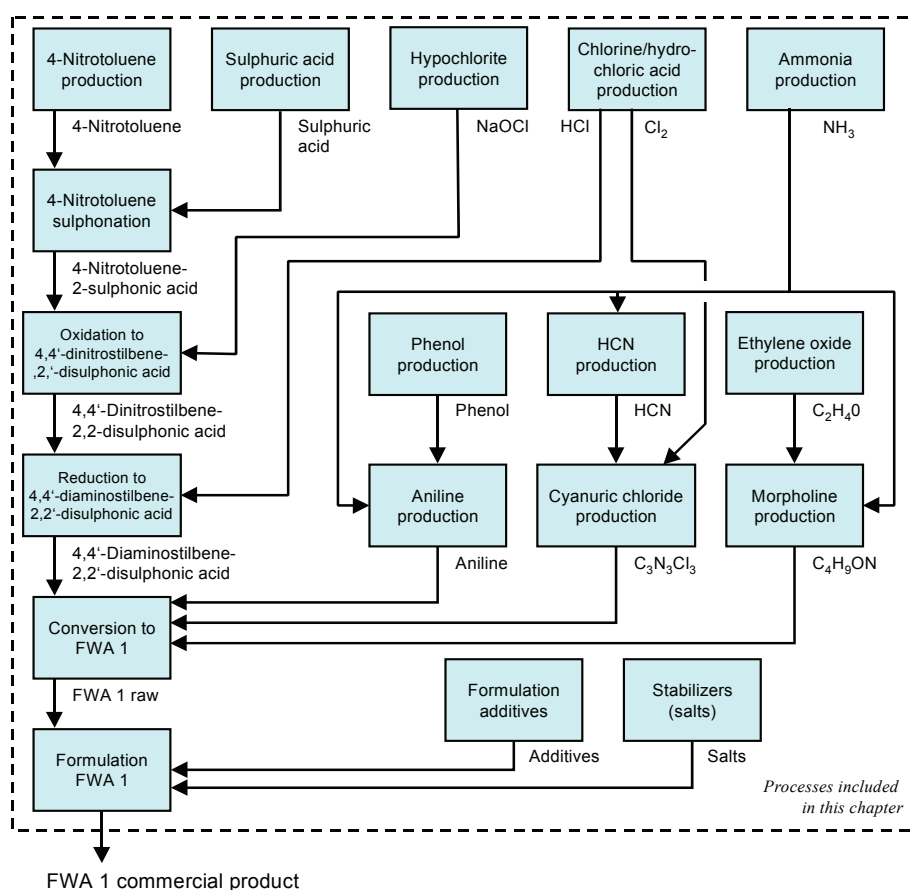


Fig. 7.2 Flow diagram of the production of the fluorescent white agent DAS-1 (triazinylaminostilbene type).

7.3.3 Fluorescent Whitening Agent Distyrylbiphenyl Type (DSBP)

Production processes

The production of distyrylbiphenyl FWAs starts with biphenyl, that is produced along with other aromatic compounds during the refining of crude oil. Reacting it with chlorine and formaldehyde pro-

duces the intermediate 4,4'-bis(chloromethyl)biphenyl, which reacts with trimethylphosphite to give 4,4'-bis(dimethoxyphosphonomethyl)biphenyl. This symmetrical biphenylphosphonate is reacted with two molecules of an aromatic aldehyde. In the case of the DSBP under consideration, benzaldehyde-2-sulphonic acid is used, a compound which is produced from 2-chlor-benzaldehyde and sodium sulphite. The two sulphonic acid groups thus introduced give the final product DSBP the desired affinity to adsorb onto cellulose fibres. The commercial product contains 83% active substance.

Systems characterization

Natural gas, oil, and hard coal needed for process heating were all transformed to process heat. For the transport processes and for the infrastructure of a chemical plant, standard values for the production of organic chemicals have been multiplied by 3, due to the many aggregated processes.

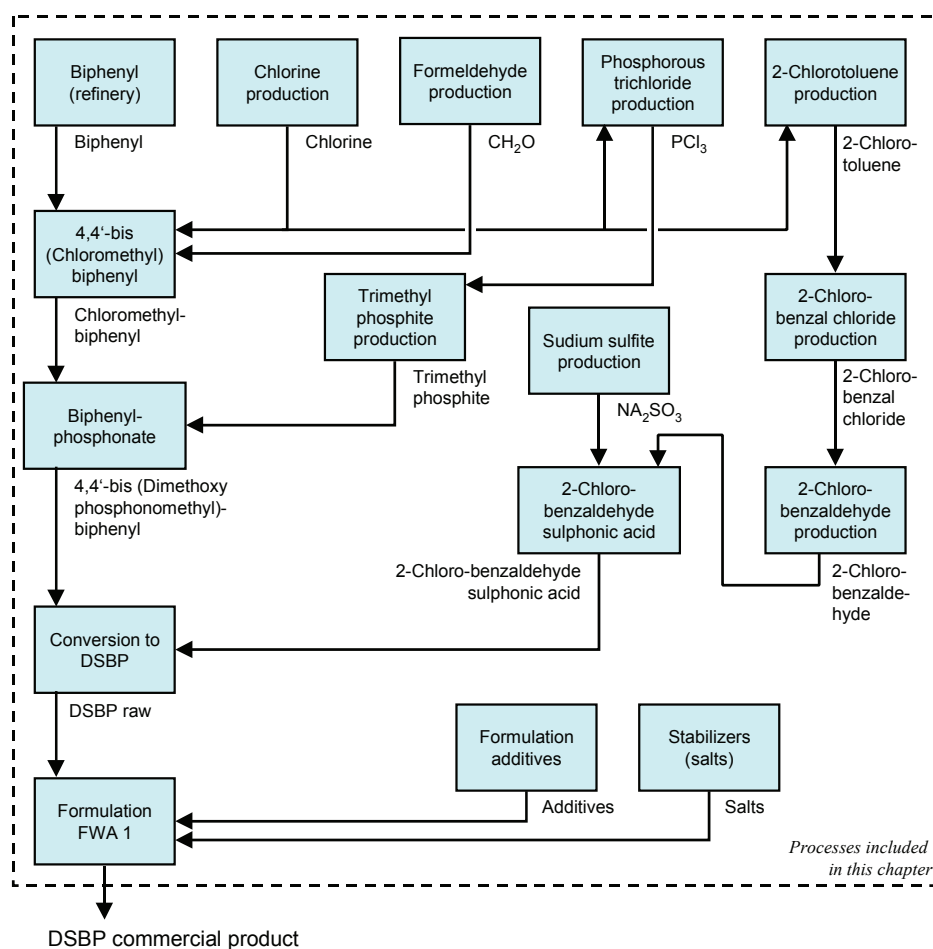


Fig. 7.3 Flow diagram of the production of the fluorescent white agent DSBP (distyrylbiphenyl type).

7.3.4 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. Input data are based on the data used in the EMPA-study on detergents (Dall'Acqua et al. (1999)) and are based on data from one European producer. Although these data are only from one producer, they are reasonably representative (there is only one producer of dis-

trylbipheyl FWAs in Europe and the United States, resp. there are just four important suppliers of triazinylaminostilbene FWAs) and the uncertainty can be assumed to be low. Differences in the uncertainty are then due to the different basic uncertainties used within this study. According to Dall'Acqua et al. (1999) it should be noted that besides the production way for triazinylaminostilbene FWAs shown in Fig. 7.3, other methods of synthesising exist. Those other methods are not covered by the here shown data.

Due to the fact that the data had been calculated by the company providing them for the mentioned study, we don't have any information about the unit processes of these two substances. Therefore the translation rules explained in chapter 3 have been used for the integration of these datasets into the ecoinvent database. Nevertheless, the most important fields of the ecospol meta information from this dataset are listed in chapter 7.4.

7.4 EcoSpold Meta Information

ReferenceFunction	Name	carboxymethyl cellulose, powder, at plant	DAS-1, fluorescent whitening agent triazinylaminostilben type, at plant	fluorescent whitening agent distyrylbiphenyl type, at plant
Geography	Location	RER	RER	RER
ReferenceFunction	InfrastructureProcess	0	0	0
ReferenceFunction	Unit	kg	kg	kg
ReferenceFunction	IncludedProcesses	This module contains material and energy input, production of waste and emissions for the production of carboxy methyl cellulose. Transport and infrastructure have been estimated.	This module contains material and energy input, production of waste and emissions for the production of the mentioned FWA (fluorescent whitening agent). Transport and infrastructure have been estimated.	This module contains material and energy input, production of waste and emissions for the production of the mentioned FWA (fluorescent whitening agent). Transport and infrastructure have been estimated.
ReferenceFunction	Amount	1	1	1
ReferenceFunction	LocalName	Carboxymethylcellulose , Pulver, ab Werk	DAS-1, optischer Aufheller Triazinylaminostilben-Typ, ab Werk	optischer Aufheller Distyrylbiphenyl-Typ, ab Werk
ReferenceFunction	Synonyms	CMC		DSBP
ReferenceFunction	GeneralComment	data based on company information for a former detergent study of EMPA	data based on company information for a former detergent study of EMPA	data based on company information for a former detergent study of EMPA
ReferenceFunction	CASNumber	9000-11-7	16090-02-1	27344-41-8
TimePeriod	StartDate	1993	1997	1999
TimePeriod	EndDate	1993	1997	1999
TimePeriod	DataValidForEntirePeriod	1	1	1
TimePeriod	OtherPeriodText	year of reported data	year of reported data	year of reported data
Geography	Text	data from 1 European producer, used as European average	data from European main producer, used as European average	data from European main producer, used as European average
Technology	Text	Processing of alkaline cellulose with mono acetic acid, used as average technology for the mid 90s	Typcial production process for end of 90s	Typcial production process for end of 90s
Representativeness	Percent		40	98
Representativeness	ProductionVolume	150000000 kg	150000000	250000000
Representativeness	SamplingProcedure	company data	aggregated data, calculated by the producing company itself	aggregated data, calculated by the producing company itself
Representativeness	Extrapolations	see technology and geography	see geography and technology	see geography and technology
Representativeness	UncertaintyAdjustments	none	none	none

8 Production of Steam

Steam made from water serves as process reactant or heat carrier in many chemical processes. This module comprises data on steam production from 11 European plants that produce sodium silicates or percarbonate. This steam module is therefor mostly valid for the production of fine chemicals like e.g. washing ingredients.

8.1 Production technologies for steam

Two principal types of boilers are found in steam producing facilities, the **fired tube** type and the **water tube** type AFG (1998).

8.1.1 Fired tube boiler

These boilers are cylinder-shaped, and closed at both ends by tubular plates enclosing the following items:

- a large diameter furnace tube passing longitudinally through the cylindrical body, and in which combustion occurs.
- flue gas tubes (or smoke tubes) in which the flue gas circulate.

The flue gas passes through two or three paths within the boiler (Fig. 8.1). The first path corresponds to the furnace tube, the second to a first series of tubes (the flue gas is then returned to the front section of the boiler) and, for the other powers, the third to a second series of tubes returning the flue gas to the back of the boiler, to be evacuated through the chimney. Their operation is typically automatic, and the boiler and accessories set is contained in a compact format. The large volume of water enables meeting with instant flow rate peaks. In the case of low NO_x boilers, the furnace has dimensions which are slightly larger than that of traditional furnaces. This enables reducing the thermal load losses and lowering the average temperature, hence supporting the reduction of the formation of thermal NO_x .

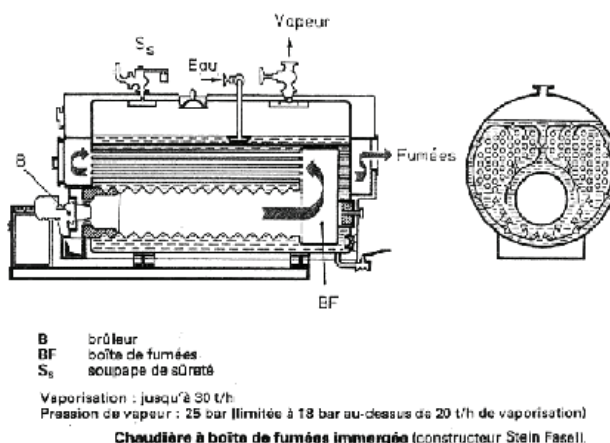


Fig. 8.1 Schematic view of a fire tube boiler

These boilers produce steam and high pressure hot water, and their power ranges from 3 to 20 MWt. They are currently most often equipped with low NO_x type burners, which use primary measures (natural gas staging technique or/and flue gas recirculation). For a new installation, the additional cost for the NO_x low burner can be considered as negligible when compared with a typical burner. The fired tube boilers equipped with the most effective low NO_x burners enable achieving a 92% PCI yield, and NO_x emission rates below 150 mg/Nm³ in normal operation.

8.1.2 Water tube boilers

The basic idea for the design of water tube boilers is that bringing a quantity of water in a single container to boiling point takes a long time; by replacing the latter with many tube shaped elements, the same result is obtained but faster; a water tube boiler comprises many tubes connected to 1 or 2 tanks, the upper tank containing the water-steam emulsion, and the lower tank containing water (Fig. 8.2). In large high pressure boilers, there is no tube bundle: the entire vaporizing process is performed in the chamber, and the lower tank disappears. The low water volume provides this type of boiler with considerable flexibility, but also strongly reduces its accumulating power.

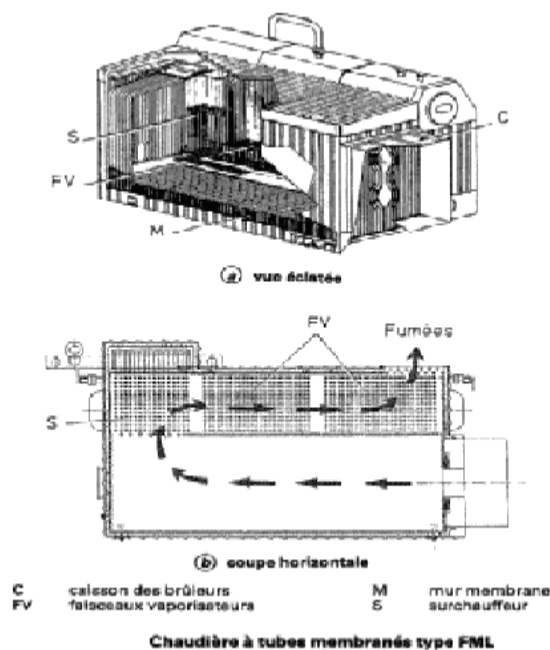


Fig. 8.2 Schematic view of a water tube boiler

Most often this type of boiler is built specifically according to the power and specifications required. The range of powers starts at 15 or 20 MWt, up to several hundreds of MWth, and even beyond. They usually require premises designed specifically for the boiler's size. They are currently most often equipped with low NO_x type burners, that use natural gas staging technique or/and internal flue gas recirculation.

8.1.3 Fuel requirements

Most of the time "natural gas" boilers are fitted to use also different fuel input than natural gas, mainly heavy fuel oil. The operation is often linked with economic conditions (cost of gas and fuel oil energies at a given time) and with the gas supply contract between the operator to the gas supplier. Most often, this mixed operation is performed with boilers fitted with a gas-fuel oil mixed burner.

8.1.4 Lifetime of installed plants

The average age of the boilers installed is estimated at 25 years for shell-type, and more than 35 years for water tubes. The average lifetime remaining ranges well beyond 10 to 15 years.

8.2 Systems characterization

Datasets from eleven plants in Middle Europe that have been evaluated for a former detergents report Dall'Acqua et al. (1999) have been used for this module. The datasets have been derived from the production of sodium silicates and percarbonate and date from 1995 to 1996. Main input for steam production is process water (Fig. 8.3). One out of eleven plants used additional chemicals. This use of additives was omitted in our module. Both, natural gas and oil burners were used. 76% of the heat was produced using natural gas (Tab. 8.1).

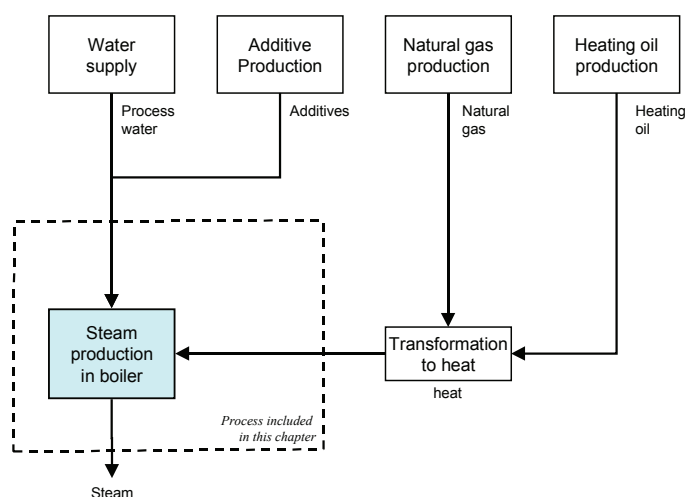


Fig. 8.3 Considered processes and system boundaries for the production of steam in the Ecoinvent database.

8.3 Data Quality and Input Data for the Database ecoinvent

The uncertainty scores established according to the method used in this study (see Frischknecht et al. (2007b)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The data represent industrial data from 11 different datasets, resulting in a quite low uncertainty. The following tables summarize the input data as well as the uncertainties for 1 kg of steam.

Tab. 8.1 Input / Output and uncertainty data for the process “steam, for chemical processes, at plant (RER)”

Explanation	Name	Location	Unit	steam, for chemical processes, at plant	UncertaintyType	StandardDeviation95%	GeneralComment
	Location			RER			
	InfrastructureProcess			0			
	Unit			kg			
Input	tap water, at user	RER	kg	1.17E+00	1	1.17	(2,3,3,2,1,4); average from 11 data sets
	natural gas, burned in industrial furnace >100kW	RER	MJ	2.60E+00	1	1.17	(2,3,3,2,1,4); 76% of heat generation
	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	RER	MJ	6.28E-01	1	1.17	(2,3,3,2,1,4); 24% of heat generation
Output	steam, for chemical processes, at plant	RER	kg	1			

8.4 EcoSpold Meta Information

ReferenceFunction	Name	steam, for chemical processes, at plant
Geography	Location	RER
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	kg
ReferenceFunction	IncludedProcesses	Input of water and energy for the production of steam. No further infrastructure is included, as the heating infrastructure is part of the respective heating modules used.
ReferenceFunction	Amount	1
ReferenceFunction	LocalName	Dampf, für chemische Prozesse, ab Werk
ReferenceFunction	Synonyms	
ReferenceFunction	GeneralComment	
ReferenceFunction	CASNumber	
TimePeriod	StartDate	1992
TimePeriod	EndDate	1995
TimePeriod	DataValidForEntirePeriod	1
TimePeriod	OtherPeriodText	
Geography	Text	value represents average steam production of 11 European chemical sites
Technology	Text	average steam production out of gas and heavy fuel oil by 11 European chemical producers
Representativeness	Percent	
Representativeness	ProductionVolume	unknown
Representativeness	SamplingProcedure	see technology
Representativeness	Extrapolations	see technology
Representativeness	UncertaintyAdjustments	none

9 Conclusion

Within this report here, average datasets for the production of the most important detergent ingredients – shown as European average data at the production site (at plant) – are established. Most of the datasets are based on information from the first part of the 1990s, but they are nevertheless considered as valuable for the representation of the production situation in the year 2000.

Most of these datasets are in accordance with the present quality guidelines of the ecoinvent project, although quite a big amount of the here shown datasets are only accessible on a cumulated level (due to confidentiality reasons). An exception are the two datasets about fluorescent whitening agents (FWA), as these two datasets have been calculated directly by the producer. This means:

- they are not harmonized with the remaining part of the ecoinvent database (e.g. they are not using the same electricity mixes, fuel chains – they have no infrastructure included – etc.);
- a in-deep analysis of these data (e.g. to see which part of the total environmental burden is due to transports, due to the electricity consumption, etc.) is not possible;

For future improvements, one objective in the area of detergent ingredients should be to get average datasets that can be published on a unit process level. Furthermore, the development in this area (new production processes but also new ingredients) should be observed and taken into account for future update work.

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